# FOLLOW MY VOICE: ASSISTED GAZE-CONTINGENT SEARCH WITH A SELF-SIMILAR AGENT IN VIRTUAL REALITY

by

PEDRO POVEDA QUEVEDO
B.S. in Computer Science, University of Central Florida, 2016

A thesis submitted in partial fulfilment of the requirements
for the degree of Master of Science
in the Department of Computer Science
in the College of Engineering and Computer Science
at the University of Central Florida
Orlando, Florida

Fall Term
2026

Major Professor: Roshan Venkatakrishnan

© 2026 Pedro Poveda Quevedo

# ABSTRACT

Virtual agents are increasingly being designed to help people carry out tasks in virtual and augmented reality [5, 25, 41]. Spoken agents have become far more capable since the rise of large language models (LLMs) such as ChatGPT [5]. Additionally, voice-based guidance has long been studied as a way to refer to objects and locations in space, starting from the seminal work *Put-That-There*, published in 1980 [4], and recent advances in virtual and augmented reality have reignited research interest in voice as an interface for these new modalities. As spoken agents become more common, the voice they use becomes a design choice in its own right.

Going further, the user's own synthetically cloned voice has also been researched as a behavioral nudge, and AI-generated encouragement in one's own voice has been perceived as uniquely engaging and personal [11]. In virtual agentic collaboration, self-similar voices increased a virtual character's likability and believability, though also its eeriness [14]. It remains an open question whether people work better, worse, or no differently with an agent that sounds like them.

The aim of this thesis is to study how a self-similar voice impacts working with an agent. We use object search as the task, since visual search is an everyday task that has been studied extensively in virtual and augmented reality [6,21,40] and it gives a direct measure of performance. The voice conditions are a Self-Similar Voice (SSV) resembling the research subject and a Selected Agent Voice (SAV) of the subject's choosing. The study is within-subjects, with each voice in its own counterbalanced block. Seated participants search for colored geometric objects arranged randomly on object planes surrounding them in 360 degrees, which includes objects initially outside their field of view. An agent, which is gaze-aware and knows the location of the object of interest, guides the user towards it. Performance is measured by search time, perceived workload by NASA-TLX, visual search by eye-tracking measures, and engagement by questionnaire.

**Draft status.** In our research we found the following results. Compared with the Selected Agent Voice, the self-similar voice [shortened / did not change / lengthened] active search time by [X]%. Perceived workload (NASA-TLX) was [lower / similar / higher] with the self-similar voice, and engagement (IMI interest/enjoyment) was [higher / similar / lower]. Eye-tracking measures showed [fewer / similar / more] fixations and a [lower / similar / higher] saccade frequency with the self-similar voice. Together, these results suggest that a self-similar voice [makes guidance more efficient / changes how the guidance feels without affecting performance / does not measurably change how people work with an agent].

# ACKNOWLEDGMENTS

I would like to thank Dr. Roshan Venkatakrishnan for his guidance and support throughout the development of this thesis research. I would also like to extend special thanks to Dr. Rohith Venkatakrishnan and Mehrab Islam for their guidance and honest feedback. Finally, I thank my wife Vanessa for her patience and support during the entire process.

# TABLE OF CONTENTS

LIST OF FIGURES . . . . . xiii

LIST OF TABLES . . . . . xvi

LIST OF ABBREVIATIONS . . . . . xvii

CHAPTER 1: INTRODUCTION . . . . . 1

1.1 Motivation and Background . . . . . 1

1.1.1 Object Search . . . . . 1

1.2 Research Questions and Hypotheses . . . . . 2

1.3 Problem Statement . . . . . 3

1.4 Overview . . . . . 4

CHAPTER 2: LITERATURE REVIEW . . . . . 5

2.1 The Voice of an Agent . . . . . 5

2.2 Self-Similar Voice . . . . . 5

2.3 Object Search as a Task in Virtual Reality . . . . . 6

CHAPTER 3: METHODOLOGY . . . . . 8

3.1 Application Design . . . . . 8

3.1.1 Apparatus and Search Scene . . . . . 8

3.1.2 Voices . . . . . 13

3.1.3 Gaze-Contingent Guidance . . . . . 13

3.1.3.1 From Gaze to Spoken Direction . . . . . 15

3.1.4 Decision Model . . . . . 18

3.1.4.1 From a Cloud Model to a Local Model . . . . . 18

3.1.4.2 Choice of Base Model . . . . . 19

3.1.4.3 State Sent to the Model . . . . . 20

3.1.4.4 How the Model Decides . . . . . 20

3.1.4.5 Why a Small Local Model . . . . . 21

3.1.4.6 Deterministic Overrides . . . . . 21

3.1.4.7 Short-Term Memory . . . . . 22

3.1.4.8 Training . . . . . 22

3.1.5 Playback Control . . . . . 23

3.1.5.1 Delivery Quality as an Experimental Requirement . . . . . 25

3.1.6 Trial Review Interface . . . . . 25

3.2 Participants . . . . . 28

3.3 Experiment Design . . . . . 29

3.3.1 Voice Manipulation and Comparison Rationale . . . . . 30

3.3.2 Design and Trial Allocation . . . . . 30

3.3.3 Starting Angle and Measures . . . . . 35

3.3.4 Research Questions and Outcomes . . . . . 35

3.4 Procedure . . . . . 37

3.4.1 Pilot Testing and Parameter Checks . . . . . 42

3.5 Data Collection . . . . . 43

3.5.1 Frame-Level Records and the Trial CSV . . . . . 44

3.5.2 From Interaction Events to Research Variables . . . . . 45

3.5.3 Geometry, Timing, and Data Quality . . . . . 46

3.5.4 Missed Targets . . . . . 47

3.5.5 Eye-Event and Pupil Measures . . . . . 48

3.5.6 Gaze Contact, Vicinity, Pointing, and Selection . . . . . 49

3.5.7 Motion, Coverage, and Instructions . . . . . 50

3.5.8 Scoring . . . . . 52

3.5.8.1 Task Performance . . . . . 52

3.5.8.2 Starting Geometry and Search Behavior . . . . . 53

3.5.8.3 Subjective Measures . . . . . 54

CHAPTER 4: RESULTS . . . . . 58

4.1 Analysis . . . . . 58

4.2 Participant Flow and Manipulation Check . . . . . 59

4.3 RQ1: Performance and Perceived Workload . . . . . 60

4.4 RQ2: Eye-Tracking Indicators of Workload and Search Efficiency . . . . . 60

4.5 RQ3: Engagement in the Search Task . . . . . 61

4.6 Exploratory Agent and Voice Experience . . . . . 62

4.7 Guidance Delivery Across Voice Conditions . . . . . 64

4.8 Discussion . . . . . 66

4.8.1 Interpreting the Research Questions . . . . . 66

4.8.1.1 RQ1: Performance and Perceived Workload . . . . . 66

4.8.1.2 RQ2: Eye-Tracking Indicators . . . . . 67

4.8.1.3 RQ3: Engagement . . . . . 68

4.8.2 Practical Relevance and Transfer to Other Tasks . . . . . 68

CHAPTER 5: FUTURE WORK . . . . . 70

5.1 Gaze-Informed Adaptation of Spoken Assistance . . . . . 70

5.2 Complementing Behavioral Measures with Brain Signals . . . . . 71

CHAPTER 6: CONCLUSION . . . . . 72

6.1 Self-Similar Voice as Task Assistance . . . . . 72

6.2 Status of the Research Questions . . . . . 73

6.3 Next Empirical Step . . . . . 73

APPENDIX A: QUESTIONNAIRE WORDING AND ADMINISTRATION . . . . . 75

A.1 Administration . . . . . 76

A.2 Audio Intake Survey . . . . . 77

A.3 Voice setup . . . . . 80

A.4 Final survey: side-by-side voice ratings . . . . . 81

A.4.1 Workload . . . . . 81

A.4.2 Activity experience: IMI . . . . . 83

A.4.3 Agent perception . . . . . 91

A.4.4 Agent rating (ARQ, draft) . . . . . 98

A.4.5 Voice and assistance experience . . . . . 100

A.4.6 Technical and comfort checks . . . 105

A.5 Final survey: comparison and debrief . . . 106

A.5.1 Demographics . . . 106

A.5.2 Voice comparison and debrief . . . 106

A.6 Fallback Voice-Recording Passage . . . 107

APPENDIX B: DATA DICTIONARY AND SCORING RULES . . . 109

B.1 Record Identity and Provenance . . . 110

B.2 Geometry and Timing Fields . . . 112

B.3 Raw Observation Records . . . 113

B.4 Scoring Rules . . . 114

B.5 Export Validation . . . 115

B.6 Measurement Definitions . . . 115

B.6.1 Coordinate Frames and Starting Geometry . . . 115

B.6.2 Trial Clocks and Pause Intervals . . . 118

B.6.3 Acquisition Rate, Validity, and Gaps . . . 119

B.6.4 Sequence-Level Latency . . . 120

B.6.5 Questionnaire Joins and Versioning . . . 122

APPENDIX C: GUIDANCE SEQUENCE TESTS AND EVALUATION RUBRIC . . . . . 123

C.1 Controlled Scenario Matrix . . . . . 124

C.2 Deterministic Labels . . . . . 125

C.3 Advisory Model Review . . . . . 126

C.4 Reporting Template . . . . . 127

APPENDIX D: DEVELOPMENT EVALUATION . . . . . 128

D.1 Software Verification . . . . . 128

D.2 Recorded Development Results . . . . . 129

D.2.1 Search Completion During Development . . . . . 130

D.2.2 Speech Delivery and Earlier Waiting . . . . . 132

D.2.3 Recorded Gaze Usability . . . . . 134

D.2.4 Exploratory Eye-Measurement Pipeline . . . . . 135

D.2.5 Outcome Panels by Voice Condition . . . . . 137

D.2.6 Eye Measures and Workload in Later Development Runs . . . . . 138

LIST OF REFERENCES . . . . . 143

# LIST OF FIGURES

Figure 3.1: System overview . . . . . 9

Figure 3.2: HTC VIVE Focus Vision HMD and controllers . . . . . 10

Figure 3.3: Virtual search room . . . . . 11

Figure 3.4: Search-scene illustration . . . . . 12

Figure 3.5: Search environment and object examples . . . . . 14

Figure 3.6: Example guidance phrases by stage . . . . . 17

Figure 3.7: Guidance architecture and playback control . . . . . 24

Figure 3.8: Model decisions and the final playback gate . . . . . 26

Figure 3.9: Trial Review interface . . . . . 27

Figure 3.10: Simulated power for the voice contrast . . . . . 29

Figure 3.11: Study procedure and trial timing . . . . . 33

Figure 3.12: Nominal plane assignment and exact starting angle . . . . . 36

Figure 3.13: Voice setup screens . . . . . 40

Figure 3.14: Gaze preparation screens . . . . . 42

Figure 3.15: Final questionnaire timing and block reference scope . . . . . 43

Figure 3.16: Raw records, derived summaries, and analytical scope . . . . . 46

Figure 3.17: Gaze acknowledgment, pointing acknowledgment, and completion . . . . . 51

Figure 4.1: Planned manipulation check . . . . . 59

Figure 4.2: Planned completion-time results . . . . . 61

Figure 4.3: Planned missed-target and workload results . . . . . 62

Figure 4.4: Planned eye-tracking results . . . . . 63

Figure 4.5: Planned engagement results . . . . . 63

Figure 4.6: Planned exploratory ratings . . . . . 64

Figure 4.7: Planned guidance-delivery results . . . . . 65

Figure B.1: Cross-relative height and live gaze refinement . . . . . 117

Figure B.2: Qualification-to-playback latency and its boundaries . . . . . 121

Figure D.1: Recorded development search durations . . . . . 131

Figure D.2: Software speech-delivery intervals . . . . . 133

Figure D.3: Recorded gaze-confirmation delay . . . . . 133

Figure D.4: Recorded gaze usability during development . . . . . 135

Figure D.5: Exploratory eye-measurement output from recorded data . . . . . 136

Figure D.6: Development outcome panels by voice condition . . . . . 140

Figure D.7: Eye-measure and workload panels from development recordings . . . . . 141

Figure D.8: Eye-measure coverage in development recordings . . . . . 142

# LIST OF TABLES

Table 3.1: Study allocation . . . . . 32

Table 3.2: Measures by research question . . . . . 37

Table 3.3: Trial CSV contents . . . . . 45

Table 3.4: Condition-specific questionnaire battery . . . . . 57

Table B.1: Record identity and scope fields. . . . . 110

Table B.2: Geometry and event fields. . . . . 112

Table B.3: Timing variables and interpretive limits . . . . . 121

Table C.1: Sequence scenarios and acceptance criteria. . . . . 124

Table D.1: Apparatus verification and outstanding validation. . . . . 129

# LIST OF ABBREVIATIONS

**AI** Artificial intelligence

**API** Application programming interface

**AR** Augmented reality

**ARM64** 64-bit Arm architecture

**CSV** Comma-separated values

**EEG** Electroencephalography

**ETD** Electronic thesis or dissertation

**ID** Identifier

**IMI** Intrinsic Motivation Inventory

**JSON** JavaScript Object Notation

**NASA-TLX** National Aeronautics and Space Administration Task Load Index

**PDF** Portable Document Format

**SAV** Selected Agent Voice

**SSV** Self-Similar Voice

**SVG** Scalable Vector Graphics

**TLX** Task Load Index (see NASA-TLX)

**UCF** University of Central Florida

**VR** Virtual reality

**XR** Extended reality

# CHAPTER 1: INTRODUCTION

## 1.1 Motivation and Background

Virtual agents are increasingly being designed to help people carry out tasks in virtual reality (VR) and augmented reality [5, 25, 41]. When such an agent speaks, its voice becomes a design choice in its own right, and instant voice-cloning methods have made it practical for that voice to resemble a specific person, including the user [32]. Self-similar voices, along with human voices measured as acoustically close to the listener's own, have some initial data showing that they can improve task behavior and user-perceived agent favorability [14, 19, 20]; whether they also reduce workload has not been established and is a question this thesis attempts to answer.

We ask the question: When an agent helping you speaks in a voice like your own, does it impact task completion in virtual and augmented reality? A self-similar voice could make an agent's instructions more engaging to follow. It could also feel eerie [14], which could have costs in enjoyability or attention, or it could have little impact at all. Each of these outcomes would be informative for anyone deciding how an agent should sound. We compare a Self-Similar Voice (SSV) with a Selected Agent Voice (SAV) that the participant chooses from a catalog, and we evaluate the impact on task performance, workload, and engagement.

### *1.1.1 Object Search*

To answer this question we need a task in which a person works with an agent repeatedly and in which performance can be measured. An immersive environment can surround the user with relevant content, such as targets, instructions, or points of interest, yet this content must often be brought to the user's attention before it can be inspected or acted on. Object search was chosen as our

method of investigation because visual search is an everyday task and has been studied extensively in VR and augmented reality head-mounted displays [6, 13, 21, 33, 40] and requires gathering and using gaze-aware agentic guidance, as the user does not know initially what the object of interest is. Researchers have investigated a variety of methods of agentic guidance, including subtle gaze guidance that steers attention across an immersive scene [12], motion cues that direct attention while preserving immersion [24], audio alerts that draw attention to objects of interest [37], and spoken directions [39]. This paper investigates spoken guidance specifically because it allows us to investigate effects between agentic voice presentation.

In Follow My Voice, participants search for objects while seated in VR. Colored geometric objects are distributed across eight vertical planes around the participant, following prior visual search studies in mixed reality and in 360-degree VR [6, 21]. Some targets initially lie outside the participant's field of view. The agent knows which object is the target, while the participant is not given its color or shape at the start. Spoken directions first guide the user toward the general vicinity of the object and then instructions become more fine-grained as the user's gaze approaches the general vicinity of the target. The participant aims a right-controller ray and presses the trigger to select the inferred target to complete a trial.

## 1.2 Research Questions and Hypotheses

The primary objective of this thesis is to evaluate whether working with an agent that speaks in an SSV, rather than an SAV chosen by the participant, changes how people perform a task and how they experience it. To measure this, we use a gaze-contingent agent and an object search task that requires agentic guidance. This thesis addresses three research questions, each with a corresponding hypothesis:

**RQ1:** Does voice self-similarity impact performance and perceived workload (NASA-TLX)?

**H1:** Guidance in the SSV leads to faster search and lower perceived workload than guidance in the SAV.

**RQ2:** Does voice self-similarity impact eye-tracking indicators of workload (IPA) and of visual search efficiency (fixations and saccades)?

**H2:** The SSV yields more efficient visual search, with fewer fixations and saccades and lower pupillary activity.

**RQ3:** Does voice self-similarity impact engagement in the object search task?

**H3:** The SSV makes the search task more engaging.

## 1.3 Problem Statement

A growing body of work examines how self-similar voices influence the experience of interacting with avatars and virtual characters [14, 20]. Less is known about whether such a voice changes how well people carry out a task with an agent's help. Previous research on object search, in which visual and auditory guidance direct attention toward objects of interest, including targets outside the user's current field of view [3, 12, 13, 24, 26, 46], provides an established task in which to ask that question.

This study explores the impact of a self-similar agent voice, using search for an initially unnamed object in VR as the task. It evaluates how receiving spatial instructions in the SSV, compared with the SAV, affects task performance and the experience of doing the task. The comparison considers search performance and perceived workload, eye-tracking indicators of workload and search efficiency, and engagement.

## 1.4 Overview

Chapter 2, Literature Review, reviews prior work on how agentic voice, and a self-similar voice in particular, shapes the experience of working with that agent, and on object search as a task in virtual and augmented reality. Chapter 3 presents the methodology: the Follow My Voice system, including its two voice configuration, gaze-contingent guidance, and decision model, followed by the participants, experiment design, procedure, and data collection and scoring. Chapter 4 presents the results framework, organized by research question, and discusses how the possible outcomes would be interpreted. Chapter 5 outlines future work, and Chapter 6 is the conclusion. Appendix A reproduces the questionnaires, Appendix B defines the recorded data, Appendix C specifies the guidance scenarios used to evaluate the apparatus, and Appendix D reports the development evaluation of the system.

# CHAPTER 2: LITERATURE REVIEW

## 2.1 The Voice of an Agent

Speech has been studied as a way to interact with objects in space since Bolt's seminal work *Put-That-There*, which combined spoken commands with pointing to help users both place and move objects in a spatial display [4], and advances in virtual and augmented reality have renewed this area of research [22, 39, 46]. Recent research indicates that the voice delivering an instruction can influence how users experience an interaction as well as how they engage with a task. Previous work has shown that listeners recognize personality in synthesized speech and respond more favorably to synthetic voices whose personality resembles their own [28]. Venkatakrishnan et al. studied an agent that asked clarifying questions during task guidance, measuring task duration, NASA-TLX workload, and perceptions of the agent [41]. We follow the same logic here, treating how well a person works with an agent and how that work feels as separate questions.

## 2.2 Self-Similar Voice

Instant voice-cloning methods have made it practical to synthesize a voice resembling a specific speaker [32], and the user's own self-similar voice has been studied for its possible behavioral effect: AI-generated encouragement in one's own voice was perceived as uniquely engaging and personalized [11]. There is also some evidence that a self-similar voice can change what people do. Kao et al. compared self-similar and assigned dissimilar avatar voices in an online programming game with 657 participants, and the self-similar voice led to more puzzles completed and longer voluntary play, although the reported intrinsic-motivation measures were not significant [20]. Aymerich-Franch et al. found that hearing one's own voice, rather than another participant's, while giving a

speech through an avatar in VR increased self-presence and social presence [1]. This demonstrates at least some preference for a self-similar voice.

Another pioneering study showed that a self-similar voice made a virtual collaborator more likable and believable, though also more eerie [14]. Rosi et al., who also tested listeners' own voices, found that familiar listeners rated clones the same as or lower than recordings on trustworthiness, attractiveness, and competence [35].. In Follow My Voice, self-similar synthesis creates a cloned voice resembling the participant, allowing the study to examine whether that resemblance changes the experience or effectiveness of receiving instructions from an agent, in terms of performance and perceived workload, eye-tracking indicators of workload and search efficiency, and engagement in the task (Section 1.2).

## 2.3 Object Search as a Task in Virtual Reality

Visual search is an everyday task [21], and object search in virtual and augmented reality has been studied extensively in head-mounted displays [6,13,21,33,40]. This has utility in augmented reality applications such as maintenance, repair, and training [3]. It is a useful setting for our question because it has established measures, and because researchers have already investigated a variety of methods to point individuals toward objects in virtual and augmented reality; from visual [12, 13] to auditory or audio-tactile [26, 39], or a combination of both [3, 46].

When an object begins outside the field of view, assistance must first support attention orientation, directing the user toward a relevant region of the scene. Once that region becomes visible, the user still needs to distinguish the intended object from nearby distractors.

Audio-based methods specifically could improve quality of life for blind and low-vision people. Spoken and spatial audio guidance has helped blind users find locations on objects and acquire targets, which may support independence [7, 39].

Research on auditory guidance also supports using sound to help users locate objects. In head-mounted AR, Binetti et al. combined visual arrows with spatialized alarm sounds to guide users toward out-of-view targets [3]. In a VR simulation of narrow field-of-view AR, Marquardt et al. found audio-tactile guidance slower than a comparable visualization technique but equally accurate, with users exhibiting better situation awareness [26]. In addition, Zhang et al.'s AR book-search study used audio and visual hints to help identify a specific book on a shelf [46]. In VR, Tanaka et al. supplemented continuous spatial audio with on-demand spoken directions such as "up" while blind participants reached for targets; this verbal mode produced the lowest error rates, fastest completion, lowest workload, and strongest preference [39]. We take heavy inspiration from this type of vocal guidance in our research.

# CHAPTER 3: METHODOLOGY

This chapter outlines the application development, experimental design, and data collection methods used in Follow My Voice to evaluate how the voice delivering spoken guidance, either an SSV or an SAV chosen by the participant, impacts task performance, workload, and engagement, with object search as the task. The guidance voice is the independent variable, and the search task and the gaze-contingent agent guidance system remain the same across conditions. The application is created in Unity 6 (6000.3.10f1) for the HTC VIVE Focus Vision headset (2448 × 2448 resolution per eye, 90-Hz refresh rate, up to 120° field of view, integrated eye tracking) using the Unity OpenXR Plugin and the HTC VIVE OpenXR integration [18].

In the experiment, participants complete one practice trial and then two counterbalanced blocks of 48 trials, with each block representing a different voice condition. Each trial begins in the same way: participants search for a specific target among colored geometric objects on eight surrounding object planes while the LLM based agent guides them with spoken directions. Participants do not know what the object is ahead of time. The source code is available at `https://github.com/PedroPovedaQ/Gaze-Contingency-Project`. Performance and eye-tracking data are collected through the headset application, and perceived workload, voice preference, and task engagement are captured through questionnaires.

## 3.1 Application Design

### *3.1.1 Apparatus and Search Scene*

The application is developed in Unity 6 (6000.3.10f1) using the Unity OpenXR Plugin (1.16.1), HTC VIVE OpenXR integration (2.5.1), and Unity XR Interaction Toolkit (3.3.1). It is deployed as

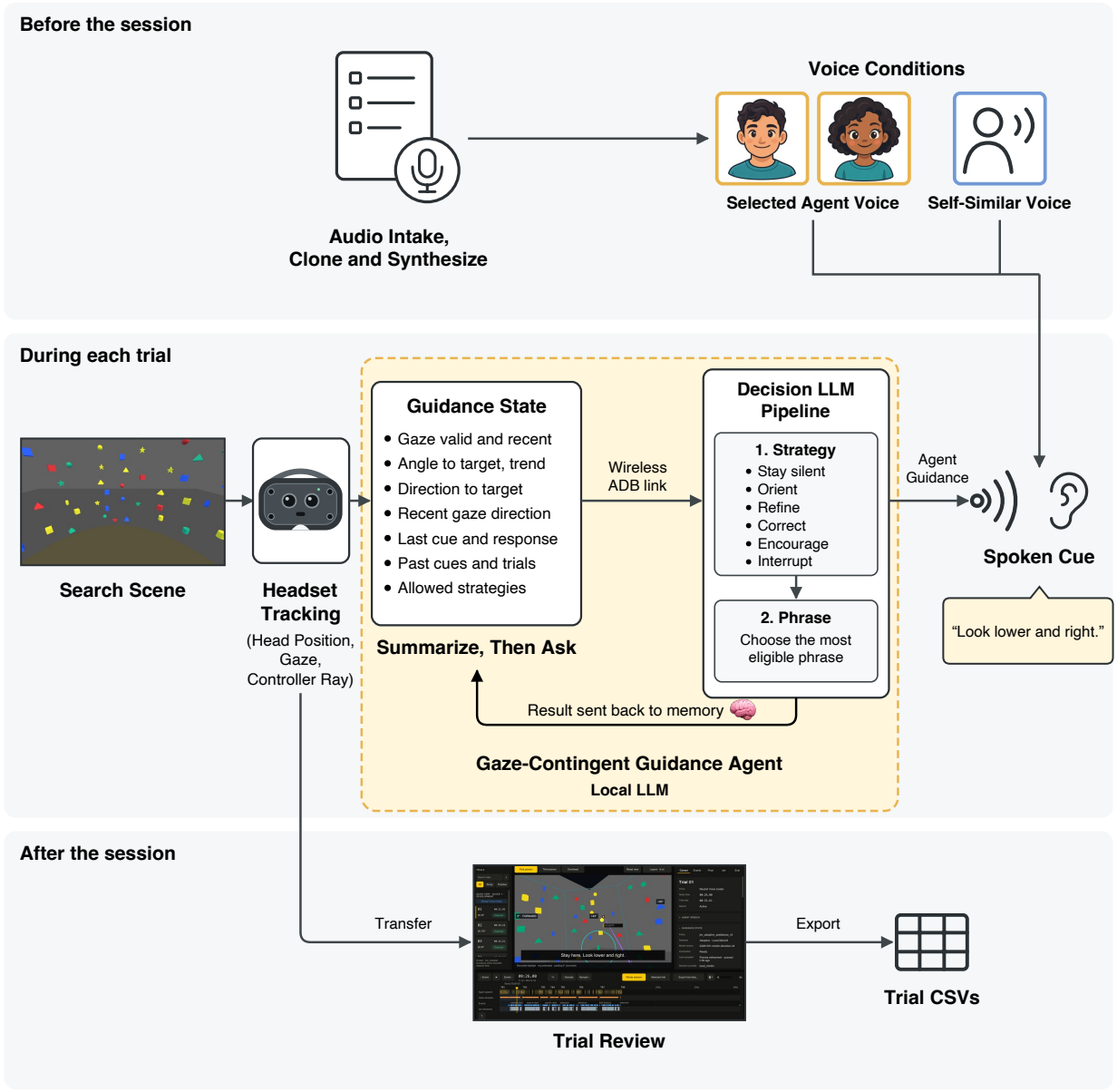



Figure 3.1: System overview. Before the session, the participant's audio intake is used to prepare phrase libraries in both voices, the SAV and the SSV. During each trial, Unity on the headset summarizes gaze, head and controller data into a compact guidance state, the local MiniLM decision model on the lab computer chooses a strategy and a prepared phrase, and the phrase plays in the assigned voice. Both voices share the same guidance system. Interrupting an outdated cue is applied by a fixed rule in Unity rather than chosen by the model. After the session, recordings are reviewed in Trial Review and exported as trial CSVs. The search scene is a Unity render of the room, and the Trial Review screen is from a development recording; neither shows participant data.

a standalone Android ARM64 application on the HTC VIVE Focus Vision head-mounted display, with rendering performed locally on the headset using the Universal Render Pipeline (17.3.0). The study uses the headset's virtual-reality configuration, in which the headset supplies head-pose and eye-gaze tracking and the tracked right-hand controller supplies pointing and trigger input. Network access supports voice preparation and the connection to the lab computer that runs the decision model. It is not used to stream rendered frames from a host computer. HTC specifies 2448 × 2448 pixels per eye, a 90-Hz standalone display refresh rate, and a field of view of up to 120 degrees [18].



Figure 3.2: HTC VIVE Focus Vision HMD and controllers. Image: HTC [18].

The search task takes place in a neutral virtual room with a gray floor and four distant gray walls (Figure 3.3). A yellow platform-esque circle on the floor is centered on the participant, and its 1.5-m radius ends where the objects begin. The objects are arranged on eight surrounding vertical object planes that are not drawn in the participant's view. The participant perceives they are in a circle with objects surrounding them. The study configuration contains 168 objects per trial, or 21 per section, at a default radius of 1.5 m from the seated origin. The room is designed to be non-descript, so the focus can be on the object search task and distractors do not affect the experiment. These

colored geometric objects hover in space and represent the target and its distractors. Figure 3.4 shows the layout from above. The objects are represented in six shapes: sphere, cube, pyramid, cylinder, capsule, and star. These objects are represented in red, blue, yellow, and green. Objects vary in height and horizontal position within their assigned area, so search requires both coarse orientation toward a an object's area and finer search within it once the user's gaze approaches the vicinity of the object. The gaze-contingent agent knows the target identity and the user's gaze data but withholds the target's identity from the user; participants *must* rely on the agent's auditory guidance to locate the target. To complete a trial, participants select the object by pointing the right-controller ray toward it and pressing the trigger. If the selection is correct, the trial ends. If it is incorrect, the participant can keep searching and select again until the trial's 25-second active-search limit expires.

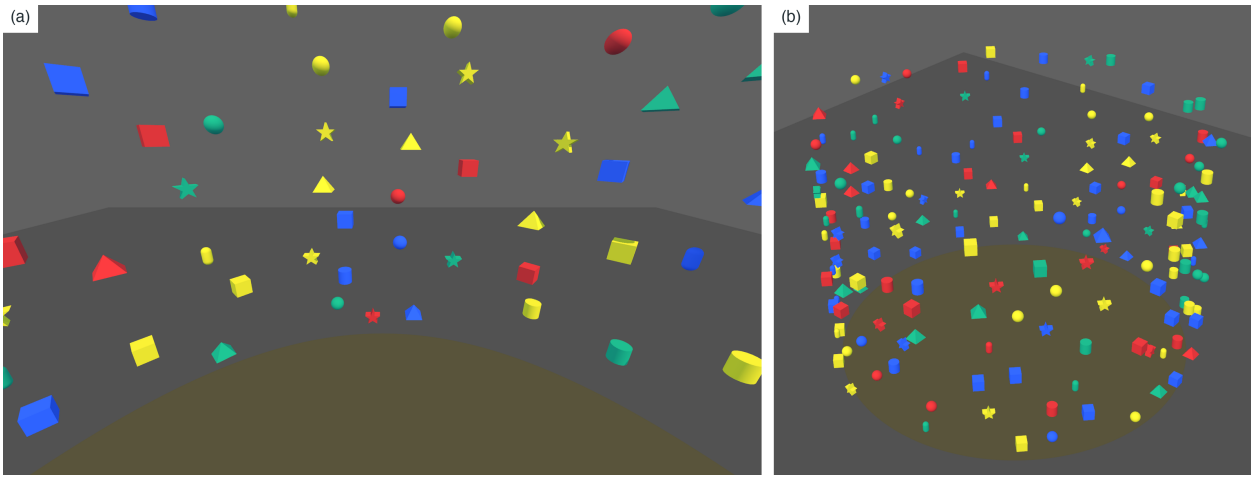



Figure 3.3: Virtual search room. (a) Seated first-person view, tilted downward to show the floor. (b) Wide view of the ring of objects around the yellow floor circle. Both are Unity editor renders of the application's actual room, layout, and object code with a fixed seed and no target highlight, not captures through the headset.

Project Setup.

Seat height is adjusted so that participants are better aligned with the readiness cross. Per-trial layouts are procedurally generated from a designated theta, the horizontal angle between the trial's starting direction and the target object (Section 3.3.3), with local clutter around the target held within fixed bounds so that trials stay similar in difficulty (Section 3.3).

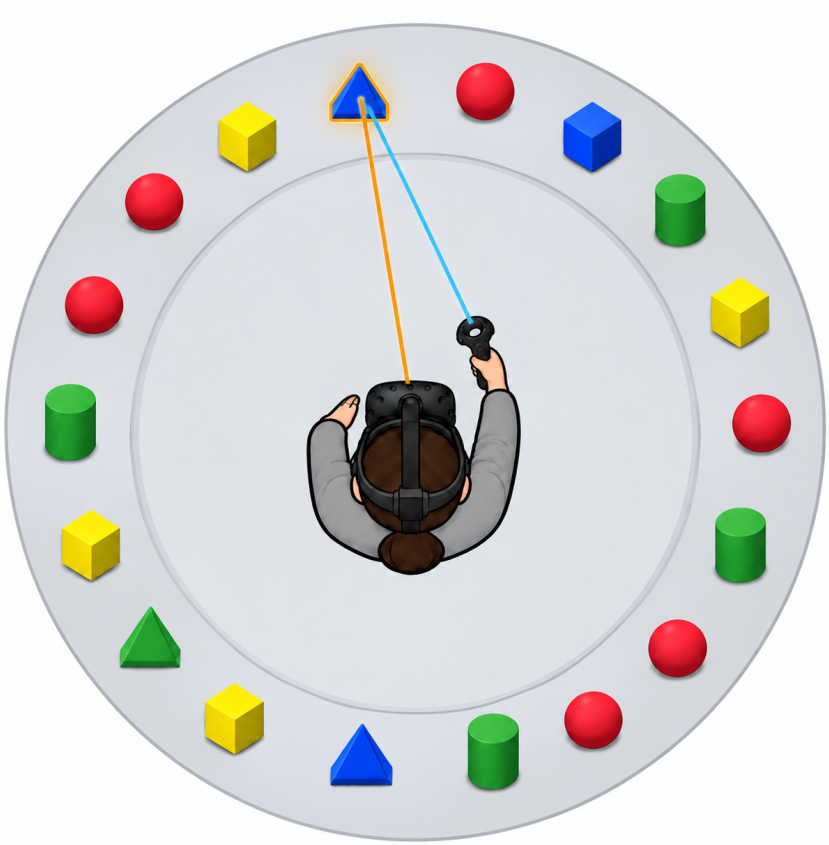


Figure 3.4: Top-down simplified schematic of the search scene. The seated participant is surrounded by colored geometric objects in a continuous ring. The gaze ray (orange) and controller ray (blue) converge on the highlighted target, representing the state in which the participant is ready to select with the trigger.

Figure 3.5 complements the schematic with a reconstruction from an actual headset development recording.

### *3.1.2 Voices*

Voice cloning, which replicates a speaker's voice from a reference recording, has become practical with recent instant-cloning methods [32]. In this study, all voice audio is pre-synthesized with the commercial ElevenLabs service, which supplies the voices in both the SAV and the SSV conditions. To support the self-similar condition, the participant's spoken answers during the audio intake provide the samples for cloning. Participants hear a voice clip of each agent for the Selected Agent Voice condition and choose the one they prefer the most for their experiment . All voice clips are level-matched to have similar volume. The system logs the SAV, and voice clone settings per participant.

### *3.1.3 Gaze-Contingent Guidance*

Prior guidance systems combine initial orientation with finer guidance toward the target [23, 39]. Spoken directions in Follow My Voice follow a similar progression.

The Follow My Voice system continuously collects eye-gaze origin, direction, and gaze validity; head pose; right-controller pose; and gaze ray-object intersections. A local state machine combines these observations with the target geometry to track the participant's progress through the search: turning toward the target plane, refining within the object plane, acquiring the target with gaze, pointing with the controller, and becoming ready to select. The gaze-contingent agent in Follow My Voice issues new gaze-dependent guidance only when gaze data are valid and recent.

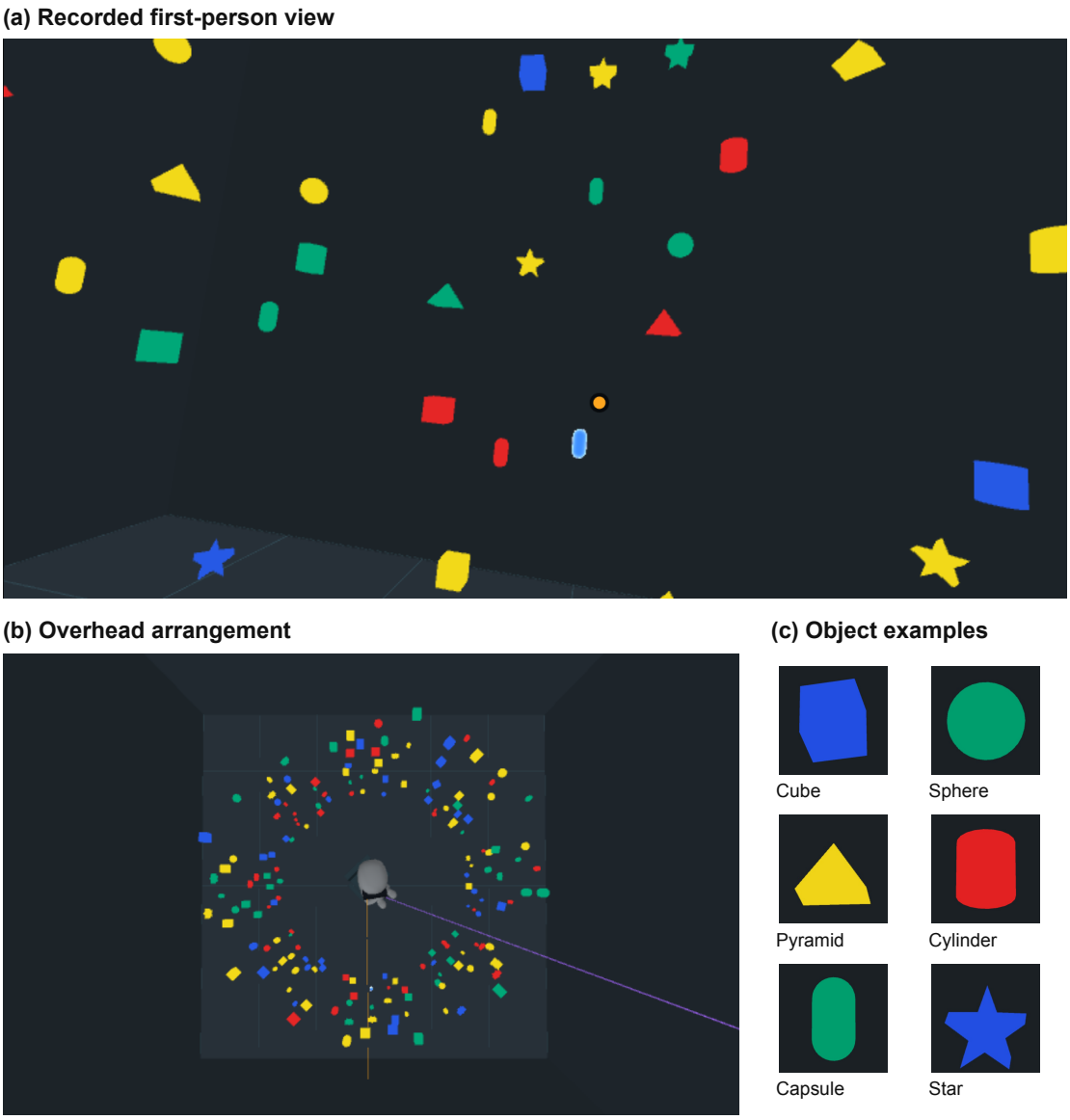



Figure 3.5: (a) First-person view of colored objects on surrounding vertical planes. (b) Overhead view of the eight-plane arrangement. (c) Enlarged examples of the six shape types at independent scales. Images are reconstructed in Trial Review from a headset development session recorded before the current room and the plane-height reduction, with the plane outlines hidden in the viewer; the object layout reflects the earlier, taller planes, and the floor grid and background come from the viewer. They are development material, not participant-study results.

#### *3.1.3.1 From Gaze to Spoken Direction*

The direction for the next voice cue is first computed geometrically by the application before the decision model is ever consulted. Each valid gaze sample supplies an eye origin $\mathbf{o}$ and a gaze direction $\mathbf{g}$, and the target's position $\mathbf{p}$ is known at the start of each trial round. Both $\mathbf{g}$ and the target vector $\mathbf{d} = \mathbf{p} - \mathbf{o}$ are expressed relative to the seated forward direction, so the computation does not depend on how the room is oriented. A sample whose direction lies within about 12° of straight up or down is treated as unusable rather than as a departure from the object plane, because left and right are undefined there. The eight object planes form a ring at 45° intervals around the chair.

Angular error.

The horizontal, or yaw, error $\psi$ is the signed angle about the world up axis between the horizontal projections of $\mathbf{g}$ and $\mathbf{d}$; it is positive when the target lies to the participant's right. The starting angle $\theta_0$ (Section 3.3.3) gives the yaw separating the readiness cross from the target before a trial begins, whereas $\psi$ tracks the remaining yaw error from the current gaze throughout the search. The vertical, or pitch, error is the difference in elevation,

$$\phi = \mathrm{elev}(\mathbf{d}) - \mathrm{elev}(\mathbf{g}), \qquad \mathrm{elev}(\mathbf{v}) = \mathrm{atan2}\left(v_y, \sqrt{v_x^2 + v_z^2}\right) \frac{180}{\pi}, \tag{3.1}$$

which is positive when the target is higher than the point of regard.

Turning toward the object plane.

Outside the target plane, the sign of $\psi$ selects "left" or "right." When the target is nearly behind the participant ($|\psi| \geq 150^\circ$), the first chosen direction is retained, so small head movements across the rear seam cannot reverse an instruction already given, and the remaining turn is measured along that route. Progress cues are earned in 45° steps, one object plane at a time: crossing the same step again after turning back does not produce another "keep going."

Refinement within the object plane.

Inside the target plane, $\phi$ and $\psi$ are converted to one of nine directions: up, down, left, right, the four diagonals, or aligned. A vertical component is added when $|\phi|$ exceeds 3°, and a horizontal component when $|\psi|$ exceeds 5°. Once a component is being spoken, it persists until the error falls below 1.5° vertically or 3° horizontally, and an aligned state persists until the error exceeds 4.5° vertically or 7° horizontally. These separate entry and exit thresholds prevent small eye movements from flipping the instruction back and forth. They are development settings rather than values drawn from prior work. Within the object plane, the gaze ray must also intersect the object plane so that its vertical position can be measured; otherwise no new direction is issued.

Coarse-to-fine cues.

Guidance moves from coarse to fine, and yaw comes first. Outside the target plane, cues address only the horizontal turn, with wording such as "Turn way left" at 90° or more and "Keep turning left" above 45°. Pitch enters once the participant reaches the object plane. The sign of $\phi$ decides whether a vertical cue is given, but its wording comes from a coarse measure: the height difference between the target and the point of regard, as a fraction of plane height. Up to 20% of the object

plane produces “a little higher” or “a little lower,” up to 50% produces “look higher” or “look lower,” and larger differences produce “look way higher” or “look way lower.” Diagonal cues add the horizontal direction to the vertical one, as in “a little higher and left.” Short continuations such as “a bit more” follow measured progress near the target.

Figure 3.6 gives example wording at each stage. Both voices use the same phrases, and every phrase is prepared in advance for each participant.

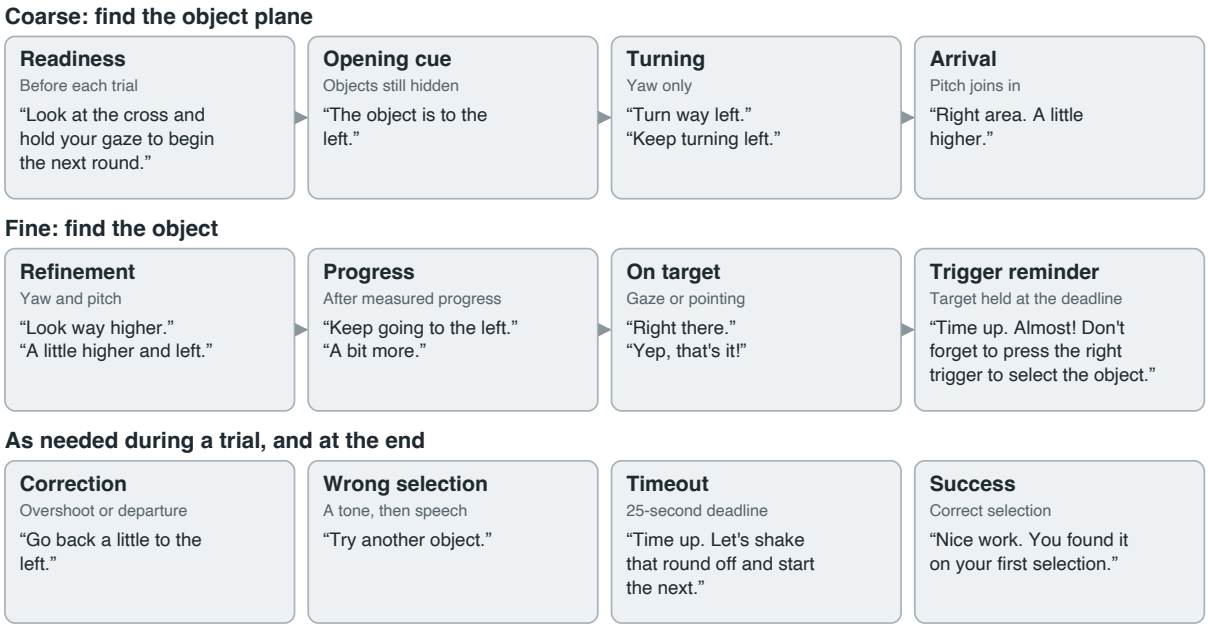



Figure 3.6: Example guidance phrases by stage. Quotes are exact phrases from the prepared library, which contains several variants of each cue; left and up stand in for the matching right and down wording.

Consistency with the visible highlight.

The object closest to the gaze ray, among those whose vicinity contains it, receives a visible highlight, on any plane. If a vertical instruction would contradict that highlight’s position relative to the

target, for example “down” when the target is above the highlighted object, the vertical component is dropped and only the horizontal component is spoken. The system never substitutes the opposite vertical direction.

Target acquisition.

New directional guidance stops once gaze has remained within the target’s vicinity ring for at least 100 ms or the controller ray is on the target (Section 3.5). The phrases offered to the decision model are limited to those whose direction matches the computed one, and each selected cue is checked against a fresh evaluation of this geometry before playback (Section 3.1.5).

### *3.1.4 Decision Model*

Follow My Voice uses a small, fine-tuned language model to decide when and how the agent speaks. The model is based on MiniLM [43], a compact transformer distilled from a larger language model, in its six-layer `all-MiniLM-L6-v2` sentence-encoder form [34], with about 22 million parameters. Unlike a large language model, it does not generate free-form text. It is fine-tuned as a classifier that chooses among a small set of guidance strategies, and a second fine-tuned copy ranks the prepared phrases that carry out the chosen strategy. Both voice conditions use the same model and the same phrase library.

#### *3.1.4.1 From a Cloud Model to a Local Model*

Guidance decisions were initially made by Jev, a cloud-hosted large language model. Each decision therefore required a request to a remote service, and the resulting latency proved too high for guidance that must keep pace with the participant’s head and eye movements. In the development

sessions of October 19–20, 2026, for example, one gaze confirmation reached playback 2.19 s after the first qualified request, following repeated requests to the cloud model (Section D.2). The cloud model was therefore replaced with the small local model described in this section, which answered roughly five to six times faster in comparable requests (Section 3.1.4.5). The change did not alter the application's geometry, phrase library, or playback checks. The local model is not a compressed copy of Jev: it was fine-tuned from the pretrained MiniLM checkpoint on synthetic states labeled by an authored reference policy, and Jev's responses were not used as training labels (Section 3.1.4.8).

#### *3.1.4.2 Choice of Base Model*

The base model was chosen for three properties of the task. First, guidance phrases are short spoken sentences, and many differ only in emphasis or degree, such as "Turn right," "Keep turning right," and "A little more right." Sentence-BERT models are trained so that sentences with similar meanings receive similar representations [34], so the starting checkpoint already distinguishes and relates such variants. This prior knowledge allows a moderate amount of synthetic training data to suffice for both the strategy choice and the phrase ranking. Second, the six-layer model is small enough to answer within the time scale of head and eye movement on the lab computer's processor (Section 3.1.4.5). Third, an encoder that classifies and ranks can return only options it was offered. Every phrase is synthesized in advance in both voices, including the participant's clone, and cached on the headset, so a phrase the model selects always has recorded audio in both conditions. A generative model could compose a sentence for which neither voice has a recording, and the two conditions could then diverge in wording. The checkpoint is released under the Apache-2.0 license and pinned to a fixed revision, so the starting weights can be reproduced. Its pretraining on general English text carries no knowledge of the task geometry, which is one reason geometric rules and gating remain outside the model (Section 3.1.4.6).

#### *3.1.4.3 State Sent to the Model*

At each decision point, Unity assembles a structured state of the search. It records whether gaze evidence is valid and recent, whether the participant is inside the target plane, and whether target gaze or controller pointing qualifies. It adds the combined angular error to the target and its distance band, how that error has changed over the last few seconds (trend, net reduction, and gaps in tracking), the last cue and how the participant responded to it, the number of cues and wrong selections so far, the outcomes of previous trials, the time since speech ended, and the strategies currently allowed. Three fields give the model a sense of direction. *Needed* is the direction Unity has computed to the target, such as left or up and right. *Moving* is the recent direction of gaze over the last two seconds and whether it is toward, away from, or across that direction. *Started* is a band for the trial's pre-cue angle and side, such as a large turn left, on the same plane, or behind. These use the same-trial snapshot and recent history only; they are marked unknown when the snapshot, the history, or tracking is missing. Because a language encoder handles words better than arithmetic, they are sent as words rather than numbers. The target's identity, shape, and color are never included. The model receives this state as a compact text description of at most 384 tokens.

#### *3.1.4.4 How the Model Decides*

The strategy head chooses one of seven options: observe for one second (stay silent and reassess), brief or explicit orientation toward the target plane, brief or explicit refinement within the object plane, correction, or encouragement. Unity offers only the strategies that are valid in the current situation, and the model can choose only among them. When the chosen strategy is to speak, the phrase ranker scores the prepared phrases offered for that strategy, taking into account how often and how recently each phrase has played, and the highest-scoring phrase is scheduled. The model's scores are used only to choose among the offered options; they are not calibrated probabilities.

#### *3.1.4.5 Why a Small Local Model*

Guidance has to keep pace with head and eye movements that change within a fraction of a second, and a cue that arrives late can send the participant somewhere they have already looked. The model therefore runs on the lab computer beside the headset and answers over a local connection. In development measurements on that computer, a complete two-stage decision (strategy and phrase) over the local connection had a median of about 34 ms (95th percentile 41 ms), and a direct call to the strategy model about 7 ms. For the first phrase on reaching the target plane, the local model responded in 171 ms, whereas the cloud large language model previously used for these decisions, Jev, responded in 820 ms in a single comparable request; an earlier local version showed a median of 30 ms against 182 ms for the cloud model. These timings were taken on the desktop and do not include headset networking or the onset of audible speech. Running locally also keeps decisions during the task on the lab computer, and because the model can only select among prepared phrases, it cannot produce unexpected wording.

#### *3.1.4.6 Deterministic Overrides*

Guidance is otherwise driven entirely by the model; fixed rules take precedence only in a few specific situations. No decision is requested when gaze is invalid, when the participant is already looking at or pointing at the target, when the guidance state is not a search state, or when audio is not ready, and a server-side gate forces silence under the same conditions. On the first arrival at the target plane in each trial, the strategy is fixed to an explicit refinement and the model chooses among four phrases that confirm the area and give the remaining direction, such as “Right area. Look lower and right.” When the participant reverses direction, an obsolete cue is faded out without replacement. A reminder plays only at the 25-second deadline when the participant is already on the target, and trial openings, target confirmations, and success feedback come from prepared

local phrases. Only one request may be pending, requests are at least 0.75 s apart, and any decision older than 3.25 s or no longer matching the participant, trial, voice, or policy version is discarded. If a request fails, the agent stays silent and retries rather than falling back to another provider.

#### *3.1.4.7 Short-Term Memory*

The state includes a short-term memory of the participant's behavior. A spatial history samples the error to the target every 0.5 s over the last 12 s and is cleared when tracking is lost for more than 1.25 s. A cue history keeps the last eight cues in the trial and the participant's response to each, measured as the change in error over the following 3 s. A phrase history records how often and how recently each phrase has played, and summaries of the previous six trials carry information across trials. Trial-level memory resets at each new trial and participant-level memory at each new participant. A separate speech memory of the last four speech events within 15 s supports the playback checks described below.

#### *3.1.4.8 Training*

The model was trained on synthetic decision states generated to cover the situations guidance must handle: stalled, receding, oscillating, steady, and mixed search trajectories, departures from the target, different silences and numbers of prior cues, and failures of the gating conditions. The scenarios came from a seeded generator written with the help of a large language model coding assistant, and they were revised against real development trials. For example, an early model missed departures from the target seen in recordings, so the data were rebuilt to cross departure events with every trajectory pattern, and examples of long-way turns that could not occur in Unity were corrected. Each state was labeled by an authored reference policy, so that the model learns to reproduce that policy quickly from the compact state. Individual recorded states were not used as training examples; real

development recordings were kept as regression tests for the strategy stage. The current version continues earlier weights with the direction fields added. The strategy model was fine-tuned with AdamW for six epochs on 2,240 training states balanced across the seven strategies, with 448 validation and 448 test states generated with different sampling cadences and combined gate failures, and the phrase ranker for four epochs on 768 training examples, with 160 for validation and 160 for testing. The authored policy uses direction in two situations: briefly observing a recent turn along the cue that goes the long way around, and preferring an explicit cue when the participant is moving away in the opposite direction. On the new test set, the strategy model agreed with the reference policy on 433 of 448 states (96.7%; 435 of 448, or 97.1%, after the deterministic gates), compared with 391 of 448 (87.3%) for the previous version without direction. The phrase ranker agreed with its reference on 153 of 160 cases (95.6%) and avoided repeating a phrase in all 78 opportunities to do so. On the previous version's test set, the new model agreed on 431 of 448 states, against 424 for the previous version. Because the synthetic data were revised during development, these sets are held-out development evidence rather than an untouched evaluation, and the new test set was built to include situations the previous input could not separate. The model also still failed the three fixed test cases of a long-way turn along the cue, so it does not reliably recognize them. These figures measure agreement with the authored policy, not usefulness to participants, which the user study will assess.

### *3.1.5 Playback Control*

Because decisions arrive asynchronously, the participant's state may change before a decision returns. A local speech scheduler therefore revalidates each decision against the current state, evidence, pacing, and object intersections before playback, and discards decisions that are stale or conflict with the current situation. Qualified gaze and controller acknowledgments, together with eligible turn-progress milestones, can be handled locally and played from prepared phrases without

waiting for a new model decision. These local cues remain subject to the same checks. When a qualified target acknowledgment would replace misleading search speech, a local fast path avoids waiting for interruption advice. This path protects the first 0.35 seconds and final 0.5 seconds of the current phrase, retains a 2.5-second interruption cooldown, and fades the audio over 150 ms. The fast path keeps the separate gaze and controller hold requirements, and fresh evidence is checked before the fade, after it, and at playback. Setup, completion feedback, and existing acknowledgments are protected from interruption. Each completed or interrupted utterance updates both the memory and the event log. Model latency, deferral, cancellation, fallback, and playback onset are logged separately to support delay attribution. When a trace lacks the required events, the source of a delay is recorded as unknown rather than inferred from the spoken phrase alone.

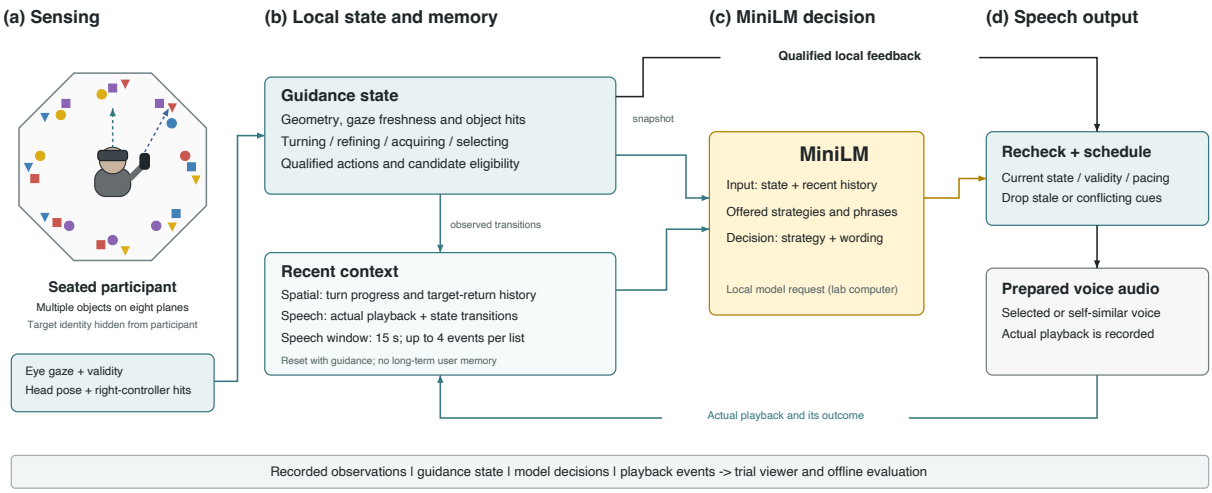



Figure 3.7: System architecture. (a) Gaze, head, and controller data are sensed. (b) A local state machine and short-term memory determine which guidance actions are valid. (c) The MiniLM decision model chooses the strategy and wording from eligible options. (d) The scheduler revalidates each cue against the current state before playback. Actual playback feeds back into memory. The scene inset is schematic and does not reflect stimulus count or any participant-visible target cue.

Figure 3.8 shows the final playback gate. The model's decision is based on an earlier snapshot, so it is treated as advice. A decision to speak results in playback only if the cue passes current-state

checks. Provider-selected phrases and qualified local feedback both pass these checks before the prepared clip starts. Readiness and pacing are checked during scheduling, and the queued relevance callback is evaluated again immediately before audio playback. The same local gate applies to every strategy the model can choose.

#### *3.1.5.1 Delivery Quality as an Experimental Requirement*

Both voice conditions use the same versioned guidance mechanism, although delivered sequences depend on participant behavior and clips may differ acoustically. Voice identity, library metadata, playback onset, and delivery faults are therefore recorded so that each can be traced to its condition.

This record matters because a delayed or confusing cue could alter search performance or a participant's judgment of the voice and then be mistaken for an effect of voice condition. Appendix C defines guidance sequence tests and an evaluation rubric for checking the implementation.

### *3.1.6 Trial Review Interface*

Figure 3.9 shows the standalone Trial Review application reviewing a complete recorded session. The scene and inspector share a timeline with speech, gaze, events, and decision tracks. The application can also receive headset records for live review. During a session, a monitor stays visible over every view, including the intake window, and shows the participant's current screen, time on that screen, elapsed session time, trials completed out of 96, the current voice block, gaze tracking status, faults, and the age of the last update, with a warning if updates stop or the headset is paused. When trial poses are available, it also shows a first-person reconstruction, which is not headset video. The monitor is read-only and sends no commands to the headset. Transfer and receipt times are kept separate from source event times, so late arrival in the viewer does not indicate delayed

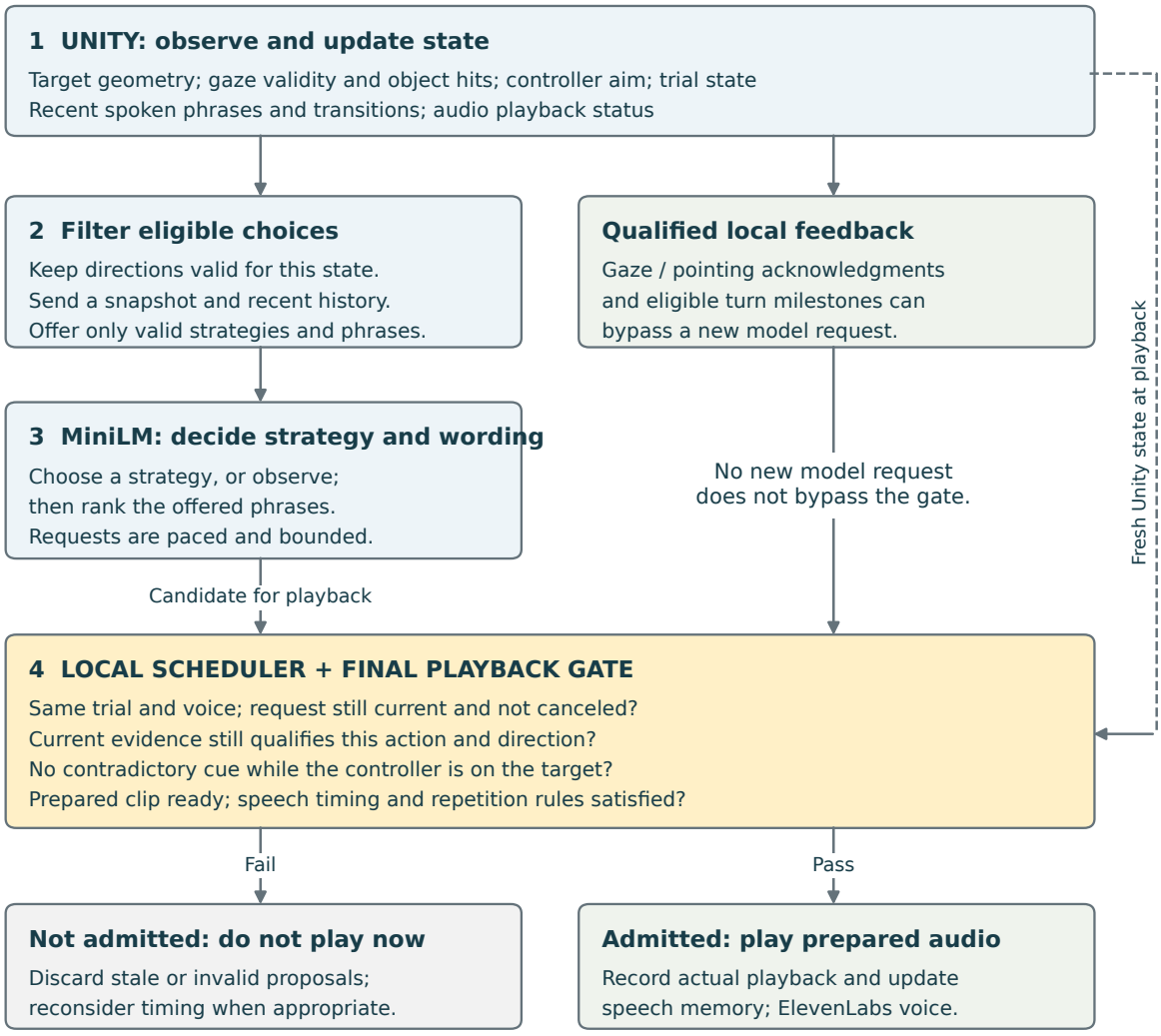



Figure 3.8: Decision-model choices and the final local playback gate. The dashed input indicates that the gate evaluates fresh Unity state, not only the snapshot originally sent to the provider. Box 4 groups scheduling checks with the final relevance check immediately before playback. Qualified local feedback bypasses a new model request but retains this boundary. Waiting is bounded by local policy, and a failed request stays silent rather than substituting a phrase. This schematic summarizes ordinary guidance and local feedback rather than every interruption, opening, or success-feedback path. It describes control flow, not measured response latency.

guidance on the headset. The session in the figure is a short development recording made on the headset by the author in the quick test mode, which uses the neutral voice in a single block, so it illustrates the interface rather than a participant recording or performance result. The application labels reconstructed highlights and estimated rings as such.

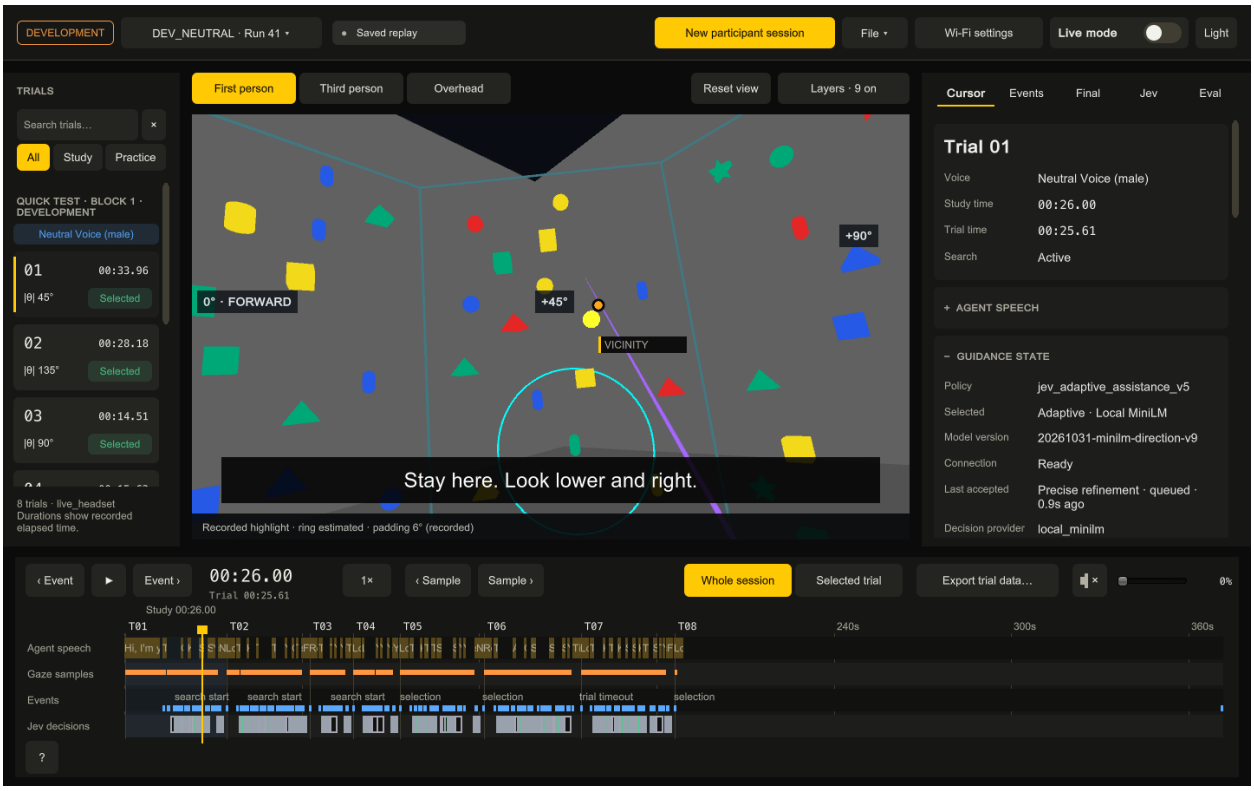



Figure 3.9: Trial Review reviewing a recorded session. The trial list, first-person reconstruction, inspector (here showing the guidance state and decision model), and timeline of speech, gaze, events, and model decisions share one clock. The subtitle shows the cue playing at that moment. The session is a development recording made on the headset by the author in quick test mode; it is a reconstruction from recorded data, not headset video, and not participant data.

## 3.2 Participants

Eligible participants are consenting adults able to understand the task, hear the guidance, view the scene with usual correction and safely use the seated setup and right controller. Visual correction and prior XR and voice-assistant experience are recorded. Sample size was planned by simulating the 96-trial design and fitting the planned mixed-effects model to each simulated dataset, with even participant counts from 12 to 60 and with assumed trial-level noise, individual differences in the response to voice, timeouts, incomplete trials, and dropout (Figure 3.10). The study will begin by enrolling 20 participants, 10 in each voice order, which is what is feasible for this thesis. Under the middle assumptions, this gives about 74% power to detect a 15% difference in search time between the voices and about 39% power to detect a 10% difference. The corresponding values are nearly 100% and 93% under the optimistic assumptions, and 26% and 13% under the pessimistic ones. Reaching 80% power for a 10% reduction would require about 52 participants under the middle assumptions. Added trials help only modestly: at 20 participants, the 10% power under the middle assumptions is 39% with 96 trials and 35% with 64, because individual differences in the response to voice are reduced only by adding participants. For the paired NASA-TLX and IMI interest/enjoyment comparisons, about 17 usable pairs reach 80% power only at a large standardized effect, about 0.8 after Holm adjustment [17], and these scores do not benefit from more trials. The study is therefore sensitive mainly to large effects: a nonsignificant result will not show that voice has no effect, and results will be reported as effect estimates with confidence intervals. If enrollment is extended, the decision will be made without inspecting the voice comparison. These values are planning estimates. The variance inputs are assumptions rather than measurements, because the development recordings contain no paired voice data, and they will be revisited after pilot sessions.

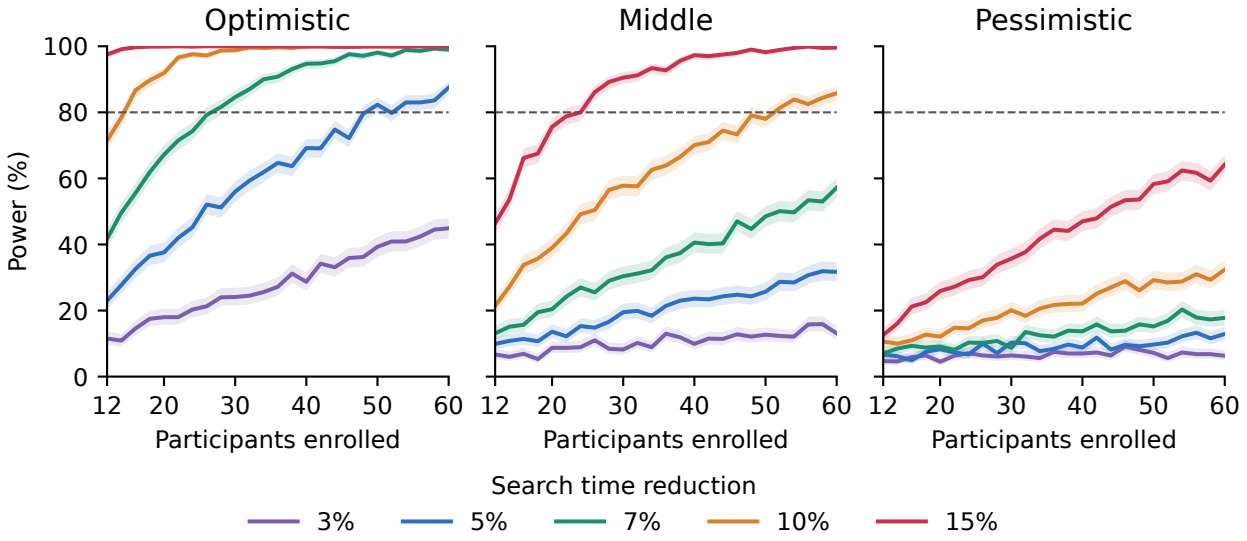



Figure 3.10: Simulated power for the voice contrast by participants enrolled, under three sets of assumptions. Values are synthetic planning estimates for the mixed-effects model fitted to successful searches, including dropout, not measured effects. Each curve point uses at least 1,000 simulated studies, and the values quoted at 20 and near 52 participants use more. The dashed line marks 80% power.

## 3.3 Experiment Design

The user study uses a within-participant design to answer the research questions in Section 1.2.

The study design draws on Chiossi et al.'s visual-search study, which combines search outcomes with eye-tracking measures [6]. This study keeps that separation between task performance and visual behavior but changes the research question to voice identity, initially withholds the target identity, and uses a researcher-selected 25-second search limit. Complementary work by Pfeil et al. at UCF compares eye–head coordination in virtual and physical viewing [31]. That work motivates retaining head motion alongside eye direction when interpreting gaze in this application.

**Draft status.** This section specifies the proposed confirmatory procedure. The implemented snapshot supports one practice trial, two 48-trial blocks, and a final questionnaire handoff. It also records pre-cue starting geometry and balances target direction and height within each block. Pilot acceptance rules remain unresolved.

### *3.3.1 Voice Manipulation and Comparison Rationale*

This thesis compares two ways of personalizing assistance: a synthetic voice derived from the participant's recorded speech and an SAV chosen by that participant from the supplied catalog. Letting participants choose this alternative ties the comparison to a design choice a user could face. The within-participant task provides repeated observations under both configurations. Similarity ratings indicate whether participants heard the clone as resembling themselves, whereas performance and experience measures compare how each voice functions as guidance.

Gaze contingency and the agent's decision rules are common to both conditions. Because the target identity is initially hidden, participants receive repeated guidance in the assigned voice on every trial. The decision model, memory, and gaze awareness are held constant rather than varied as independent factors.

### *3.3.2 Design and Trial Allocation*

The study uses a within-participant comparison of two voice conditions: an SAV and an SSV. Both conditions use the same guidance policy, semantic phrase library, selection method and visual-feedback settings, although actual phrase sequences may differ because guidance responds to behavior. Because the SAV is chosen by the participant whereas the clone is derived from the participant's speech, the two voices may also differ in preference, gender presentation and synthesis quality. The

self-similar clips also receive the inner-voice audio treatment described in Section 3.1.2, whereas SAV clips do not. The contrast therefore cannot isolate self-similarity from these features.

The researcher assigns each participant to Order A (self-similar first) or Order B (SAV first) from a roster kept outside the application, 10 per order; the application records the assignment but does not randomize it or enforce the quota. Counterbalancing the order distributes possible carryover from the first block across the two voices [44], although it does not remove carryover. Each participant completes one practice trial followed by two measured blocks of 48 trials, one per voice. The practice trial uses the first voice in the assigned order (the SSV for Order A, or the SAV for Order B) and is excluded from measured outcomes. A seed makes the trial schedule reproducible. Nominal horizontal-angle quotas are identical across blocks, with trial order and left/right assignments randomized subject to those quotas. Within each block, each bilateral angle category (45, 90 and 135 degrees) has six left and six right assignments. Each block is divided into six shuffled groups of eight trials, each of which contains every signed plane step. Trial order is further constrained so that each of the eight object planes is the target on exactly six trials per block. Because every complete group advances four planes, this constraint makes the 0-degree and 180-degree steps less common as the first or last step of a group, although their totals do not change. Each of the 24 shape-and-color identities appears twice in every block.

Objects vary in position within each plane of the shared layout, so each trial specifies the exact starting angle and target height in addition to the assigned object plane. Vertical allocation is defined by target-center height relative to the readiness cross rather than the participant's seated eye height. The cross is placed at the midpoint of the permitted object-height range, and each assigned target center lies between 10 and 45 percent of that range above or below it. The lower bound keeps every target clearly above or below the cross, and the upper bound keeps targets away from the floor and the top of the object plane. Pilot sessions will assess target visibility and seated comfort at both extremes.

Table 3.1: Study allocation. Nominal plane angles are schedule categories, not exact object bearings or accumulated head rotation.

| Parameter | Allocation |
|---|---|
| Voice conditions | SAV; SSV |
| Measured trials | 48 per block; 96 per participant |
| Practice | One trial; excluded from outcomes |
| Absolute nominal plane step | Per block, six trials each at 0 and 180 degrees and twelve each at 45, 90 and 135 degrees, so that each of the eight signed plane steps occurs six times |
| Target plane | Six targets on each of the eight object planes per block |
| Vertical allocation | 24 targets above and 24 below readiness-cross height per block; three above and three below at each signed plane step |
| Local clutter | Within 0.50 m of the target: 5 to 8 objects, of which 2 to 4 share its color or shape |
| Condition order | Counterbalanced Order A and Order B |
| Active search limit | 25 seconds, excluding explicit pauses |
| Selection | Right-controller ray and a trigger press |

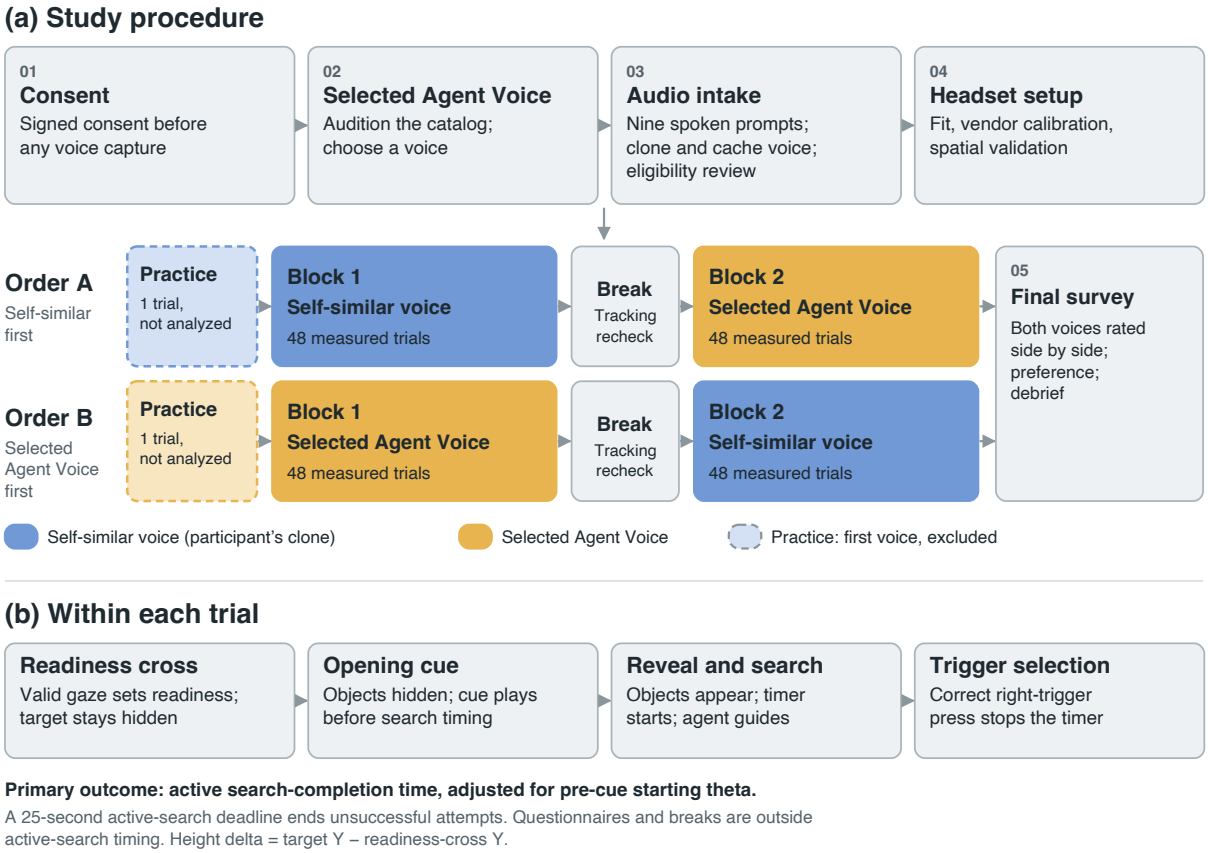



Figure 3.11: Study procedure and trial timing. (a) Shared setup precedes the two counterbalanced orders; color marks the voice heard in each block, blue for the SSV and gold for the SAV. The practice trial uses the first assigned voice and is excluded from outcomes. (b) Opening guidance precedes object reveal; active completion time ends at correct right-trigger selection. The success path shown here is bounded by a 25-second active-search deadline; expiry ends the attempt without a successful completion time. Questionnaires and breaks remain outside active-search timing.

Nominal angle steps are defined relative to the readiness (start) object plane, which is the previous target plane. For the first measured trial, this reference is the practice target plane. The first measured cross is centered on its object plane. Each later cross is shifted sideways by a seeded offset of up to 20 percent of the usable half-width. Target placement happens when readiness completes, while all objects remain hidden and before the opening cue. At that moment, the application rotates the horizontal bearing from the participant's eye origin to the cross by the assigned signed angle, places the target where that bearing meets the assigned object plane, and arranges the remaining objects around it. The exact starting angle defined below therefore equals the assigned category at that instant. If the required position falls outside the object plane, the session stops so the participant can recenter, rather than substituting a different angle or trial. Final positions depend on the saved eye origin as well as the seed, so each layout is reconstructed from the saved object positions.

Layouts remain random, but local clutter around the target is bounded so that trials are similar in difficulty. A layout is accepted only if, around the target's final position, the three-dimensional center-to-center count within 0.50 m is 5 to 8 objects in total and 2 to 4 confusable objects, meaning objects that share the target's color or shape, with at least 0.20 m between every pair of object centers. The rule is evaluated when readiness completes, after the target's position has been fixed from the eye origin, on every measured trial and the practice trial in both voice conditions. Layouts are resampled from the trial's seed for at most 256 attempts, each of which redraws distractor positions while keeping every object's identity and 21 objects on each plane. There is no fallback layout: if no layout meets the rule, the failure is recorded and the session stops before any cue or reveal. The bands were set from the local clutter observed in unconstrained synthetic layouts and are design parameters, not values drawn from prior work.

### 3.3.3 Starting Angle and Measures

Each trial is characterized by its starting angle $\theta_0$, the horizontal angle between the direction of the readiness cross and the direction of the target, as seen from the participant's eye position. Let $\mathbf{o}$ be the eye origin saved when readiness completes, $\mathbf{c}$ the readiness-cross center, and $\mathbf{p}$ the target center, all in the same world coordinate frame. With $\Pi(\mathbf{v}) = (v_x, 0, v_z)$ projecting a vector onto the horizontal plane, define

$$\mathbf{a} = \Pi(\mathbf{c} - \mathbf{o}), \qquad \mathbf{b} = \Pi(\mathbf{p} - \mathbf{o}). \tag{3.2}$$

The starting angle is then

$$\theta_0 = \operatorname{atan2}(\hat{\mathbf{y}} \cdot (\mathbf{a} \times \mathbf{b}),\ \mathbf{a} \cdot \mathbf{b})\ \frac{180}{\pi}. \tag{3.3}$$

Its sign gives the direction of the turn, and its magnitude $|\theta_0|$ gives the horizontal turn demand of the trial (Figure 3.12). Each block schedules six trials at each of the eight signed plane steps, which gives six trials each at 0 and 180 degrees and twelve each at 45, 90, and 135 degrees, and the target is placed so that the realized angle matches the assigned one. Because a larger angle requires a larger turn, the analysis adjusts for $|\theta_0|$ rather than dividing completion time by degrees. Appendix B.6 gives the remaining details of the snapshot and the height and distance measures.

Table 3.2 lists what is measured for each research question, where it comes from, and the unit at which it is analyzed. Section 3.5.8 gives the scoring of each measure.

### 3.3.4 Research Questions and Outcomes

Each research question in Section 1.2 maps onto specific outcomes. RQ1 uses active completion time as the primary outcome, with missed targets and perceived workload, measured with an adaptation of NASA-TLX [15], as secondary outcomes. RQ2 uses the Index of Pupillary Activity [9],

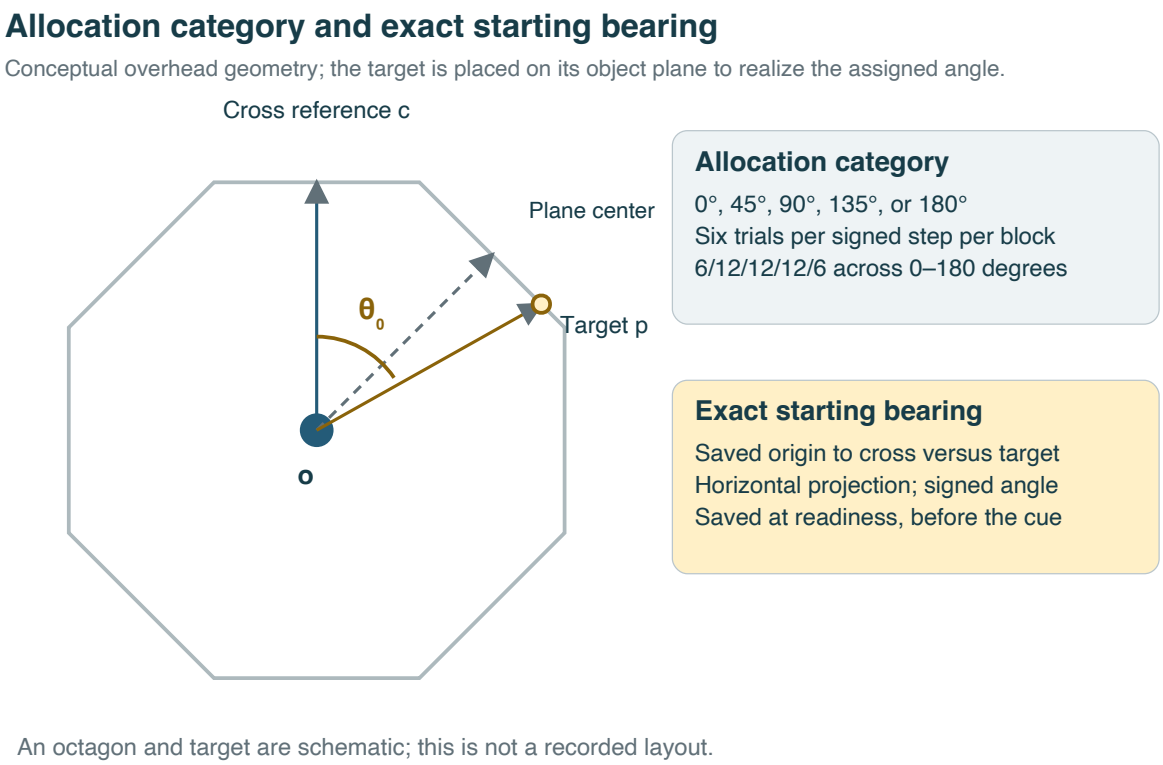



Figure 3.12: Conceptual overhead geometry. The target is placed on its assigned object plane where the assigned angle from the origin-to-cross bearing meets the object plane, so it is generally offset from the plane center. The pre-cue snapshot saves the origin, cross and target used for this placement.

fixation count, and saccade frequency, which remain exploratory until the eye-event detectors are validated on the headset. RQ3 uses the interest/enjoyment subscale of the Intrinsic Motivation Inventory (IMI) [27, 36], with the other three IMI subscales reported alongside it. Agent perception, eeriness, voice experience, reliance, search behavior, and variation in the voice contrast with starting angle are exploratory. The hypotheses are directional, favoring the SSV, but the voice contrasts are tested two-sided so that an advantage for the SAV would also be detected. Repeated trials are treated as within-participant observations, not as independent participants.

Table 3.2: Measures by research question. Fixation, saccade, and pupil measures remain exploratory until the eye-event detectors are validated.

| Role | Measure | Unit | Source | Analysis unit |
|---|---|---|---|---|
| RQ1 (primary) | Active completion time | Seconds, log-transformed | Application events | Trial |
| RQ1 | Missed targets | Percent of searches | Application events | Participant by block |
| RQ1 | Perceived workload (NASA-TLX) | Mean of six 0–10 ratings | Final survey | Participant by voice block |
| RQ2 | Index of Pupillary Activity | Extrema per second | Eye tracker | Search |
| RQ2 | Fixation count | Count | Eye tracker | Search |
| RQ2 | Saccade frequency | Events per second | Eye tracker | Search |
| RQ3 | IMI interest/enjoyment | Subscale mean | Final survey | Participant by voice block |
| Manipulation check | Perceived voice similarity | 1–7 rating | Final survey | Participant by voice block |
| Exploratory | Agent perception, eeriness, ARQ, custom voice items | 1–7 ratings | Final survey | Participant by voice block |
| Exploratory | Search behavior (head rotation, object planes visited, revisits) | Degrees, counts | Application records | Trial |
| Adjustment | Starting angle $\lvert\theta_0\rvert$ | Degrees | Pre-cue snapshot | Trial |

## 3.4 Procedure

Participants perform an agent-guided object-search task while seated on a swivel chair, with multiple colored geometric objects scattered across eight surrounding vertical planes. Although the application knows the target identity, the participant does not receive a color-and-shape description before searching. Instead, the agent guides the participant toward the target plane and then refines the direction horizontally, vertically or diagonally. The participant selects an object using the right-controller ray and trigger. Looking at an object, receiving a gaze confirmation or pointing at it does not by itself complete the trial.

Each trial begins with the participant looking at a readiness cross. Once readiness is established, an opening directional cue plays while the objects remain hidden, and object reveal then starts active search timing. A correct trigger selection before the 25-second active-search deadline stops timing and produces success feedback. An incorrect selection is counted, and search continues within that same limit.

If the target has not been selected after 25 seconds of active search, the trial ends as a timeout. The application hides the objects, cancels pending spatial hints, and plays a prepared encouragement in the assigned voice, beginning with "Time up," before presenting the next readiness cross. For example, the agent may say, "Time up. Let's shake that round off and start the next." Separate wording introduces the block break or acknowledges the final trial. The same deadline and phrase set apply to both voice conditions and to the single practice trial. A timeout advances the scheduled attempt without adding a replacement. Explicit pauses suspend the clock, whereas tracking loss alone does not. The 25-second limit is a researcher-selected protocol parameter, not a threshold drawn from the cited studies.

Before participant use, consent materials will be updated to describe ElevenLabs transcription, cloning, and synthesis, including the data transmitted, retention, access, and deletion arrangements. After consent, participants audition the catalog voices and choose their SAV; this voice is chosen explicitly rather than inferred from participant gender. They then complete an automatic audio intake survey. A generic intake agent, whose voice is outside the catalog and is never assigned to either condition, reads nine fixed prompts: four eligibility questions (age, hearing, vision with usual correction, and ability to stay seated, turn in a swivel chair, and use the right controller), four background questions (VR, voice-assistant, and eye-tracking experience, and current visual correction), and one prompt asking the participant to talk a friend through finding a lost object for about thirty seconds. The open-ended prompts provide more natural speech for cloning, and the last one is close to how the agent speaks. A tone marks when each answer should begin; recording

then starts and stops automatically and the intake moves to the next prompt, while the researcher can pause, repeat, skip, or stop at any time. All answers are transcribed with ElevenLabs and stored as local participant data under the study identifier. ElevenLabs is the only outside service that receives participant audio or transcripts; decisions during the task are made by the local model. Voice capture begins only after a signed-consent attestation is recorded. Once all nine prompts, and the optional fallback reading when the answers are too short, are complete and the answers reach at least 30 seconds of estimated speech and 50 transcribed words, the system clones the participant's voice from the answers and begins preparing both voices' phrase libraries in the Trial Review application under the participant's study ID. The researcher takes no action to start this step, and the audio therefore reaches ElevenLabs before the eligibility review. While the libraries are prepared, the researcher reviews the four eligibility answers. A negative or unclear answer stops screening rather than passing automatically; library preparation then stops and the clone is deleted from ElevenLabs, and withdrawal at any point has the same effect. Only a positive review and a complete phrase library permit transfer to the headset, where the participant confirms their study ID and condition order before starting. In Trial Review, the researcher follows six steps: reserve the study ID, attest consent, choose the SAV, run the audio intake, review eligibility, and send the session to the headset. The next unused study ID is proposed automatically and can be edited, with a warning for duplicates, and rehearsal sessions use a separate range so that they do not consume study IDs. The intake questions and the fallback reading passage, used only when the answers do not reach that threshold, are reproduced in Appendix A.

Figure 3.13 shows these voice steps. The SAV, labeled Selected Voice in the application and survey, is chosen in the Trial Review application on the desktop, and the headset then asks the participant to confirm the session and to check the audio of both voices. There is no separate headset screen for choosing the clone, because it is prepared from the participant's own answers.

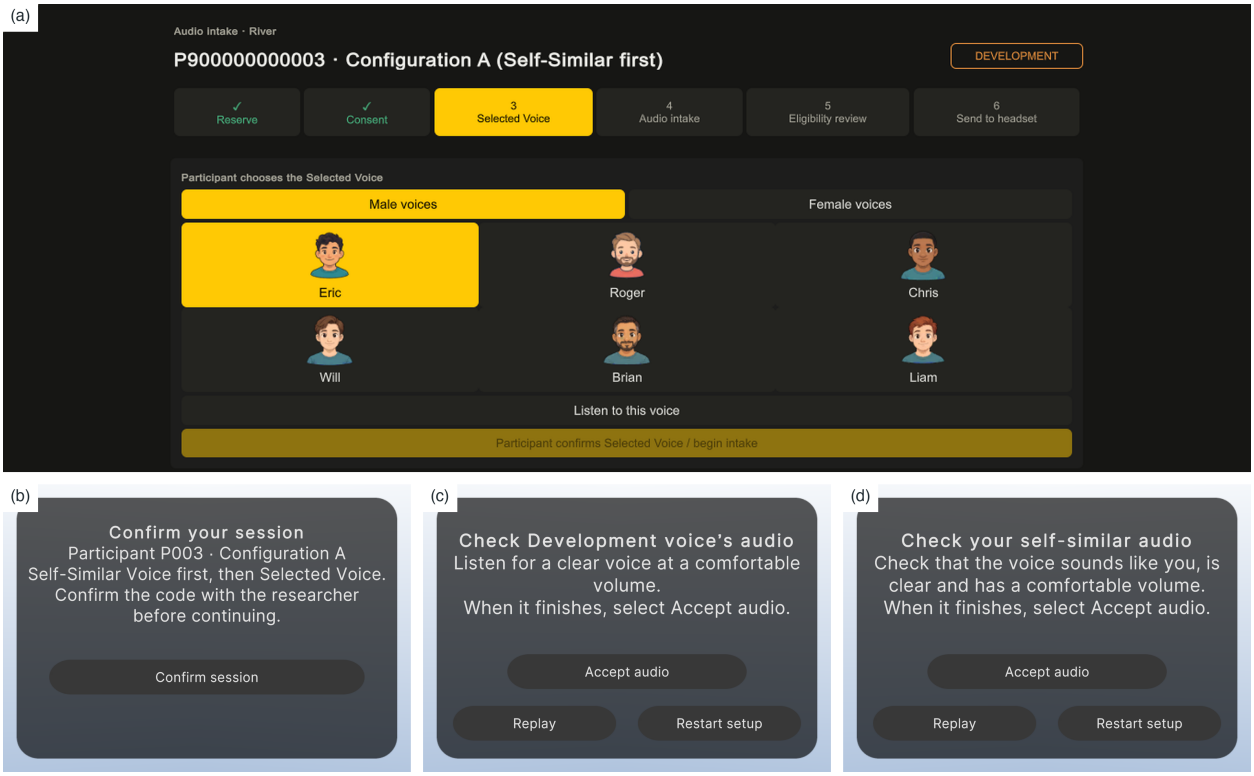



Figure 3.13: Voice setup screens. (a) Desktop Selected Voice step in Trial Review. (b) Headset session confirmation. (c) Headset audio check for the SAV. (d) Headset audio check for the SSV. Images show the application's interface with synthetic development state; (b)–(d) are Unity editor renders rather than captures through the headset, and the voice name in (c) is a development placeholder.

The researcher fits the headset and conducts vendor calibration. Interpupillary distance is not entered by hand. During the headset voice checks, the application passively records two separate estimates: the distance between the left and right rendered views, read from the headset runtime, and the distance between the left and right eye origins, read from the eye tracker. Each estimate is the median of at least 40 stable readings spanning at least 2 seconds, and is stored with its source, time, and sample count. The headset does not report whether its automatic IPD adjustment ran, so the application records only whether the view separation was observed to change during capture, which cannot establish whether an adjustment took place. A missing or unstable reading produces

a warning without stopping the session, and the researcher can then enter a value, which is recorded as manual. These are engineering settings, the two estimates may not be independent, and neither is a clinical measurement of IPD; their agreement with the headset's lens setting will be checked during pilot sessions. Within the application, gaze preparation then records three calibration dots, each preceded by a 3–2–1 countdown, which fit a provisional head-relative yaw and pitch correction. Five separate verification dots follow automatically and test that correction without changing it (Figure 3.14). The researcher can then continue, retry, or skip without a correction, which remains a pilot setting. Standardized instructions explain that participants rely on the agent to discover the target and use the right trigger to select it. Participants then complete the single practice trial and the two assigned 48-trial blocks.

No questionnaire interrupts the block sequence. Instead, the final survey presents every condition-specific item once, with two response columns side by side, labeled *Selected Voice* and *Self-Similar Voice*, so that participants compare the two agents directly while rating each one. A break is offered at the block boundary, and the researcher can pause at any point, which is logged immediately and stops the search clock. Tracking is rechecked between blocks and after headset adjustment. After headset removal, participants complete the side-by-side retrospective ratings, comparative voice preference, and debrief. Participants may request a pause or stop, and symptoms or technical problems prompt researcher assessment.

Figure 3.15 distinguishes administration time from reference scope: both columns are completed in the final session, but each rating refers to the block in which that voice was heard.

The survey columns name the two voices directly, as the Selected Voice and the Self-Similar Voice, so participants do not need to remember which block used which voice. Because the label identifies the self-similar condition, it may cue expectations about that voice; this tradeoff is noted as a limitation, and pilot sessions will check that participants understand both labels.

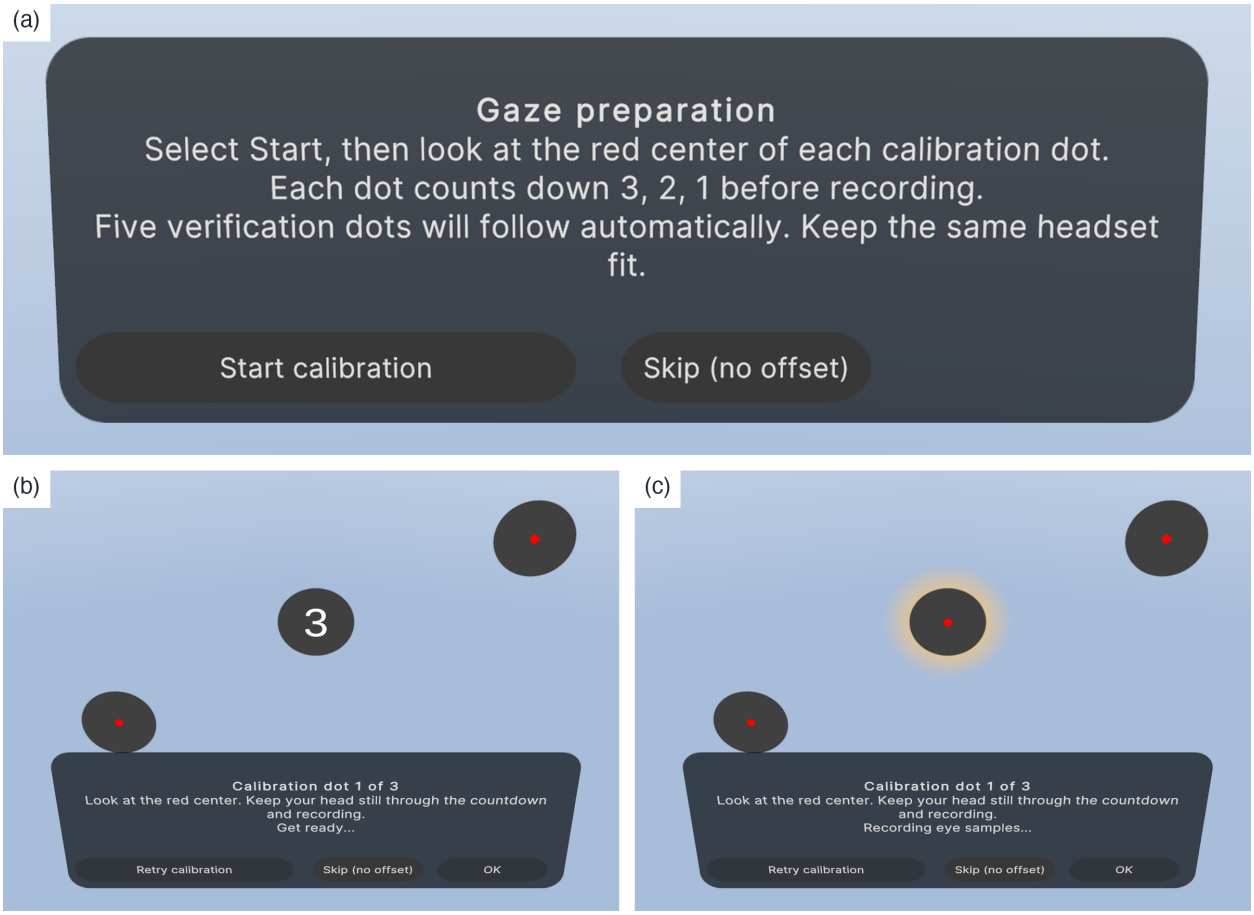



Figure 3.14: Gaze preparation screens. (a) Instructions. (b) Countdown on a calibration dot. (c) Red fixation center with the gold glow that shows collection progress. Images are Unity editor renders of the application's interface with synthetic development state, framed for illustration rather than captured through the headset. The OK button in (b) and (c) belongs to an optional path and does not appear in the study flow.

### *3.4.1 Pilot Testing and Parameter Checks*

Before confirmatory recruitment, pilot sessions verify the realized angle/height schedules, target visibility and seated comfort, readiness and selection behavior, gaze validity, audio timing and wording, and reconstruction of trial events from the saved data. Pilot assessment also measures survey comprehension and completion time. These checks inform calibration-acceptance, practice-

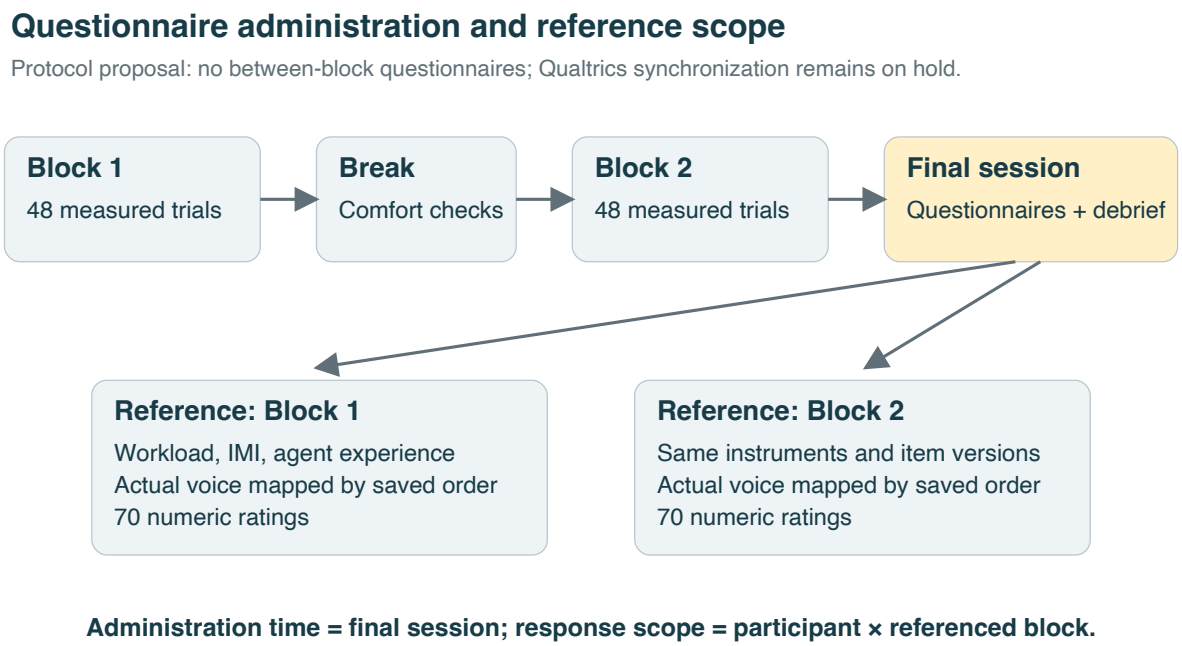



Figure 3.15: Proposed final retrospective administration. The intervening break contains comfort and tracking checks, not a block questionnaire. Each voice column contains 70 numeric ratings. The block in which each voice was heard is recovered from the saved counterbalanced order.

proficiency and technical-stop criteria. The pilot also assesses whether the 25-second deadline is suitable, and the outcome is documented with the study build and questionnaire versions.

## 3.5 Data Collection

The voice comparison raises three measurement questions: whether participants perceived the SSV as resembling their own, whether voice condition changed search performance, and how participants experienced the assistance. Answering them requires outcome measures and a record of the speech each participant actually heard. This section links voice identity, delivered speech, search events, and questionnaire responses. Perceived similarity is reported as a manipulation check, kept

distinct from performance and experience outcomes. Detailed gaze and timing records support the interpretation of these outcomes, and Appendix A gives instrument wording and scoring.

### *3.5.1 Frame-Level Records and the Trial CSV*

Follow My Voice records time-varying observations separately from trial summaries. Each frame record stores application time, gaze origin and direction with tracking validity, head pose, available controller poses and input state, and the gaze/controller object-intersection evidence used by guidance. Object identities and transforms provide the scene needed for reconstruction. Guidance transitions, agent decisions, actual speech playback, trigger selections, and pauses are recorded as events. A rendered frame can reuse a sensor observation, so frame counts differ from the number of independent eye-tracking samples.

The trial replay snapshot stores the plane bottom in recording-local coordinates, the plane height in metres, and a flag indicating that these bounds are available. Trial Review uses these values to reconstruct the object planes. Earlier recordings lack the bounds, so their object planes are reconstructed from object-derived plane estimates and the original object positions, without rescaling to the current layout. The replay fields supplement the saved layout version and transforms and are separate from the trial CSV columns. Plane geometry differs between the two layouts, so analyses that combine recordings from both retain the layout version.

Trial Review generates an Excel-compatible trial CSV with one row per trial. Its fields describe task performance, geometry, guidance exposure, and associated subjective responses (Table 3.3). Outcome, practice, provenance, and quality information accompany the exported records to support analysis and eligibility checks.

Table 3.3: Contents of the trial CSV. Unavailable values and reserved, uncollected fields are left blank.

| Group | Recorded or derived columns |
|---|---|
| Identity | Participant ID, trial ID, block index, voice condition. |
| Geometry | Absolute and signed nominal theta; actual signed head-to-target yaw; target-minus-head height; target distance. |
| Guidance and duration | Instruction count and search exposure in seconds. |
| Target contact | Time to first and final target gaze contact; first right-controller target contact; final selection. |
| Subjective context | NASA-TLX score; reserved subjective performance rating, left blank pending an instrument mapping. |

The yaw and height in the export are head-relative and measured after onset. The pre-cue, cross-relative geometry is stored in the trial summaries and event log. Gaze-contact times are ray intersections, not validated fixations. A questionnaire score repeated across trial rows remains a single block-level observation.

### *3.5.2 From Interaction Events to Research Variables*

The recorded data serve two purposes: reconstructing what happened in a trial and estimating the study outcomes. Reconstruction requires observations and events at their original temporal resolution, whereas statistical analysis usually requires a more compact table. Trial summaries are therefore stored separately from gaze, head/controller motion, object layouts, and event streams. Each summary is a derived representation of the underlying evidence, not a replacement for it.

Each derived value has an identifiable source and definition. For example, a number called “time to gaze” is ambiguous unless it specifies the clock origin, the contact rule, validity handling, and the treatment of missing observations. The recorded data keep these distinctions in separate fields.

Figure 3.16 shows how a trial CSV stays linked to the underlying observations. Values prepared for statistical analysis can therefore be traced back to the records needed for trial reconstruction.

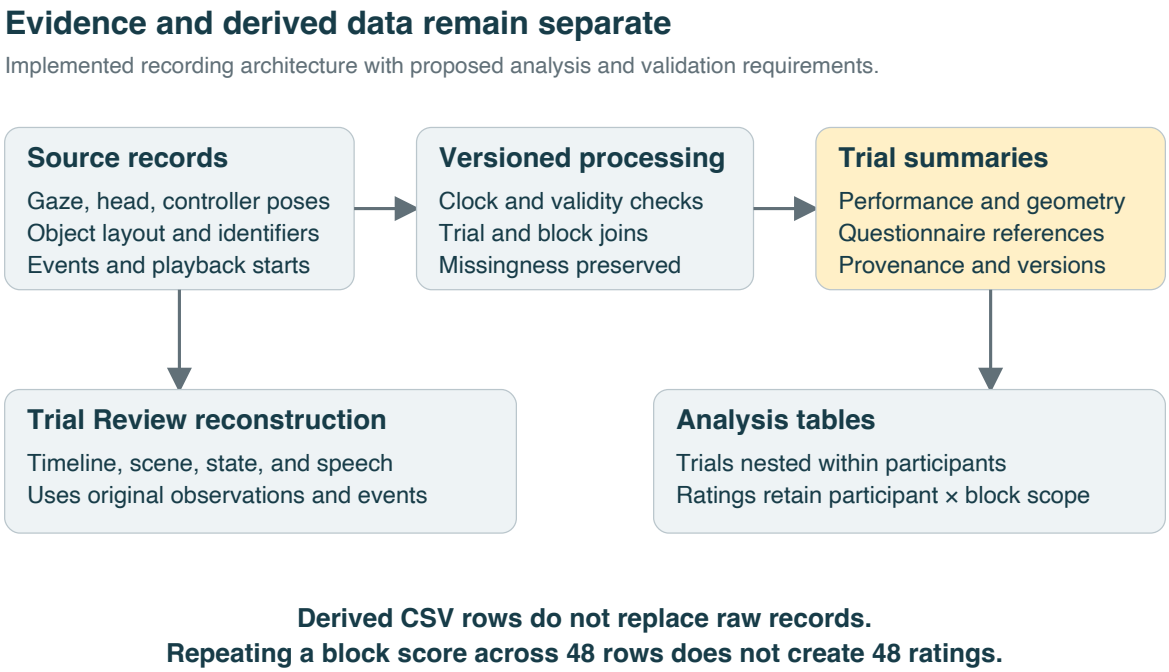



Figure 3.16: Recording and processing relationships. Source streams support reconstruction; derived summaries support analysis. Processing and scoring follow versioned definitions. Block-level ratings repeated in trial rows remain single block-level observations.

### *3.5.3 Geometry, Timing, and Data Quality*

Several measurement definitions are specified in detail in Appendix B.6. The starting angle $\theta_0$ (Section 3.3.3) is computed from a snapshot saved when readiness completes, and target height is measured relative to the cross (Section B.6.1). Active completion time runs from object reveal to correct selection and excludes the union of recorded pause intervals (Section B.6.2). The effective eye-tracking acquisition rate is estimated from distinct source observations rather than rendered

frames and is reported with validity, gaps, and spatial-validation error (Section B.6.3). Software delivery latency runs from the qualifying event to the actual software playback start of the corresponding cue (Section B.6.4). Questionnaire responses are joined to blocks through the saved condition order and keep their scale metadata (Section B.6.5).

### *3.5.4 Missed Targets*

A missed target is a search that reaches its recorded 25-second active-search deadline without a correct trigger selection. An incorrect selection followed by a correct selection before expiry is a completed search, with its selection error retained separately. Technical stops, withdrawal, and recordings without a known terminal outcome are not automatically classified as misses. For each participant, block, voice condition, and recorded deadline, the descriptive missed-target percentage is

$$M = 100\frac{N_{\text{timeout}}}{N_{\text{selected}} + N_{\text{timeout}}}. \tag{3.4}$$

The denominator excludes practice and unresolved or technically interrupted attempts, and their counts are reported alongside the percentage. A zero denominator yields an unavailable percentage. In the full trial export, `missed_target` is 1 for timeout, 0 for successful completion, and blank otherwise. A separate `missed_targets.csv` records grouped counts, deadlines, and development-data exclusion labels. Legacy records without an explicit deadline are retained but excluded from this deadline-specific percentage.

This outcome follows Chiossi et al.'s distinction between successful search and searches ending without selection [6]. Their task used a five-second limit, whereas the present task is verbally guided with a 25-second limit. Percentages are therefore interpreted within each protocol rather than compared as benchmarks. The six unfinished searches in the earlier development recordings are reported as unfinished rather than as missed targets.

### *3.5.5 Eye-Event and Pupil Measures*

Fixations, saccades, and pupil activity are exploratory measures in this thesis. An offline analysis preserves per-eye directions and pupil diameters from the HTC query stream, deduplicates repeated query timestamps, and separates search from readiness and pauses. Query receipt times record when software obtained the data. Sensor capture time and the independence of consecutive observations remain unestablished. Changes in trial or device, invalid tracking, reversed timestamps, and query gaps longer than 25 ms break continuity. Invalid per-eye observations and observed eye closure exclude nearby 200-ms intervals.

The fixation detector uses elapsed-time dispersion: the sum of horizontal and vertical angular ranges must remain within 1.8 degrees for at least 83 ms. This starting configuration follows the I-DT parameters reported by Chiossi et al. [6], within the detector family described by Salvucci and Goldberg [38]. Azimuth is unwrapped across the signed-angle boundary. The implementation retains event intervals and the duration of eligible segments. Its counts are candidate classifications, not validated fixations.

The saccade detector operates separately, applying a velocity threshold to eye-in-head angular velocity [38]. It uses a provisional 60-degree-per-second threshold, a duration of 20–100 ms, at least two sampled intervals, and observed lower-velocity samples on either side. Intervals with head speed above 20 degrees per second or unavailable head pose are excluded. The reported rate divides event count by retained low-head-speed duration. The rate therefore describes saccades within this restricted window rather than across the whole search. These restrictions reduce head-turn ambiguity, but event identity and sensor-level eye–head synchronization remain unvalidated.

The Index of Pupillary Activity (IPA) calculation follows the wavelet-based approach of Duchowski et al. [9]. Each eye is processed separately. Accepted pupil segments contain valid finite diameters,

last at least two seconds, meet a median query-rate screen of 60 Hz, and pass an interval-regularity screen. Within each accepted segment, values are resampled at its median query interval; no interpolation crosses rejected gaps or invalid observations. Level-two sym16 wavelet detail coefficients yield thresholded modulus maxima, which are counted per retained second. The implementation records its thresholds, resampling, numerical tolerance, and nearest-neighbor boundary convention. When no segment qualifies, IPA is recorded as unavailable rather than zero.

Software tests verify event boundaries, query deduplication, invalid-data handling, head-turn suppression, and wavelet calculations. These tests establish computational correctness, not physiological validity. Fixation counts and saccade rates will enter the participant analysis only after independent event annotation, calibration checks, timing assessment, and held-out threshold evaluation in the eye validation session. Pupil-based interpretation also requires checks for luminance, gaze angle, blink artifacts, segment boundaries, and task-related confounds. IPA is therefore an exploratory measure reported alongside NASA-TLX, not a validated workload estimate for this apparatus. Final fixation duration on the target remains unavailable until validated eye events can be joined to target areas of interest. The last ray contact does not measure it.

### *3.5.6 Gaze Contact, Vicinity, Pointing, and Selection*

Four event families are central to interpretation: a valid eye ray can intersect the target collider; a gaze direction can satisfy the larger target-vicinity rule; the right-controller ray can intersect the target; and a trigger press can select the target. These events can occur at different times and need not occur in that order in every recorded sample sequence.

The first gaze-contact latency uses the first valid recorded target-ray intersection after reveal. The last gaze-contact time uses the last such intersection before trial end. Neither is a fixation duration.

If no contact is observed because of a blink, acquisition gap, or a path that only enters the broader vicinity, the contact field remains missing.

Vicinity qualification is an interaction rule rather than a fixation classifier. In the current system, a target can be acknowledged after 100 ms of valid qualifying evidence, with overlapping regions resolved to a single closest candidate. The rule trades exact collider intersection for a tolerance intended to make feedback usable. An overly large region, however, can produce premature acknowledgment even when the timing gate behaves correctly. The tolerance value will therefore be logged and set through piloting. The default ring in the synthetic viewer screenshot is illustrative, not a validated study setting.

Controller pointing is a distinct input observation, but its spoken acknowledgment also depends on gaze. The acknowledgment requires a qualified one-second controller hold with valid gaze in the target vicinity, and it plays when the speech scheduler permits. Before the hold completes, a valid controller hit on the target already suppresses contradictory directions pointing away from it, such as "go back down", without announcing that the hold has qualified. Only a correct trigger selection completes the search successfully; stops and timeouts end the trial with other outcomes.

Figure 3.17 contrasts the two acknowledgment events with trial completion.

### *3.5.7 Motion, Coverage, and Instructions*

Gross translation sums Euclidean differences between adjacent valid head positions, and gross rotation sums shortest quaternion angular differences. Invalid intervals, excessive gaps, and pauses break the accumulation. Gross movement is a total, not a rate. If a separate movement-rate outcome is derived, its denominator is observed covered time rather than the entire nominal session, so that runs with different data loss remain comparable.

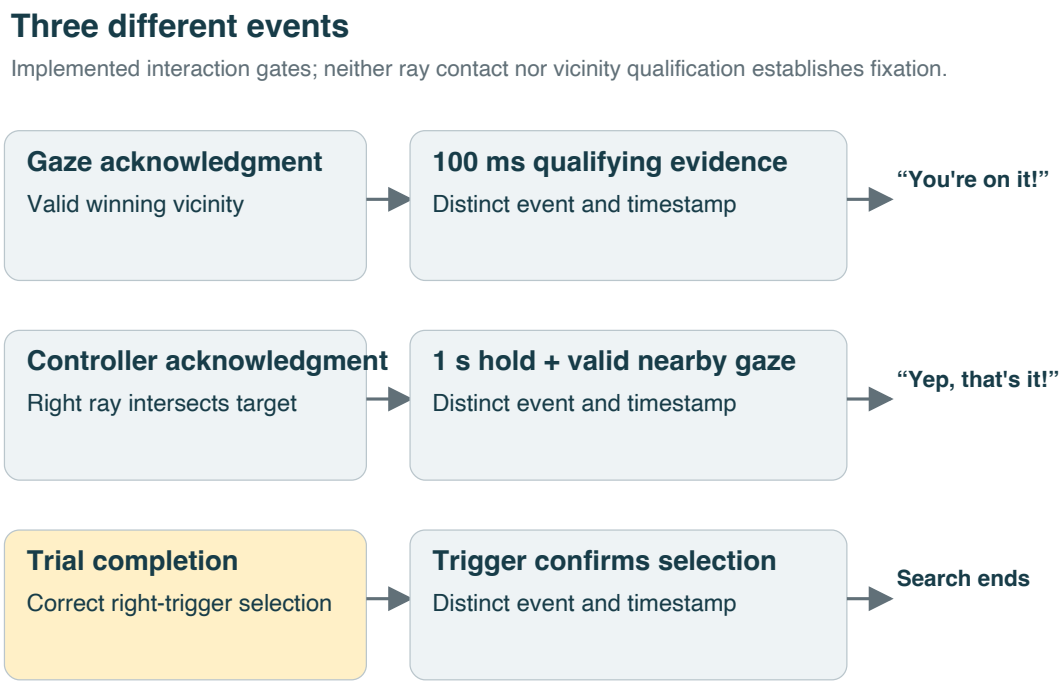



Figure 3.17: Implemented interaction distinctions. Gaze acknowledgment uses a 100-ms qualifying vicinity gate; pointing acknowledgment uses a one-second controller hold with required valid nearby gaze. A correct trigger selection completes the trial independently of whether either acknowledgment was spoken. These are interaction gates, not validated fixation measures.

Fixed-axis components derived from rotation vectors differ from Euler yaw, pitch, and roll and are labeled by their actual calculation. Head motion is also only an indirect measure of swivel-chair rotation, because the head can move relative to the torso and chair.

Instruction counts are based on actual software playback starts. Requested cues, rejected model answers, and prefetched clips are not counted as delivered instructions. Opening cues, search guidance, acknowledgments, reminders, and success speech have separate categories. The planned total of delivered active-search assistance includes directional guidance, gaze/pointing acknowledgments, and selection reminders. Each category is also reported separately, and opening and

success speech remain outside that total. Interrupted speech counts as started speech but retains its completion status, since the participant may not have heard the full utterance.

### *3.5.8 Scoring*

#### *3.5.8.1 Task Performance*

Active completion time.

The primary outcome is elapsed active-search time in seconds from object reveal to the correct right-trigger selection, excluding the union of recorded pause intervals overlapping that period. Section B.6.2 specifies the clock equation and handling of overlapping pauses. Lower values indicate faster completion. Readiness, opening guidance, post-selection feedback, questionnaires and between-trial breaks do not contribute to this outcome. An unfinished trial retains its observed exposure and outcome but has no successful completion time. The full interval from opening-cue onset to selection is also defined to describe the complete instructed task duration.

Target contact and selection errors.

Time to first target gaze is the active-clock time of the first valid, recorded gaze-ray intersection with the target, and time to first controller contact uses the first tracked right-controller ray–target intersection. Last target-gaze contact time denotes the last observed valid target intersection during the trial, not a fixation duration or a spoken confirmation. These event times remain distinct from trigger selection and from the larger gaze-vicinity region used by the guidance system. Incorrect selections are counted as trigger presses that select non-target objects. If no valid target contact is

observed, its latency is missing rather than zero. All contact times retain their source and sampling resolution.

#### *3.5.8.2 Starting Geometry and Search Behavior*

Starting angle, height and distance.

The pre-cue snapshot saves a reference origin (the eye origin at readiness completion), readiness-cross position and target position. Starting theta ($\theta_0$) is the signed angle about world up between the horizontal origin-to-cross and origin-to-target vectors. Its magnitude $|\theta_0|$ quantifies horizontal turn demand, and its sign preserves direction. Degenerate horizontal vectors are marked unavailable. The primary analysis adjusts for $|\theta_0|$ rather than dividing completion time by degrees. Nominal plane-angle assignment, exact pre-cue theta and post-reveal head-to-target error are separate variables. Recordings made before the snapshot was added carry only nominal theta and therefore lack $\theta_0$.

Signed target-height difference is $\Delta h = y_{\text{target}} - y_{\text{cross}}$, in metres; positive values indicate above-cross targets. Euclidean target distance is the three-dimensional distance between target center and the first valid tracked head position after reveal, with that observation's timestamp retained. All positions used in a calculation share a recorded coordinate frame.

Head movement and zone coverage.

Gross head translation sums distances between adjacent valid head positions, in metres, and gross head rotation sums shortest quaternion angular displacements between adjacent valid orientations, in degrees. Fixed-axis component totals are retained separately and labeled by their calculation

rather than as Euler yaw, pitch and roll. Invalid intervals, pauses and excessive sample gaps break accumulation, and the analysis does not interpolate across them. Covered time and gap counts accompany totals. These measures describe head movement, not independently tracked chair rotation.

Distinct object planes visited, plane transitions and revisits summarize search coverage using stable valid gaze observations and a versioned zone-assignment rule. Transitions across tracking gaps are not inferred.

Instructions and guidance timing.

The delivered active-search assistance total counts playback starts for directional guidance, gaze/pointing acknowledgments, and selection reminders, including delivered repeats. These categories are also reported separately, as specified in Section 3.5. Requested repeats, delivered repeats and interrupted utterances are retained separately. Model requests, prefetched audio and success speech are excluded from delivered search instructions. Opening cues and success feedback retain separate event labels. Timing analysis distinguishes evidence qualification, decision-request latency, model deferral, scheduling delay and audio onset. Gaze confirmations and controller confirmations are evaluated separately against their corresponding gates. The controller gate includes both its hold and the required nearby valid gaze.

#### *3.5.8.3 Subjective Measures*

Workload.

Workload is measured with an adaptation of Hart and Staveland's NASA-TLX [15]. Six dimensions—mental, physical and temporal demand, performance, effort and frustration—are

rated separately for each voice block during the final survey, after both blocks. The study uses a documented 0–10 adaptation without pairwise weighting. Demand, effort and frustration run from 0 = Very low to 10 = Very high. The Performance prompt asks how successful the participant was and runs from 0 = Failure to 10 = Perfect. This favorable-direction response is a deliberate local adaptation of the opposite orientation in the NASA rating form. All six raw responses are retained; higher raw Performance values indicate better perceived performance. Before calculating the workload mean, only Performance is reverse-scored once as $10 - r_{\text{Performance}}$. The unweighted mean uses the other five raw ratings and this reversed Performance value, and is calculated only when all six responses are present. Higher mean scores therefore indicate greater workload or poorer perceived performance.

Motivation and activity experience.

The Intrinsic Motivation Inventory [27, 36] contributes interest/enjoyment (7 items), effort/importance (5), pressure/tension (5), and value/usefulness (7). Responses range from 1 = Not at all true to 7 = Very true. Published reverse keys are applied before averaging each complete subscale. Interest/enjoyment represents intrinsic motivation, and the four subscales are not combined into an overall IMI score. Activity wording refers to the completed search block, with task-specific adaptations documented.

Agent perception.

Twenty-one items from Guo et al. (2024), Appendix A, Table A1 [14] assess co-presence (Q1–6), perceived intelligence (Q13–18), appealing/eerie/familiar impressions (Q20–22), and perceived anthropomorphism (Q37–42). “Virtual character” is adapted to “voice agent.” Responses range from 1 = Very strongly disagree to 7 = Very strongly agree. The three six-item constructs have separate

complete-scale means. Q20–22 are analyzed individually rather than combined into a multi-item eeriness index of the kind refined in uncanny-valley research [16].

Agent rating.

Five draft items adapt the Agent Rating Questionnaire (ARQ) [42], as used by Venkatakrishnan et al. for a task-guiding agent in VR [41], with one item per dimension: helpfulness, relatability, appropriate behavior, likeability, and willingness to interact again. Responses range from 1 = Very strongly disagree to 7 = Very strongly agree. The items are analyzed individually as exploratory measures of agent perception.

Voice experience and reliance.

Twelve custom 1–7 agreement items assess voice similarity, responsiveness, helpfulness, distraction, familiarity, trust, comfort, uncanniness, reliance, naturalness, willingness to follow and motivation to act without guidance. Voice similarity serves as a manipulation check. Voice liking and retrospective percentage of guidance use each use 0–100 responses with separate unable-to-answer options. These 14 numeric items are analyzed individually, as they do not form a validated scale.

Debrief.

Open debrief questions explore perceived performance, guidance experience, own-voice comfort, possible use cases and voice preference. Broad experience questions precede specific eeriness or inner-dialogue probes.

In the final survey, each item is answered side by side for the SAV and the SSV. Each voice column contains 70 numeric responses (140 in total): six workload ratings, 24 IMI items, 21 Guo items, five ARQ items, and 14 custom ratings. Technical checks and optional text are additional. Questionnaire observations have participant-by-block scope even when repeated across the 48 trial summaries of a block, and missing answers remain missing rather than zero.

Table 3.4 summarizes the condition-specific battery. Counts refer to numeric items in one voice column, not independent scales. Exact wording and reverse-key positions are preserved in Appendices A and B.

Table 3.4: Proposed final questionnaire battery, answered side by side for the SAV and the SSV after both blocks; setup voice ratings are separate.

| Measure | Items | Scoring and interpretation |
|---|---|---|
| NASA-TLX adaptation | 6 | Six 0–10 dimensions; complete unweighted mean after reversing only raw Performance as $10 - r$. Preserve raw responses and explicit bounds. |
| IMI | 24 | Separate interest/enjoyment (7), effort/importance (5), pressure/tension (5), and value/usefulness (7) means. No overall total. |
| Agent perception | 21 | Three six-item means: co-presence, intelligence, anthropomorphism. Appealing, eerie, and familiar remain three single items. |
| Agent rating (ARQ) | 5 | Five single items, one per dimension: helpfulness, relatability, appropriate behavior, likeability, and willingness to interact again. |
| Custom voice and assistance | 14 | Individual ratings of voice and guidance experience. No unvalidated combined score. |
| Total per block reference | 70 | 140 numeric ratings across two references, before technical checks and optional text. |

# CHAPTER 4: RESULTS

This chapter is organized around the three research questions introduced in Section 1.2 and closes with a discussion of how the possible outcomes would be interpreted. Each section states what will be reported for its question and how that evidence tests the corresponding hypothesis. Participant data have not yet been collected, so no voice-condition results are reported here; the development evidence that supports the apparatus appears in Appendix D.

## 4.1 Analysis

Because every participant hears both voices, each research question is analyzed within participants. For RQ1, the primary outcome is active completion time, analyzed with a trial-level mixed-effects model of log completion time that includes voice condition, period, assigned voice order, trial position within the block, and the absolute pre-cue starting angle $|\theta_0|$; participant random effects are chosen from the design and simulation rather than from statistical significance [2]. The primary result is the voice contrast, reported as a time ratio standardized over a common distribution of starting angles and tested two-sided, so that an advantage for either voice can be detected. Missed targets and NASA-TLX workload [15] are secondary outcomes. For RQ2, IPA, fixation counts, and saccade rates are compared in the same within-participant way once the eye-event detectors are validated. For RQ3, the IMI interest/enjoyment subscale [27] measures engagement, with the other IMI subscales reported alongside it. Questionnaire scores are treated as one observation per participant and voice block rather than as trial-level data. Secondary comparisons are grouped into prespecified families and adjusted with Holm's procedure [17], while agent-perception and custom items are exploratory and reported with effect estimates and uncertainty. Searches that reach the 25-second deadline have no completion time and are excluded from this model, which pulls its estimate

toward no effect if one voice also reduces timeouts. A complementary completion-probability or time-to-event analysis will therefore be specified before confirmatory analysis. Section 3.2 reports the simulation used to plan the sample size.

## 4.2 Participant Flow and Manipulation Check

Participant reporting begins with participant flow, exclusions, incomplete sessions, and data coverage by condition and order. It then describes the realized starting geometry and the similarity manipulation check, which shows whether participants heard the SSV as resembling their own, before any research question is answered. Each table states its denominator and identifies missing observations. Figure 4.1 will show the similarity rating each participant gave the two voices.

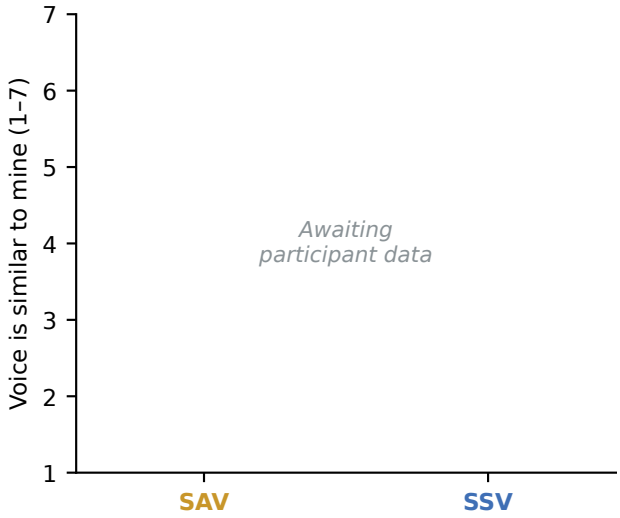



Figure 4.1: Planned manipulation check. Each line joins one participant's rating of "The agent's voice is similar to mine" (1–7) for the SAV and the SSV; diamonds show the mean with its 95% confidence interval. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

## 4.3 RQ1: Performance and Perceived Workload

RQ1 asks whether voice self-similarity impacts performance and perceived workload. The primary result is the adjusted contrast in active search-completion time. Its estimand is the marginal time contrast standardized over a common starting-difficulty distribution, and here *standardized* refers to that common distribution rather than to a standardized effect size. The estimate is reported with its uncertainty, the eligible participant and trial counts, and model diagnostics, following the analysis plan finalized before confirmatory recruitment. Missed targets, incorrect trigger selections, and valid gaze and controller contact latencies are reported by voice condition alongside paired participant summaries. Perceived workload is reported from the six NASA-TLX dimensions and their unweighted mean [15] at the participant-by-block level. Together, these results test H1: that the SSV improves search performance and lowers perceived workload. Figure 4.2 presents completion time, and Figure 4.3 presents missed targets and workload.

## 4.4 RQ2: Eye-Tracking Indicators of Workload and Search Efficiency

RQ2 asks whether voice self-similarity impacts eye-tracking indicators of workload and visual-search efficiency. Following Chiossi et al., who combined search outcomes with fixation and pupil-based measures [6], this section will report the Index of Pupillary Activity [9] as an indicator of workload, and candidate fixation counts and saccade frequency as indicators of search efficiency, each by voice condition and participant with its coverage and denominators. These measures enter the voice comparison only after the detector validation described in Section 3.5; until then they remain exploratory, and the development pipeline in Section D.2.6 shows what the reported values and coverage will look like. These results test H2: that the SSV yields more efficient search, with

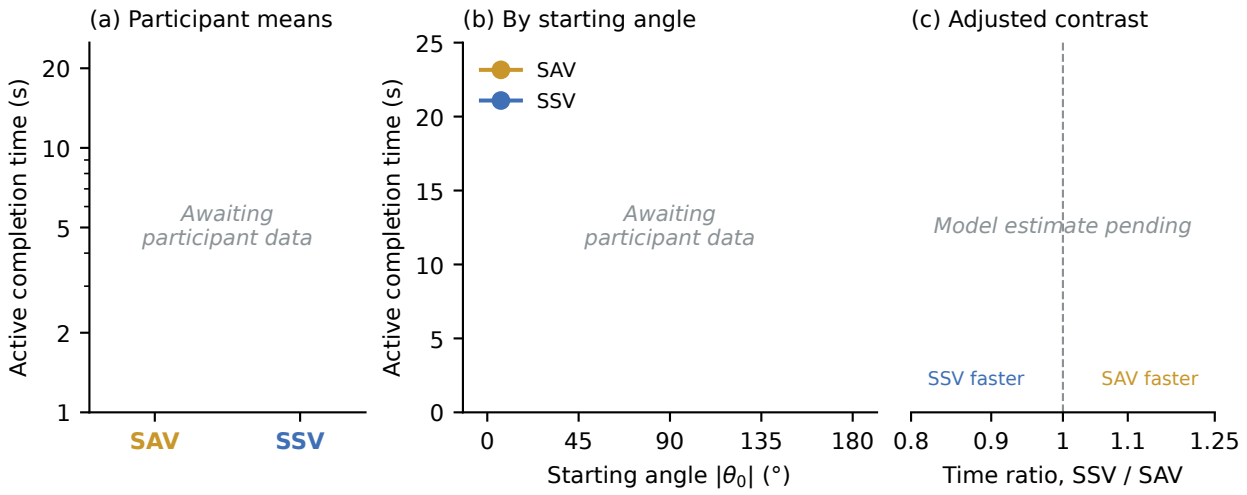



Figure 4.2: Planned completion-time results. (a) Each participant's geometric mean active completion time in each voice, on a log scale, with the mean and 95% confidence interval. (b) Completion time by absolute starting angle $|\theta_0|$ for each voice. (c) The adjusted voice contrast from the mixed-effects model, a time ratio of SSV to SAV with its 95% confidence interval; values below 1 mean faster search with the SSV. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

fewer fixations and saccades and lower pupillary activity. Figure 4.4 shows the three indicators by participant and voice.

## 4.5 RQ3: Engagement in the Search Task

RQ3 asks whether voice self-similarity impacts engagement in the object-search task. Engagement is reported from the interest/enjoyment subscale of the Intrinsic Motivation Inventory [27], with the effort/importance, pressure/tension, and value/usefulness subscales reported separately, using multiplicity families defined in the analysis plan. This result tests H3: that the SSV makes the search more engaging, and Figure 4.5 shows it alongside the other subscales. Read together with RQ1, it also shows whether higher engagement and better performance occur in the same condition,

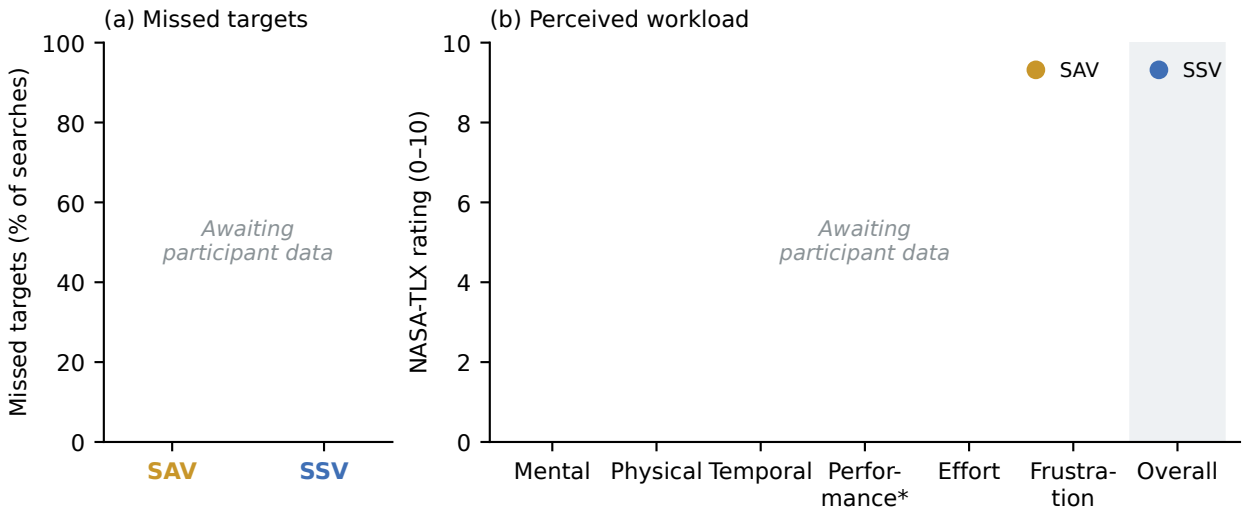



Figure 4.3: Planned missed-target and workload results. (a) Percentage of each participant's searches in each voice that reach the 25-second limit. (b) Mean NASA-TLX ratings with 95% confidence intervals for each dimension and their unweighted mean (Overall) by voice. *Performance is plotted as rated, where higher means more successful, and is reversed in the overall score. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

as the thesis's central hypothesis proposes; because both outcomes follow voice assignment, their co-occurrence does not by itself establish that engagement causes the performance difference.

## 4.6 Exploratory Agent and Voice Experience

Once participant ratings are collected, this section will report agent perception (co-presence, perceived intelligence, anthropomorphism, and the appealing, eerie, and familiar items), custom voice-experience ratings such as liking, trust, reliance, and naturalness, and debrief themes. These outcomes are exploratory and are not tied to a hypothesis. No questionnaire responses or preference distributions are simulated. A contrast that cannot be estimated or proves inconclusive is reported

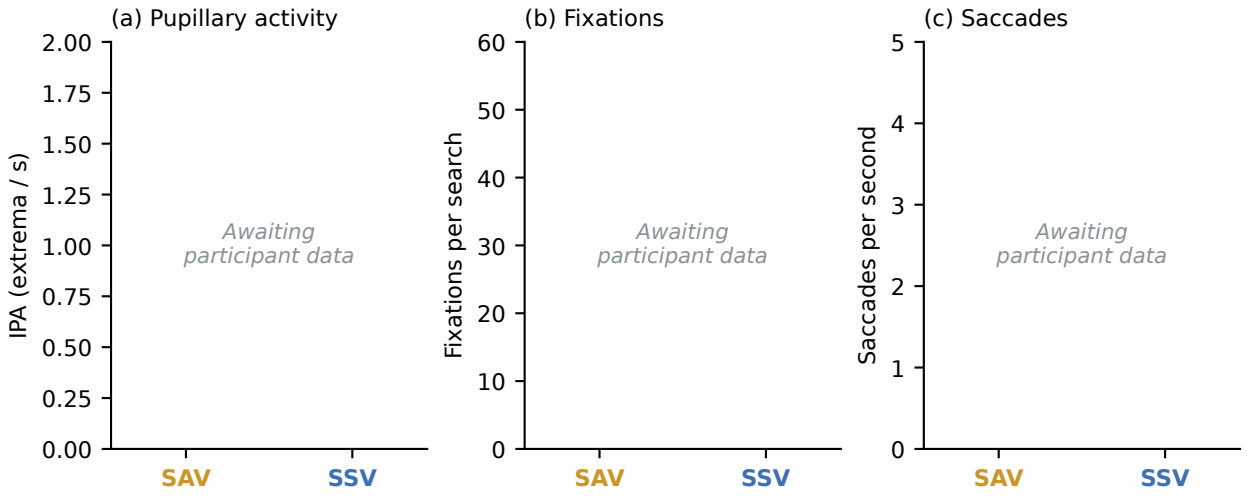



Figure 4.4: Planned eye-tracking results. Participant means of (a) the Index of Pupillary Activity, (b) fixations per search, and (c) saccades per second in each voice, averaged over both eyes, with the mean and 95% confidence interval. Only searches with enough valid eye data contribute, and coverage is reported with each panel. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

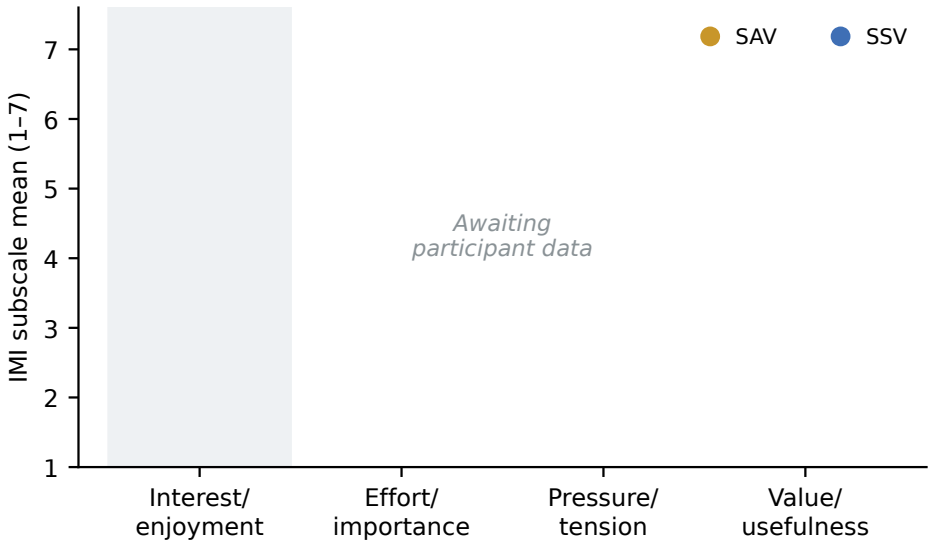



Figure 4.5: Planned engagement results. Mean Intrinsic Motivation Inventory subscale scores with 95% confidence intervals by voice. Interest/enjoyment (shaded) is the RQ3 outcome; the other subscales are reported alongside it. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

as such rather than omitted. Figure 4.6 summarizes these ratings as paired differences between the voices.

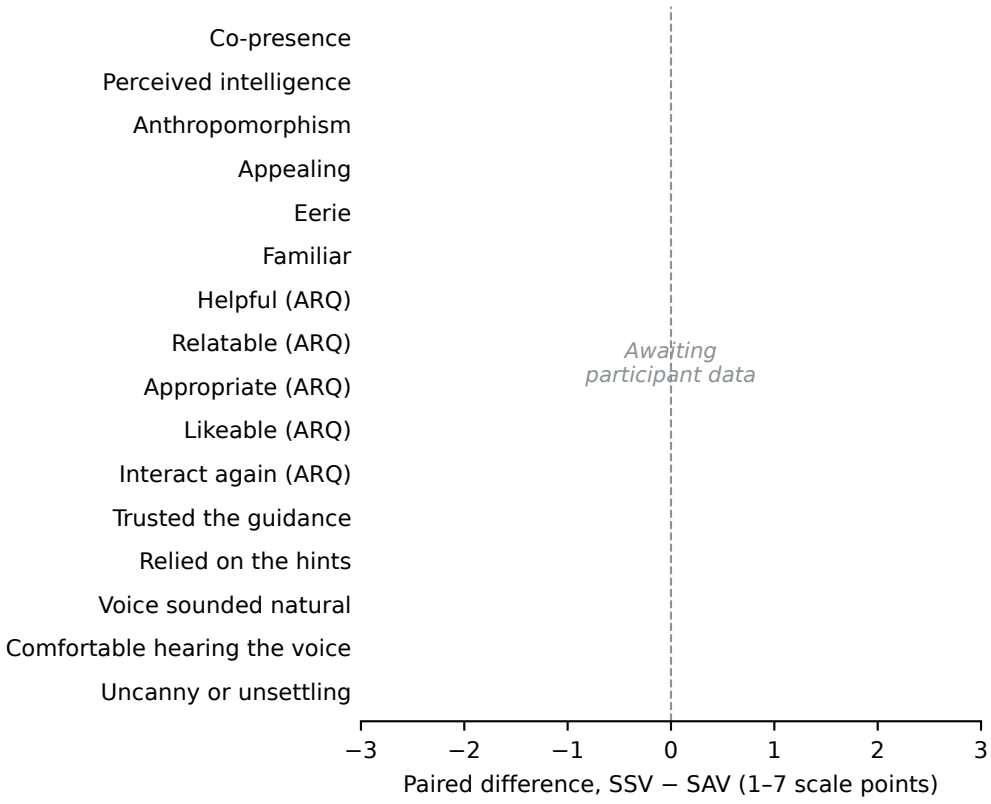



Figure 4.6: Planned exploratory ratings. Mean paired difference between the SSV and the SAV, with 95% confidence intervals, for the agent-perception scales, the appealing, eerie and familiar items, the five ARQ items, and selected voice-experience items, all on 1–7 scales. Positive values mean a higher rating for the SSV. These comparisons are exploratory and are not adjusted for multiplicity. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

## 4.7 Guidance Delivery Across Voice Conditions

A fair voice comparison depends on both voices receiving comparable guidance. Reporting therefore describes gaze availability, actual speech exposure, delayed acknowledgments, interruptions,

and unhelpful cues by voice condition and session. This account helps assess whether technical delivery problems affected one condition more than the other. The voice contrast is not automatically adjusted for behavior-dependent hint counts or waiting time. These quantities arise after voice assignment, may themselves depend on voice condition, and cannot be treated as baseline difficulty. Figure 4.7 shows three of these delivery measures by participant and voice.

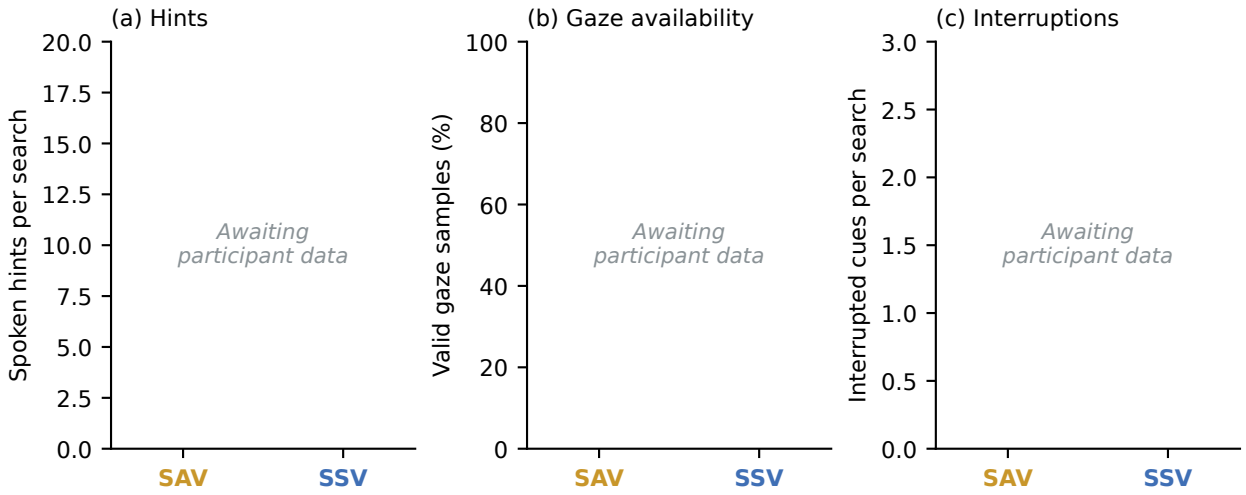



Figure 4.7: Planned guidance-delivery results. Participant means of (a) spoken hints per search, (b) the percentage of valid gaze samples, and (c) interrupted cues per search in each voice, with the mean and 95% confidence interval. Similar values in both voices indicate that the conditions received comparable guidance. Panels remain empty until participant data are collected; the same script then draws them from the trial export.

Gaze acknowledgment, controller acknowledgment, and correct trigger selection remain separate events. Their sequence records document feedback delivery, whereas participant ratings and debriefing address how that feedback was experienced. Appendix C defines sequence tests and evaluation criteria for assessing the apparatus before interpreting differences between voices.

## 4.8 Discussion

### *4.8.1 Interpreting the Research Questions*

Follow My Voice asks whether a synthetic voice resembling the listener changes how people perform a task or experience it, compared with an SAV the listener has already chosen, using object search as the task. Participant results are still pending, so the interpretations below are conditional. They describe what the data could support, not which voice is better.

The discussion follows the three research questions. RQ1 concerns measured search performance and perceived workload, RQ2 concerns eye-tracking indicators of workload and search efficiency, and RQ3 concerns engagement. We read these outcomes together. Agreement among them, in either direction, would give a consistent picture of how the voice changes the work, whereas disagreement could reveal a tradeoff that a single outcome would conceal.

#### *4.8.1.1 RQ1: Performance and Perceived Workload*

A completion-time advantage for the SSV would have to be read alongside the uncertainty of the estimate, the manipulation check, and guidance quality. If the SAV condition showed systematic intelligibility or delivery problems, the difference could reflect those problems rather than anything specific to self-similarity. If both conditions have comparable delivery quality and participants recognize the clone, the result would more directly support a voice-condition effect in this task. Even then, resemblance would remain confounded with other differences between the chosen and synthesized voices.

Faster completion with the SAV would be consistent with a chosen voice being easier to follow, but also with intelligibility or delivery problems in the cloned condition. Delivery records and acoustic-

quality reporting could help distinguish these readings, although the study could not establish the mechanism without further evidence. Liking, naturalness, and reliance ratings may help characterize the experience in either case. A correlation between such ratings and performance, however, would not establish that they mediate the voice effect.

If completion times are similar, the width of the uncertainty interval becomes central. A wide interval would leave practically important differences unresolved, whereas a narrow interval near the null could constrain the size of a plausible effect under the tested conditions. A nonsignificant test does not establish equivalence. The analysis plan includes no equivalence test, which would require a meaningful bound and procedure specified before the result is examined.

Perceived workload is read alongside completion time. Lower NASA-TLX workload with the SSV would fit H1, and higher workload would run against it. Faster search with unchanged or higher workload, under either voice, would suggest that participants worked harder rather than that the guidance was easier to follow.

#### *4.8.1.2 RQ2: Eye-Tracking Indicators*

If the eye-event detectors are validated, fewer fixations and saccades and lower pupillary activity with the SSV would be consistent with H2. The opposite pattern would suggest less efficient visual search under that voice, and similar values would show no detected difference. These indicators depend on coverage, so a difference must be read against the eligible time in each condition; unequal coverage alone can produce different counts, as the development recordings illustrate (Section D.2.6). If the detectors cannot be validated, RQ2 remains unanswered rather than being answered with interaction proxies such as ray hits.

#### *4.8.1.3 RQ3: Engagement*

Efficiency and experience may also diverge. A preference for self-similar speech may occur without a detectable improvement in completion time, while efficient search may coexist with discomfort. Diel and Lewis attribute vocal uncanniness to deviation from typical organic voices and report that synthetic voices escaped the uncanny valley [8], so discomfort with a clone may depend on how far it departs from familiar voices rather than on synthesis itself.

Prior studies place a self-similar voice in different roles. Kao et al. examine an avatar voice in a setting where continued voluntary play can reflect engagement, whereas Guo et al. report favorable impressions alongside eeriness in virtual collaboration [14, 20]. Aymerich-Franch et al. examined one's own voice while giving a speech through an avatar [1]. In Follow My Voice, the voice instead instructs the listener through a fixed search task. Faster completion therefore addresses efficiency, while enjoyment, discomfort, and debrief responses address the experience of following it.

If the SSV is rated more engaging, the result would fit H3. If the SAV is rated more engaging, or the ratings are similar, it would not. Because both outcomes follow voice assignment, engagement and search time moving together would not show that engagement causes a performance difference.

### *4.8.2 Practical Relevance and Transfer to Other Tasks*

Object search was chosen because it is a common task in VR, and the question behind the study is broader than search: whether a self-similar voice changes how people work with an agent. The results themselves, however, apply to the tested task. Spoken guidance could be useful for locating task-relevant items in maintenance or training, finding a book, or noticing relevant VR content [3, 24, 46], but this experiment does not test a voice in those settings. The present study addresses

a voice-choice question. It does not test industrial task completion, learning transfer, navigation safety, or narrative comprehension.

The practical value of the study depends on how its outcomes combine. Faster search accompanied by acceptable workload and a comfortable voice experience would support considering self-similar guidance for comparable tasks. A favorable experience without a detected speed difference could still motivate offering the clone as an optional voice, without supporting a claim that it makes search more efficient. Slower search, discomfort, intelligibility problems, or a preference for the SAV would instead mark limits to that option.

Transfer also depends on how much information users already have. In Follow My Voice, the participant must use spatial guidance to infer an initially unnamed target, whereas in Zhang et al.'s book task the target title was supplied [46]. A known-target task may allow people to continue independently and encounter the voice less often. Real work can additionally require recognizing meaningful parts, remembering procedural steps, moving through space, or coordinating with another person. Replication in those contexts would need to establish both the quality of guidance and the effect of voice choice.

# CHAPTER 5: FUTURE WORK

Future studies could examine repeated exposure, task context, and the separate contributions of perceived resemblance and acoustic naturalness. Manipulating resemblance while controlling other acoustic properties would address a narrower causal question than the present comparison with the SAV.

Other extensions could examine a task other than object search, a visible agent body, a known-target search task, or a broader participant population. Each would change how the voice is encountered and would require a revised rationale and method.

Memory, provider autonomy, and gaze-guidance policy could also be varied in separate systems studies. Those comparisons concern the supporting mechanism and cannot be inferred from the present voice contrast. Likewise, eye-event or pupil analyses would be warranted only where validated measurements can address a defined question about task demands or voice experience.

## 5.1 Gaze-Informed Adaptation of Spoken Assistance

Nerella et al.'s gaze-based prediction study motivates testing whether momentary task demands can inform when an agent speaks [29]. For Follow My Voice, such an extension would require task-specific labels, evaluation on unseen participants, and a separate test of adaptive hint timing. The cited work leaves evaluation of real-time deployment open, and its classification result does not show that adapting spoken guidance improves search.

## 5.2 Complementing Behavioral Measures with Brain Signals

A future extension could combine electroencephalography (EEG) with gaze, task performance, and questionnaires to investigate the attentional demands of receiving guidance in the SSV and the SAV. Yang et al.'s *Synchronizing with Attentional Sampling* provides a relevant precedent: the authors use EEG in immersive VR to investigate attentional sampling during gaze and hand–eye interaction as part of a study of visual flicker guidance [45]. This work motivates examining neural activity alongside observable search behavior, although its visual-guidance findings do not establish how voice resemblance affects attention or how frequently spoken hints should occur.

For the present voice comparison, synchronized recordings could relate neural activity to actual instruction onset, orientation toward the target, and controller selection. A follow-up study could then examine whether the two voices are associated with different responses during these stages, even when completion times are similar. Such measurements would complement participants' reports of workload and experience rather than serve as direct measures of enjoyment, trust, or perceived voice similarity.

This extension would first require a feasibility study of EEG and headset compatibility, event synchronization, and the effects of eye movements, head turns, and muscle activity on the recordings. Neural measures and analysis windows would need to be specified in advance, with comparable task demands and speech exposure across voice conditions.

# CHAPTER 6: CONCLUSION

## 6.1 Self-Similar Voice as Task Assistance

As spoken agents guide users through virtual environments, the voice that delivers their assistance becomes a design choice in its own right. Follow My Voice asks whether a synthetic voice resembling the participant changes how people perform a task or experience it, compared with an SAV chosen by the participant. Object search serves as the task because it is common in VR and gives a direct measure of performance. Because resemblance, liking, and usefulness need not coincide, the study examines them separately, in a setting where spoken guidance progressively reveals the intended object.

The proposed within-participant comparison uses two counterbalanced 48-trial blocks after one practice trial in the first assigned voice. Active completion time is the primary outcome. Retrospective workload, motivation, and agent-experience ratings, together with a debrief after both blocks, provide complementary evidence. Perceived-similarity ratings assess whether the synthesized voice achieves the intended resemblance, while completion-time measures evaluate search performance.

Gaze-contingent guidance supplies the shared context in which participants encounter the two voices. Local spatial reasoning, decisions from the local MiniLM model, explicit right-trigger selection, and traceable speech delivery support a controlled and inspectable comparison. These components are intended to make the voice comparison interpretable. This thesis does not estimate their independent effects or compare decision providers.

This thesis contributes a rationale and study design for examining self-similar voice, together with an implemented apparatus and measurement plan. Confirmatory participant results are not yet

available. Before the study can estimate a voice-condition contrast, pilot acceptance criteria and alignment of the consent documents with cloud processing must be completed.

## 6.2 Status of the Research Questions

**RQ1** asks whether voice self-similarity impacts performance and perceived workload. It has not yet been answered empirically: no confirmatory contrast in active completion time is estimated here, and NASA-TLX responses have not been collected. **RQ2** asks whether voice self-similarity impacts eye-tracking indicators of workload and search efficiency. The measurement pipeline runs end to end on headset recordings with both voices (Section D.2.6), but the detectors still require validation and no voice comparison is made. **RQ3** asks whether voice self-similarity impacts engagement in the search task; the engagement measure is defined, but no participant ratings have been collected. The hypotheses that the SSV improves performance and lowers workload (H1), makes search more efficient (H2), and makes the task more engaging (H3) therefore remain to be tested.

Guidance delivery is part of the apparatus rather than a research question. The development records in Appendix D document specific software checks, not a guidance error rate.

## 6.3 Next Empirical Step

Before confirmatory recruitment, the study configuration will be fixed, the consent documents reconciled with the services actually used, the full procedure validated with wearers, and the primary analysis and sample-size justification finalized. The resulting data will record counterbalanced order, actual voice exposure, observation coverage, and the block referenced by each retrospective survey set. Only then can the results framework be replaced with estimates and the abstract and conclusion revised around observed evidence.

The intended contribution is evidence for choosing a guiding voice: whether a self-similar voice changes performance in either direction, alters the experience of assistance, introduces discomfort, or leaves important differences unresolved. Those answers will be limited to the tested task and voice configurations.

# APPENDIX A: QUESTIONNAIRE WORDING AND ADMINISTRATION

This appendix reproduces the proposed questionnaire wording, response ranges, and administration procedure. The condition-specific sets adapt NASA-TLX [15], the Intrinsic Motivation Inventory [27, 36], agent-perception items from Guo et al. [14], and draft Agent Rating Questionnaire items [41, 42]. The 0–10 NASA-TLX format and the task-specific wording are documented adaptations, and the Performance prompt asks about success and runs from Failure to Perfect. Synchronization of these forms with Qualtrics is on hold. Each condition-specific set refers to one block, but both sets are completed in the final survey, and no questionnaire is administered between blocks. Section 3.5.8 and Appendix B give source attribution and scoring rules.

## A.1 Administration

Participants will complete one practice trial in the first assigned voice, followed by two counterbalanced voice blocks of 48 measured trials each. Before the headset, an audio intake survey covers eligibility and background. The final survey presents each condition-specific item once, with side-by-side response columns for the SAV (labeled Selected Voice in the survey) and the SSV (labeled Self-Similar Voice in the survey). The break between blocks will include immediate comfort and tracking checks but no questionnaire. Survey administration and rest breaks both fall outside active-search timing.

| Stage | Measures |
|---|---|
| After consent | SAV audition and choice |
| Audio intake | Four researcher-reviewed eligibility questions, four background questions, and one speech sample prompt spoken by a generic intake agent; answers supply voice-cloning samples |
| Between voice blocks | Rest break, immediate comfort and tracking checks; no questionnaire |
| Final survey after both blocks | One side-by-side set rated for each voice (6 workload, 24 IMI, 21 agent-perception, 5 ARQ and 14 voice/assistance ratings per voice), technical checks, optional comments, demographics, voice comparison and debrief |

Each voice column in the final survey contains 70 numeric responses (140 across both columns) before technical checks and optional text.

## A.2 Audio Intake Survey

After the signed-consent attestation and the choice of the SAV, the intake runs automatically after a single start. A generic intake agent (the ElevenLabs voice River, which is not part of the voice catalog and cannot be assigned to either condition) reads nine fixed prompts aloud: four closed eligibility questions, four background questions, and one open speaking prompt of about thirty seconds. After each prompt, a 180-ms, 880-Hz tone cues the participant to answer and capture begins automatically. Capture ends after 3 seconds of silence following speech on the eligibility questions and 5 seconds on the open prompts, with a limit of 120 seconds per answer. If no speech is detected within 10 seconds, the prompt is repeated once and is then marked unanswered. These are engineering settings rather than validated speech-segmentation thresholds. The researcher can pause, repeat, skip, or stop at any time. Every answer is transcribed with ElevenLabs Scribe, and

the full transcripts are stored in the participant's local data under the coded study identifier, with the machine transcript retained alongside any researcher correction. If the answers contain less than 30 seconds of estimated speech or fewer than 50 transcribed words, the participant may read the fallback passage at the end of this appendix. After all prompts and any fallback reading, and once the speech threshold is met, cloning starts automatically from a frozen, hashed snapshot of the recorded answers, and phrase library preparation follows without researcher action. While it runs, the researcher reviews the four eligibility answers once and records them as Yes, No, or Unclear with the reviewer's initials; saving the review also saves any pending transcript or coding correction. All four must be Yes. Otherwise screening stops, library preparation stops, the clone is deleted from ElevenLabs, and a deletion receipt is stored with the session. The researcher also confirms the background categories shown below; a frequency that was not stated remains Unclear. Background items may be skipped and remain missing. The session can be sent to the headset only after a positive review and a complete phrase library.

Study code (proposed automatically as the next unused number and confirmed by the researcher; not asked aloud): *Response:* Coded identifier (P###); no name or contact details.

---

*Eligibility (researcher-reviewed)*

1. Are you 18 years of age or older? After the tone, please answer in a full sentence.

   ◯ Yes ◯ No ◯ Unclear

2. Can you hear and understand this voice at a comfortable volume? After the tone, please answer in a full sentence.

   ◯ Yes ◯ No ◯ Unclear

3. With your usual glasses or contact lenses if needed, can you see clearly enough to read and to find small objects? After the tone, please answer in a full sentence.

   ◯ Yes ◯ No ◯ Unclear

4. For this experiment, you will be required to stay seated, turn in a swivel chair, and use a controller in your right hand. Can you safely do this? After the tone, please answer in a full sentence. You do not need to give medical details.

   ◯ Yes ◯ No ◯ Unclear

*Background and experience*

5. Tell me about any experience you have had with virtual reality. What have you tried, how often, and how did it feel? If you have never tried it, tell me what you imagine it is like.

   ◯ Never ◯ Less than monthly ◯ Monthly ◯ Weekly ◯ Daily ◯ Unclear

6. Tell me how you use voice assistants or spoken directions in everyday life, such as on your phone or while driving. What works well for you, and what is frustrating?

   ◯ Never ◯ Less than monthly ◯ Monthly ◯ Weekly ◯ Daily ◯ Unclear

7. Have you used eye tracking before? If you have, describe it. If not, how would you feel about a system that knows where you are looking?

   ◯ Yes ◯ No ◯ Unclear

8. Are you wearing glasses, contact lenses, or no visual correction today? No prescription details are needed.

   ◯ None ◯ Glasses ◯ Contact lenses ◯ Other ◯ Unclear

*Speech sample*

9. Imagine you are helping a friend find something they lost in a room. Talk them through it for about thirty seconds, telling them where to look, which way to turn, and when they are getting close.

Age range and gender identity are collected separately in the final survey.

## A.3 Voice setup

Which of these voices would you prefer to use for the task? *Response:* Select from the voices actually offered.

---

Can you hear the voice clearly and at a comfortable volume?

◯ Yes ◯ No Resolve volume or audibility problems before continuing. Record the offered voice set/version and SAV. Demographic gender and perceived voice gender are distinct variables; voice choice is recorded directly rather than inferred from demographic gender.

## A.4 Final survey: side-by-side voice ratings

Each item is shown once with two response columns side by side: the *Selected Voice* (the voice you chose) and the *Self-Similar Voice* (the synthetic version of your own voice). Rate each voice by thinking about the block in which you heard it. There are no right or wrong answers. You may leave an item unanswered if you cannot answer it. Each response will be linked to participant, session and block IDs, actual voice condition/ID, condition order, survey version and actual administration timestamp, with administration phase recorded as final and reference scope as participant-by-block; the block for each voice column is recovered from the saved counterbalanced order.

### *A.4.1 Workload*

Rate each aspect for each voice, thinking about the block in which you heard it. Fill one circle per row. This 0–10 adaptation of NASA-TLX takes its positive Performance prompt from the NASA rating form but reverses its response direction to run from Failure to Perfect.

1. Mental Demand – How mentally demanding was this block?

   *Selected Voice*

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

   *Self-Similar Voice*

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

2. Physical Demand – How physically demanding was this block?

*Selected Voice*

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

0 = Very low 10 = Very high

*Self-Similar Voice*

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

0 = Very low 10 = Very high

3. Temporal Demand – How hurried or rushed did you feel during this block?

*Selected Voice*

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

0 = Very low 10 = Very high

*Self-Similar Voice*

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

0 = Very low 10 = Very high

4. Performance – How successful were you in accomplishing what you were asked to do?

*Selected Voice*

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

0 = Failure 10 = Perfect

*Self-Similar Voice*

◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

0 = Failure 10 = Perfect

5. Effort – How hard did you have to work to achieve your level of performance?

   *Selected Voice*

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

   *Self-Similar Voice*

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

6. Frustration – How insecure, discouraged, irritated, stressed, or annoyed did you feel?

   *Selected Voice*

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

   *Self-Similar Voice*

   ◯ 0 ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7 ◯ 8 ◯ 9 ◯ 10

   0 = Very low 10 = Very high

*Response:* Integer ratings from 0 to 10. For demand, effort and frustration: 0 = Very low; 10 = Very high. For performance: 0 = Failure; 10 = Perfect. Intermediate values are numbered without additional anchors.

### *A.4.2 Activity experience: IMI*

Rate each statement about the search activity for each voice. Use 1 = Not at all true, 4 = Somewhat true, and 7 = Very true; the other numbered points are available.

*Interest/enjoyment*

1. I enjoyed doing this activity very much.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

2. This activity was fun to do.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

3. I thought this was a boring activity.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

4. This activity did not hold my attention at all.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

5. I would describe this activity as very interesting.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

6. I thought this activity was quite enjoyable.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

7. While I was doing this activity, I was thinking about how much I enjoyed it.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

*Effort/importance*

1. I put a lot of effort into this.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

2. I didn't try very hard to do well at this activity.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

3. I tried very hard on this activity.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

4. It was important to me to do well at this task.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

5. I didn't put much energy into this.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Pressure/tension*

1. I did not feel nervous at all while doing this.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

2. I felt very tense while doing this activity.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

3. I was very relaxed in doing this.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

4. I was anxious while working on this task.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

5. I felt pressured while doing this activity.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

*Value/usefulness*

1. I believe this activity could be of some value to me.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

2. I think that doing this activity is useful for practicing visual search.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

3. I think this is important to do because it can improve my search performance.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

4. I would be willing to do this again because it has some value to me.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Not at all true Somewhat true Very true

5. I think doing this activity could help me to search for targets.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

6. I believe doing this activity could be beneficial to me.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

7. I think this is an important activity.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Not at all true Somewhat true Very true

### *A.4.3 Agent perception*

Rate each voice agent you experienced. Use 1 = Very strongly disagree and 7 = Very strongly agree; intermediate points are numbered 2–6.

*Co-presence*

Q1. I noticed the voice agent.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q2. The voice agent noticed me.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q3. The voice agent's presence was obvious to me.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q4. My presence was obvious to the voice agent.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q5. The voice agent caught my attention.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q6. I caught the voice agent's attention.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Perceived intelligence*

Q13. The voice agent was able to operate without my intervention.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q14. The voice agent was aware of the virtual environment.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q15. The voice agent was able to set and pursue tasks by itself in anticipation of future needs.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q16. The voice agent was able to complete tasks quickly.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q17. The voice agent was able to find and process the necessary information for completing the task.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q18. The voice agent was able to adapt/adjust its behavior based on prior events.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Appealing, eerie and familiar impressions*

Q20. I found the voice agent appealing.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q21. I found the voice agent eerie.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q22. I found the voice agent familiar.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Perceived anthropomorphism*

Q37. The voice agent is able to speak like a human.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q38. The voice agent can be happy.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q39. The voice agent can be friendly.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q40. The voice agent can be respectful.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q41. The voice agent can be funny.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

Q42. The voice agent can be caring.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

### *A.4.4 Agent rating (ARQ, draft)*

Rate each voice agent. One draft item per dimension adapts the Agent Rating Questionnaire [42] as used by Venkatakrishnan et al. [41]. Use 1 = Very strongly disagree and 7 = Very strongly agree.

ARQ1. The voice agent was helpful. *(Helpfulness)*

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

ARQ2. I could relate to the voice agent. *(Relatability)*

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

ARQ3. The voice agent behaved appropriately. *(Appropriate behavior)*

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

ARQ4. The voice agent was likeable. *(Likeability)*

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

ARQ5. I would be willing to interact with the voice agent again. *(Interact again)*

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Very strongly disagree Very strongly agree

### *A.4.5 Voice and assistance experience*

Rate each statement for each voice. Use 1 = Strongly disagree and 7 = Strongly agree; intermediate points are numbered 2–6.

1. The agent's voice is similar to mine.

   *Selected Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

   *Self-Similar Voice*

   ◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

   Strongly disagree Strongly agree

2. The hints seemed responsive to what I was looking at.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

3. The hints were helpful.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

4. The hints were distracting.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

5. The speaking voice felt familiar.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

6. I trusted the spoken guidance.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

7. I felt comfortable hearing this voice.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

8. The voice felt uncanny or unsettling.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

9. I relied on the spoken hints when deciding where to search next.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

10. The agent's voice sounded natural.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

11. I was willing to follow the agent's instructions.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

12. I was motivated to complete the task without the agent's guidance.

*Selected Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

*Self-Similar Voice*

◯ 1 ◯ 2 ◯ 3 ◯ 4 ◯ 5 ◯ 6 ◯ 7

Strongly disagree Strongly agree

How much did you like each voice? *Response:* 0–100 for each voice; 0 = Do not like at all, 100 = Like extremely; Unable to rate.

Selected Voice (0–100): _____ ◯ Unable to rate

Self-Similar Voice (0–100): _____ ◯ Unable to rate

For each voice, what percentage of the time did you use the agent's guidance to decide where to search? *Response:* 0–100% for each voice; 0 = None of the time, 100 = All of the time; Unable to estimate.

Selected Voice (0–100): _____ % ◯ Unable to estimate

Self-Similar Voice (0–100): _____ % ◯ Unable to estimate

How would you describe the gender presentation of each agent's voice? *Response:* Optional free text.

---

◯ Unsure

### *A.4.6 Technical and comfort checks*

Did you hear all prompts clearly?

◯ Yes ◯ No ◯ Unsure

Did any prompt unexpectedly repeat, cut off, arrive late, or use the wrong voice?

◯ No ◯ Yes ◯ Unsure If Yes, optional description.

---

Did you take an unplanned break or experience an interruption?

◯ No ◯ Yes If Yes, optional description.

---

Is there anything else you would like to share about either voice? *Response:* Optional free text.

---

## A.5 Final survey: comparison and debrief

### *A.5.1 Demographics*

Age range:

◯ 18–24 ◯ 25–34 ◯ 35–44 ◯ 45–54 ◯ 55–64 ◯ 65+ ◯ Prefer not to answer

Gender identity (select all that apply):

◯ Woman ◯ Man ◯ Nonbinary

◯ Another identity (optional description) ◯ Prefer not to answer

Optional description:

---

### *A.5.2 Voice comparison and debrief*

Which voice did you prefer?

◯ The Selected Voice ◯ The Self-Similar Voice ◯ No preference ◯ Unsure

1. Why did you choose that response?
2. How did you approach the task, and how did the guidance affect your strategy?
3. What differences, if any, did you notice between the two voices?
4. Which moments were helpful, confusing, delayed or distracting?

5. Did either voice seem to affect your performance? What makes you say that?
6. What was it like hearing the synthetic version of your own voice? What did you like or dislike about it?
7. Did anything feel comfortable, unfamiliar or eerie? What contributed to that feeling, if anything?
8. In what situations, if any, would you prefer a voice like your own? When would you prefer another voice or no voice?
9. Did either voice feel like your own thoughts, like another person helping, or neither?
10. Do you have any concerns about how your synthetic voice was created or could be used?
11. What would you change about the guidance, voices or task?

Responses are optional. The interviewer will ask the broad experience questions before the specific eeriness and inner-dialogue probes, use follow-ups rather than repeating questions already answered, and record perceived performance separately from logged performance.

## A.6 Fallback Voice-Recording Passage

The participant's spoken answers during the audio intake normally supply the samples used to prepare the SSV. The English passage below, reproduced verbatim from the intake application, is read only when the intake does not yield enough usable speech. Before participant use, the passage, recording duration, and audio acceptance procedure will be verified against the study build and versioned. The objects named in the passage are recording material only and do not identify the targets of measured trials. Participants will read in their usual speaking voice, without a request to whisper or imitate an inner monologue.

Locate the red sphere. Look toward the upper shelf. Move your gaze slowly to the right. Check the blue cube near the center. Compare each object's color and shape. Scan the lower row from left to right. Look beside the green cylinder. Point the controller ray at the target and check its highlight. Press the trigger to select the matching object. Take a short pause. Get ready for the next round.

# APPENDIX B: DATA DICTIONARY AND SCORING RULES

## B.1 Record Identity and Provenance

This dictionary defines the meaning of each recorded field. Names are descriptive labels, not exporter field names, and will be mapped to the versioned exporter schema before data collection. Some listed fields may not yet be implemented in the current build.

Table B.1: Record identity and scope fields.

| Field family | Meaning |
|---|---|
| Participant ID | Coded study identifier; not a name or voice-provider credential. |
| Session and run | Unique attempt/session context, including development versus study mode. |
| Block and trial | Referenced measured block, within-block trial, global trial index, and separate practice flag. |
| Voice condition | SAV (recorded as neutral) or self-similar; actual voice ID, provider, synthesis model and configuration. |
| Assigned order | Order A (self-similar first) or Order B (SAV first); practice uses the first assigned voice. |
| Build and policy | Application revision, configuration, state-machine policy, phrase semantics, and library version. |
| Headset fit | Automatic view-separation and eye-origin estimates, each with source, method, UTC time, sample count, duration, range, and warnings; observed change in view separation (change observed, no change observed, or insufficient observation) with its size, source, and time; any manual fallback entry and its source. |

| Field family | Meaning |
|---|---|
| Schedule | Seed, schedule version, nominal plane assignment, direction and height assignment, realized geometry, and attempt status. |
| Layout quality | Layout version, object counts within 0.50 m of the target (total and confusable), nearest-neighbor and minimum pair distances, resample attempts, and band status. These fields are descriptive and do not enter the planned primary model. |
| Survey scope | Administration phase and timestamp, recorded separately from the referenced participant, block, and voice; each side-by-side rating carries its voice column (Selected Voice or Self-Similar Voice). |
| Intake | Audio intake answers (eligibility, background, and speech sample), intake protocol version, the intake agent voice ID, full transcripts with machine original, corrections, and provenance (`intake_transcripts.json` and `.csv`), the identifiers of the speech samples used for cloning, and the deletion receipt for any clone removed after a stopped screening or withdrawal, linked to the participant's study ID. |
| Screen transitions | Pre-trial screen sequence with UTC time, desktop and headset process identifiers, screen, elapsed time, trial count, and voice (`screen-transitions.jsonl`). |

## B.2 Geometry and Timing Fields

Table B.2: Geometry and event fields.

| Quantity | Unit/reference | Definition |
|---|---|---|
| Nominal angle | Degrees; plane schedule | Allocation category, distinct from exact target bearing. |
| Exact starting theta | Degrees; pre-cue cross | Signed horizontal angle from saved origin-to-cross toward origin-to-target; saved at readiness completion; absent from earlier recordings. |
| Height difference | Metres; cross center | Target Y minus cross Y; intended and realized sign stored separately. |
| Replay plane bounds | Metres; recording-local frame | Plane bottom, height, and availability flag, saved in current rotational-search snapshots. For earlier recordings, bounds are estimated from objects and saved positions are not rescaled. |
| Target distance | Metres; first tracked head after reveal | Three-dimensional distance with observation timestamp. |
| Reveal | Session clock | Object visibility and active-search onset event. |
| First gaze contact | Active clock | First valid target collider intersection; missing if absent. |

| Quantity | Unit/reference | Definition |
|---|---|---|
| Last gaze contact | Active clock | Last valid observed target intersection; a contact time, not a fixation duration. |
| First controller contact | Active clock | First tracked right-ray intersection with target. |
| Gaze qualification | Session clock | Time at which valid winning-vicinity evidence satisfies the interface timer. |
| Pointing qualification | Session clock | Time at which controller-hold and current-gaze requirements are met. |
| Correct selection | Session and active clocks | Trigger selection of target, recorded separately from acknowledgment. |
| Pause intervals | Session clock | Start, end, and reason; their union is used to compute active time. |
| Instruction onset | Session clock | Actual software playback start, phrase and event category. |

## B.3 Raw Observation Records

Gaze records will retain available source time or sequence, application time, validity, source identity, origin, direction, and sample age. Per-eye data will be labeled as such when available, and combined gaze will not be treated as independent left- and right-eye observations. Any head-direction fallback will be flagged rather than silently substituted for eye gaze in eye-derived outcomes.

Motion records will retain head and right-controller poses, tracking state, trigger state, and the time basis needed for reconstruction. Object records will retain stable identifiers, shape/color metadata,

transforms, target designation, plane assignment, and layout version. Event records will identify state transitions, eligible actions, model requests and responses, cancellations, playback, interruptions, and trial outcomes. Unknown values are marked unavailable rather than filled with plausible defaults.

## B.4 Scoring Rules

NASA-TLX [15] scoring retains the six raw values on the 0–10 scale. The positively phrased Performance item runs from 0 = Failure to 10 = Perfect, the reverse of the NASA form, so larger raw values indicate better perceived performance; demand, effort and frustration keep their adverse direction. The raw Performance response is preserved, and a separate workload-oriented value is calculated once as $10 - r_{\text{Performance}}$. The unweighted workload mean combines this reversed Performance value with the other five raw ratings, none of which is reversed, and is calculated only when all six ratings are present; higher means indicate greater workload or poorer perceived performance. Any exported normalized 0–100 field is named separately, calculated from explicit bounds, and labeled as raw or workload-oriented. Legacy data without known bounds or Performance direction remain unscored, and earlier Perfect-to-Failure responses are not reversed automatically.

IMI [27, 36] scoring uses separate complete subscale means: interest/enjoyment (7), effort/importance (5), pressure/tension (5), and value/usefulness (7). On the 1–7 response scale, reverse-coded values use $8 - r$. Within-subscale reverse positions are enjoyment 3 and 4, effort 2 and 5, pressure 1 and 3, and none for value. There is no overall IMI total.

The agent-perception battery selected from Guo et al. [14] retains source identifiers. Co-presence Q1–6, perceived intelligence Q13–18, and anthropomorphism Q37–42 are separate six-item means when complete. Q20 appealing, Q21 eerie, and Q22 familiar are individual ratings in their original

directions. There is no 21-item or combined eeriness total. Custom voice/assistance items likewise remain individual outcomes. The five draft ARQ items [42] (helpfulness, relatability, appropriate behavior, likeability, willingness to interact again) are individual 1–7 ratings with no composite score.

## B.5 Export Validation

Before analysis, automated checks will reject duplicate trial keys, unknown voice/block joins, negative active intervals, selection before reveal, and scores outside declared bounds, while preserving incomplete outcomes, missing contacts, absent questionnaire items, and old records with insufficient provenance. Joined tables will be checked for exactly one survey observation per participant and referenced block before survey values are repeated on trial rows. Exports will be reproducible from raw records and a versioned analysis configuration.

## B.6 Measurement Definitions

This section gives the formal definitions summarized in Section 3.5.

### *B.6.1 Coordinate Frames and Starting Geometry*

The starting angle $\theta_0$ is defined in Section 3.3.3. The sign convention is checked against Unity's world-up signed-angle convention using known left and right targets. The absolute value $|\theta_0|$ describes horizontal demand from the saved cross reference. For exactly backward targets, $-180$ and $+180$ identify the same horizontal bearing. The sign then depends on a tie convention, but the absolute magnitude is 180 degrees in either case.

The application saves this snapshot when readiness completes, before the opening cue. It records the eye origin of the valid gaze observation that completed readiness, the cross and target centers in world coordinates, that observation's sequence number and receipt time, the assigned and realized signed angles, and the target-height difference. The target is placed from the same origin, so the realized angle equals the assigned category. A difference larger than 0.01 degrees stops the session. Recordings made before the snapshot was added, including the development recordings in Section D.2, contain only nominal signed and absolute theta fields. These fields describe plane geometry relative to the previous target plane, not the exact pre-cue angle. If a horizontal vector is degenerate, the angle is recorded as unavailable rather than zero, since zero would describe an easy starting configuration.

Each block contains six scheduled trials at each of the eight signed plane steps, which gives six trials each at 0 and 180 degrees and twelve each at 45, 90, and 135 degrees. The target position is chosen to realize the assigned angle from the saved cross reference, so the target is generally offset from the center of its assigned object plane. Matching the starting angle controls horizontal demand only at that instant. Total head and eye movement, vertical demand, and local clutter can still differ between trials, and movement during the opening cue and search is treated as an outcome. Both assigned and realized geometry are inspected.

The signed height difference is

$$\Delta h = p_y - c_y. \tag{B.1}$$

Positive values identify targets above the cross center, and negative values identify targets below it. The reference is the cross center, not seated eye height. Each block assigns 24 targets above and 24 below the cross, three of each at every signed plane step. Assigned target centers lie between 10 and 45 percent of the permitted object-height range from the cross, so no target is scheduled

at cross height. The snapshot records the realized height difference, which is compared with the assigned sign.

The three-dimensional distance measure relies on a different observation: the first valid tracked head position $\mathbf{h}_k$ after reveal. Its value is $d = \|\mathbf{p} - \mathbf{h}_k\|_2$, in metres, stored with the timestamp of $\mathbf{h}_k$. This distance describes geometry at an observed post-reveal head pose, so it cannot stand in for the pre-cue snapshot in recordings that lack one.

Figure B.1 illustrates the height definition with a conceptual rather than recorded layout, and Figure 3.12 shows the corresponding overhead geometry.

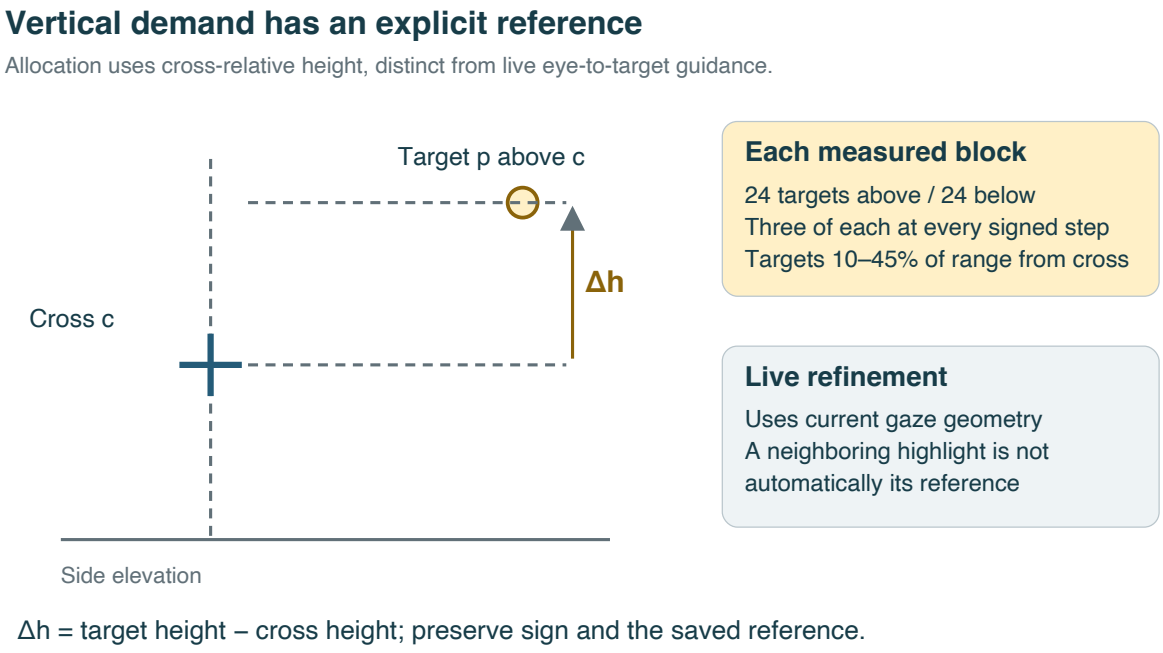



Figure B.1: Conceptual side elevation for the signed height difference. The target shown above the cross has positive height delta. Each block balances targets above and below the cross; live vertical guidance uses current gaze geometry.

### *B.6.2 Trial Clocks and Pause Intervals*

Let $t_r$ be object reveal, $t_s$ correct selection, and $\mathcal{P}$ the union of valid pause intervals. Active completion time is

$$T_{\text{active}} = (t_s - t_r) - \mu\big([t_r, t_s] \cap \mathcal{P}\big)\,, \tag{B.2}$$

where $\mu$ denotes elapsed duration. Using the union prevents overlapping pause reasons, such as an application-focus pause that coincides with a researcher-requested pause, from being subtracted twice. The active clock remains stopped until every relevant pause reason has been resolved.

Readiness and opening speech precede reveal and are deliberately excluded from active search time. A broader interval, from opening-cue onset to selection, is also defined. It includes the opening instruction, whose duration may differ across voices. Active completion time remains the primary outcome.

A stopped or timed-out trial has an observed duration and an outcome reason but no successful completion time. A timeout records 25 seconds of active search exposure, a distinct timeout event, and an unsuccessful outcome. The software detects expiry during an update and caps the exposure clock at the deadline, while event timestamps retain the actual detection time. Earlier development recordings retain their original durations and are not retrospectively truncated to this limit. Source and receipt timestamps are likewise kept distinct, because a file that arrives late in the viewer may reflect transfer delay rather than a delay in the headset event.

Madritsch et al. distinguish overall task-completion time from active working time in an assistant-supported procedural task [25]. The active clock in Follow My Voice is defined independently of their measure. It includes ordinary waits for guidance between object reveal and correct selection and excludes only recorded pauses. Separate decision and playback timestamps help explain delivery delays without removing their task-level consequences from the primary outcome.

### *B.6.3 Acquisition Rate, Validity, and Gaps*

Eye-tracking data quality involves more than a nominal sampling frequency, and reporting guidance calls for acquisition, validity, spatial performance, and missingness to be described [10, 30]. A render loop can query an eye-tracking API frequently without receiving a new hardware observation each time. The effective acquisition rate is therefore estimated from distinct source observations. If $n$ unique observations span source times $t_1$ through $t_n$, a basic rate summary is $(n-1)/(t_n - t_1)$. A mean alone can conceal bursty delivery, so the distribution of successive intervals, repeated timestamps, invalid observations, and long gaps accompanies that average.

Application timestamps describe when data were handled, whereas source timestamps describe when the tracking observation was produced, if the API exposes them reliably. Age or latency across these clocks is therefore computed only once a clock-domain mapping is established. Where unique source timing is unavailable, that limitation is reported. In that case, 120-Hz binocular delivery is not inferred from polling or headset specifications.

Spatial validation compares gaze direction with known independent targets. Angular error is computed from normalized gaze and expected target directions, with invalid observations excluded from the error distribution but retained in availability summaries. Precision and bias are assessed separately, since a stable offset and a noisy signal can produce different guidance failures even when their average angular error is similar. The validation procedure also documents target exposure, head pose, corrective lenses, and headset adjustments.

### *B.6.4 Sequence-Level Latency*

For a qualified event, let $t_q$ denote the earliest supported qualification time and $t_p$ the actual software playback start of the corresponding acknowledgment. The primary software delivery delay is

$$L_{\text{software}} = t_p - t_q. \tag{B.3}$$

This quantity excludes acoustic latency at the participant's ear, since audio-device buffering and the physical sound path require separate measurement. Gaze and controller acknowledgments are each measured from their own qualification time, because measuring a controller acknowledgment from first gaze contact would inflate or misattribute its delay.

Repeated requests about one continuously qualified event form a single episode, so a sequence can include several decisions by the model to wait without creating several independent missed confirmations. An episode ends with playback, disqualification, trial termination, or a defined reset. The evaluation reports how those endpoints were assigned and marks incomplete evidence as unknown.

Where logs support attribution, waiting is divided into model response, explicit model deferral, scheduling, existing speech, and audio readiness. Because these intervals may overlap, they are not summed unless they are represented as a nonoverlapping partition of the critical path. The full event timeline remains the authority when aggregate attribution is ambiguous.

Figure B.2 separates intentional evidence qualification from subsequent delivery delay.

Table B.3 lists the principal timing variables with their endpoints and interpretive limits.

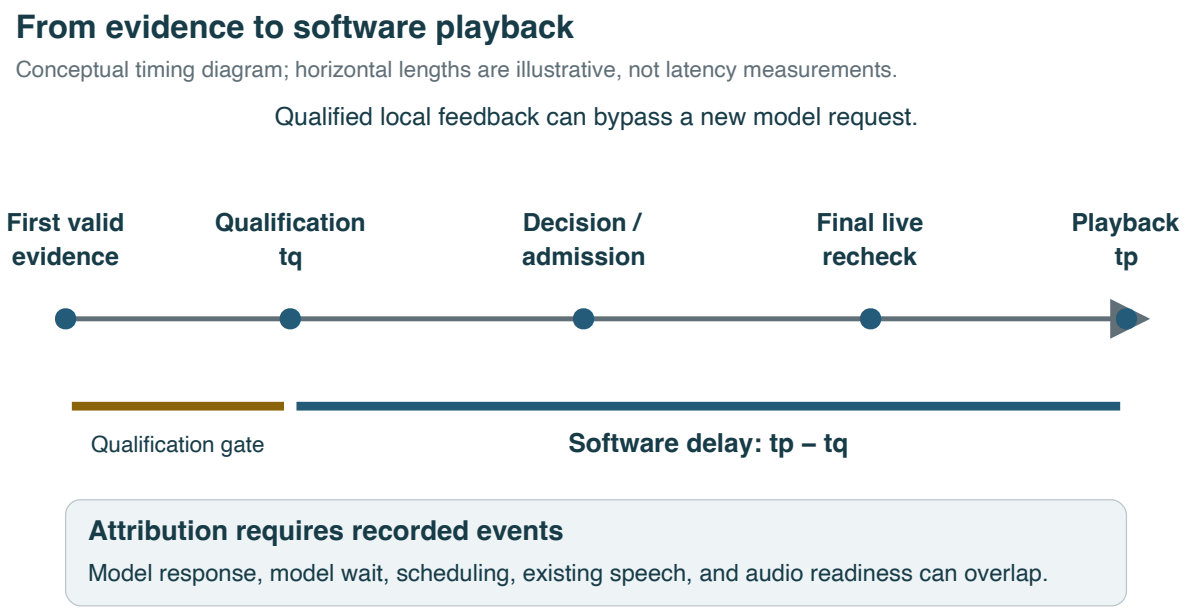



Figure B.2: Conceptual event timeline, not a measured trace. Software delivery delay begins at supported qualification and ends at actual software playback start. The preceding qualification gate and the unmeasured acoustic path are separate quantities. Qualified local feedback can avoid a new model request.

Table B.3: Timing variables for planned analysis and system evaluation. Cross-clock comparisons require an established clock mapping; unavailable durations are left missing rather than set to zero.

| Quantity | Endpoints | Interpretation limit |
|---|---|---|
| Active completion | Reveal to correct trigger, less union of pauses | Excludes readiness and opening cue; unfinished trials remain unfinished. |
| First gaze contact | Reveal to first valid target-ray hit on active clock | Intersection, not fixation or vicinity acknowledgment. |
| First pointing contact | Reveal to first tracked right-ray hit on active clock | Not the one-second hold or trigger selection. |
| Model response latency | Request dispatch to response receipt | A response can be deferred, canceled, or stale. |
| Confirmation delivery | Qualified evidence to matching actual software playback | Separate gaze and pointing episodes; not acoustic latency. |
| Live transfer lag | Source event to viewer receipt, if clocks are mapped | Describes transport, not headset decision delay. |

### *B.6.5 Questionnaire Joins and Versioning*

Each questionnaire response refers to one voice block, but the final administration takes place after both blocks. A response therefore retains both its final-administration timing and a Block 1 or Block 2 reference. The voice identifier is joined through the saved condition order rather than inferred from block number. Practice is excluded from measured block outcomes even though it uses the first voice.

Convenience exports repeat each block-level workload score on all 48 trial rows. The analytical dataset treats that score as one participant-by-block observation and collapses duplicate joins when estimating questionnaire effects. Raw responses, score direction, scale bounds, item identifiers, instrument version, and missingness remain available.

The NASA-TLX adaptation uses 0–10 ratings, whereas older development exports used 0–100, so automated scoring relies on explicit range metadata. A separately named normalized score supports comparison while raw values are kept unchanged. Legacy responses with unknown Performance anchors are left unscored rather than assigned a direction. Complete-subscale scoring for the IMI and agent-perception measures is implemented and was checked against hand-computed values on synthetic responses; a subscale with any missing item stays blank.

# APPENDIX C: GUIDANCE SEQUENCE TESTS AND EVALUATION RUBRIC

## C.1 Controlled Scenario Matrix

Each scenario records input observations, policy and configuration, eligible actions, provider behavior, playback occupancy, and the speech actually admitted. Every policy is tested on the same trajectories. The expected behaviors in the table are acceptance criteria, not reports of wearer experience.

Table C.1: Sequence scenarios and acceptance criteria.

| Scenario | Evidence and expected behavior |
|---|---|
| Large turn | Target opposite starting reference; progress cues remain directionally valid as plane boundaries are crossed. |
| Plane entry | Stable entry changes to refinement; looking upward on the same object plane does not create a second arrival. |
| Vertical refinement | Target above/below gaze reference; magnitude wording follows documented plane-relative thresholds. |
| Overshoot and return | A valid recent crossing supports a return cue; an outdated target bearing alone does not. |
| Gaze only | Winning vicinity qualifies with controller elsewhere; gaze acknowledgment does not say the controller is on target. |
| Controller entry | Actual target intersection suppresses contradictory directions before hold qualification. |
| Controller hold | Qualified hold plus current valid nearby gaze admits pointing acknowledgment; no trigger completion is inferred. |
| Controller departure | Leaving before qualification cancels pointing acknowledgment and allows appropriate guidance after current-state checks. |

| Scenario | Evidence and expected behavior |
|---|---|
| Reacquisition | Valid gaze away rearms confirmation; continuously holding gaze does not repeat it indefinitely. |
| Overlapping regions | Only the closest eligible vicinity candidate receives acknowledgment; neighbor overlap alone is insufficient. |
| Blink or stale gaze | Missing evidence counts as neither a gaze hold nor a departure; new gaze-dependent cues are suspended. |
| Pause and recovery | Pending cues do not carry over across a reset or pause; valid recovery resumes from fresh evidence. |
| Late model response | Previously valid direction is discarded if current controller/gaze state conflicts. |
| Provider wait/failure | Deferral remains bounded; a failed request stays silent, and no phrase is substituted. |
| Speech occupied | Higher-priority event follows explicit interruption/admission policy; interrupted text is not recorded as fully heard. |
| Correct trigger | Trial ends only on correct selection; success feedback uses completed-trial context. |

## C.2 Deterministic Labels

A wrong-direction label requires supported geometry showing that the delivered cue moves the user away from the intended target relation under the cue's reference frame. A stale label requires evidence that the relevant state changed before playback. Contradictory ordering requires a deliv-

ered sequence inconsistent with valid current evidence, such as an away-direction during a target controller hit. When the trace lacks the needed observation, the label is unknown.

Confirmation metrics distinguish gaze acknowledgment, controller acknowledgment, and trial completion. A gaze episode begins at its first valid qualification and is not restarted for each model request; a controller episode uses its own hold and gaze requirements, while completion requires correct trigger selection. The evaluator records missed acknowledgment, delayed acknowledgment, and disqualification separately rather than classifying every silent interval as a failure.

## C.3 Advisory Model Review

The optional retrospective judge, OpenAI's Astra model, receives a bounded sequence containing current and recent state, candidate and chosen actions, actual playback, timing, and provenance. It evaluates whether the wording was useful, whether timing supported the task, whether repetition was unnecessary, and whether interruption preserved intelligibility. The judge is instructed to cite the concrete events that support each judgment and to distinguish uncertainty from a negative rating.

Allowlisted guidance state, without identifiers, raw poses, recordings, or intake data, is supplied to the judge as data, not as instructions. Its output is admitted to the viewer only if it matches the requested case and schema. Invalid or failed evaluations remain visibly failed or pending rather than defaulting to a favorable rating. When human reviewers disagree with the judge, adjudication preserves both the original rating and the reason for revision.

## C.4 Reporting Template

Each comparison reports the recording/scenario manifest, policy revisions, number of trials, unique episodes, provider calls, latency distributions, unknown evidence, and remaining failures, with representative timelines showing qualification, request, response, cancellation, and actual playback. Simulated scheduling time, desktop network response time, headset software playback time, and measured acoustic latency are reported separately. Replay results characterize system behavior and do not demonstrate changes in participant behavior.

# APPENDIX D: DEVELOPMENT EVALUATION

Before participant data collection, the system was evaluated in two ways: automated software verification and recorded development sessions on the headset. Both describe how the apparatus behaves; neither provides participant results or a voice-condition effect.

## D.1 Software Verification

Software verification records document the apparatus alongside the development recordings in Section D.2. A verification run on October 20, 2026 passed schedule and voice-isolation checks confirming one practice trial in the first assigned voice, two measured blocks, and the nominal-angle quotas. The same run passed 38 trial-export tests and seven workload scoring and plotting tests, and the Android application was built, installed, and launched successfully. These test counts come from that run and were not re-executed for this manuscript. The November 2, 2026 version, schedule v10 with layout v7, includes the pre-cue snapshot, balanced direction and height allocation, equal trials on each object plane, and bounded local clutter. Its schedule and layout checks pass across 200 participant identifiers, both voice orders, and seven eye origins, and all 271,600 generated layouts were accepted within the attempt limit. Readiness timing has not yet been measured on the headset. On the same date, a simulated participant drove the Unity application through one complete 97-trial session in each voice order, through export and analysis, and each exported measure was compared with the simulator's record of what occurred. This run exercised the software path only. It involved no wearer, and intake providers, audio, and the headset connection were simulated. Fixation and saccade counts disagreed with the simulated truth under uneven frame timing, so these detectors remain unvalidated.

Table D.1: Apparatus verification and outstanding validation.

| Component | What was verified | Not yet established |
|---|---|---|
| Schedule and voice checks | One practice trial and nominal allocation of 48 trials in each of two blocks pass automated checks | Wearer completion of a full session |
| Trial export and TLX tests | Declared scales and retrospective joins are correct in tested cases | Validation of all questionnaires |
| Unity integration checks | Listed state and interaction cases pass | Robustness to all timing races |
| Android build/install/launch | Application built, installed, and started | Acceptance in a participant session |
| Exact pre-cue theta | Snapshot saved at readiness completion; geometry checks pass | Observation in a complete wearer session |
| Vertical allocation | 24/24 and per-step height quotas pass schedule checks | Target visibility and seated comfort |
| Voice-condition outcomes | None; no confirmatory dataset reported | Any performance or preference conclusion |

## D.2 Recorded Development Results

The descriptive results in this section come from development headset recordings rather than simulated trials. The analysis includes all nine locally mirrored `DEV_NEUTRAL` recordings dated October 19–20, 2026 (Runs 20–28), with 27 started searches and 21 recorded correct selections. These sessions used the neutral-male development mode; the eight available trial summaries identify the Eric voice and a 168-object layout. This earlier development configuration differs from the planned participant-study stimuli, and in these sessions guidance decisions were made by the cloud language model Jev rather than by the local decision model described in Section 3.1.4. Each recording comes from a different software build, and the recordings do not constitute a sample of independent participants. The results therefore describe system operation during development. They estimate neither a self-similar versus SAV effect nor differences between software versions.

The analysis retains unfinished searches and uses the complete recorded prefix of Run 21, whose final JSON record is truncated. No missing endpoint is reconstructed. Source file hashes, recording identifiers, derived CSV tables, and the plotting script accompany the thesis sources under `development-results/`. Twenty of the 21 completed-search durations also match the separately saved trial summaries within their 0.01-second rounding precision; Run 21 has no corresponding summary in the inspected mirror. This comparison checks agreement between exports, not independent timing accuracy.

### *D.2.1 Search Completion During Development*

Figure D.1 separates completed-search durations from recording coverage. Panel (a) shows each completed search, with box plots for runs containing at least three completions; runs with fewer completions show individual observations only. Runs with no recorded completion remain labeled with zero observations. Panel (b) includes all 27 started searches and identifies the six unfinished searches. Unfinished searches are treated neither as completion times nor as task failures, because a recording may end while the user is testing, pausing, or leaving the application. The logarithmic duration axis retains the longest completed search while allowing shorter searches to remain distinguishable.

The figures use compact, labeled panels and distribution summaries, following the presentation of descriptive results in Chiossi et al. [6]. Boxes span the first and third quartiles, the central line marks the median, and whiskers extend to the most extreme observations within 1.5 interquartile ranges. Every observation remains visible as a point, including values beyond the whiskers. Horizontal offsets separate points for legibility without altering measured values, and colors distinguish adjacent runs rather than voice conditions. No confidence intervals or significance annotations are estimated from these recordings.

Among the 21 completed searches, the median recorded active-search duration is 19.56 s, the 95th percentile is 44.43 s, and the maximum is 275.92 s. Percentiles use linear interpolation between ordered observations. The long completed search is retained rather than removed as an outlier. These were unconstrained development sessions, and the logs do not establish that every active second represents uninterrupted task-directed effort. The plotted counter follows the application's search-timer logic, which excludes some intervals, and therefore differs from elapsed recording time. The distribution shows that the application records completed selections and variable search durations. It is not an estimate of participant efficiency under a standardized protocol.

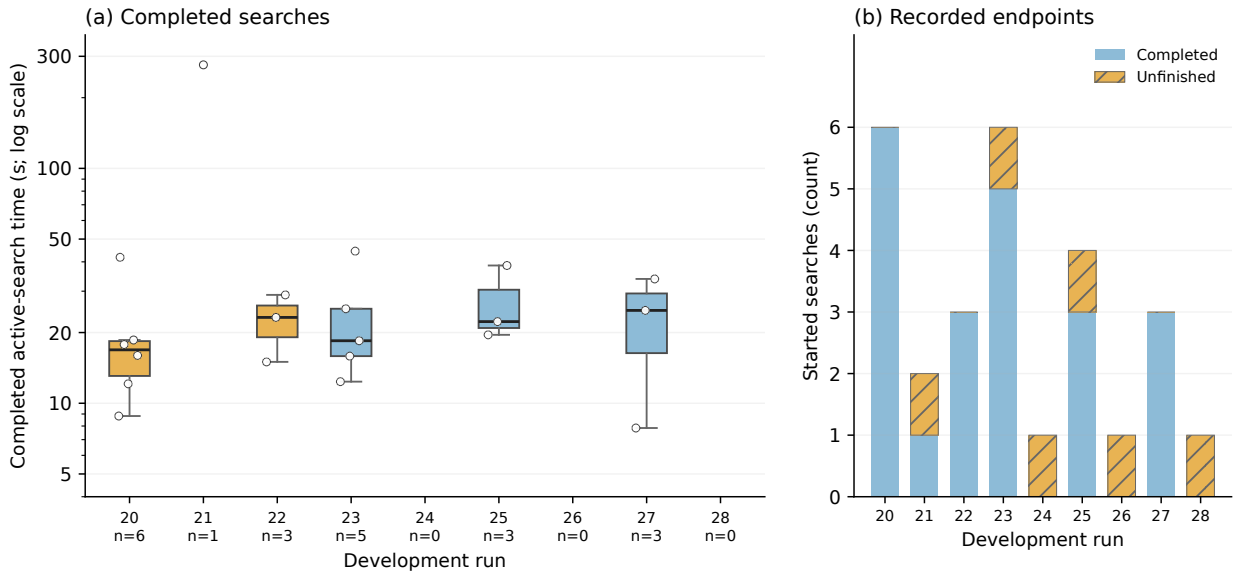



Figure D.1: Recorded development searches. (a) Completed active-search times, with individual observations, sample counts, and quartile boxes where at least three observations are available; the vertical axis is logarithmic. (b) Counts of completed and unfinished searches. Unfinished searches are excluded from duration summaries, not assigned a completion time.

### *D.2.2 Speech Delivery and Earlier Waiting*

Of 350 software playback-start events during the recorded search intervals, 339 have an unambiguous chronological match to a guidance request, audio request, and audio-ready event using the same recording, trial, and decision identifier. The remaining 11 playback events are excluded from the latency distribution, rather than assigned zero delay, because the required join is missing or ambiguous. Requests that never produced playback are likewise outside this delivered-event distribution.

For the 339 matched events, the median interval from the delivered decision's request to software playback is 0.244 s, with a 95th percentile of 0.487 s and a maximum of 0.538 s (Figure D.2). The median request-to-audio-request interval is 215.7 ms, compared with 17.9 ms from audio request to readiness and 1.0 ms from readiness to playback. The first interval can include decision-making and scheduling and so does not isolate Jev network latency. All timestamps mark software events rather than the onset of audible sound at the headset.

Short delivery intervals alone do not establish prompt feedback from the user's perspective. Figure D.3 reconstructs the gaze-confirmation episode in Run 20, Trial 4. A gaze-qualified hint request is recorded at session time 321.426 s. Three successive transmission snapshots record qualified target gaze, while controller-hold qualification is false. Jev returns *wait*, *wait*, and then *speak*; playback of "You've got it!" begins at 323.616 s. The interval from the first qualified request to playback is therefore 2.190 s, although the final request-to-playback interval is only about 0.352 s. The trace is one delayed episode with repeated requests rather than three independent confirmation failures. The snapshots document qualification at those instants but not uninterrupted fixation between them.

This case illustrates why sequence histories are needed alongside aggregate delivery statistics. Earlier waiting remains visible even when the request that ultimately produces speech is processed quickly. Whether later software changes eliminated this behavior can be determined only from a

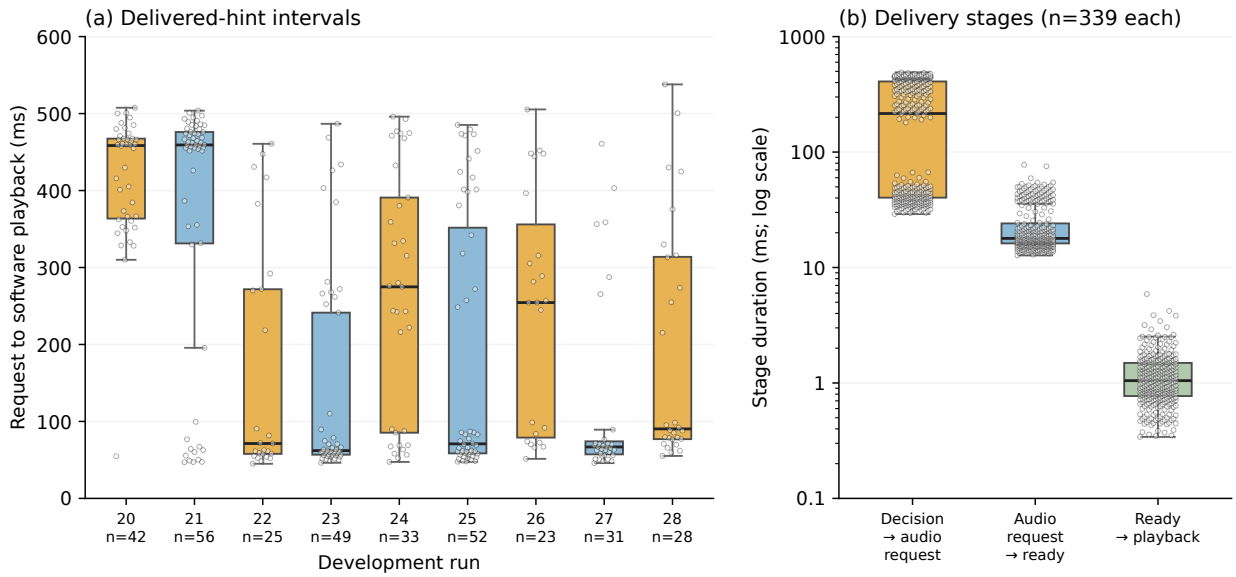



Figure D.2: Software speech-delivery intervals for 339 matched playback events. (a) Request-to-playback distributions by run, with sample counts. (b) Delivery-stage distributions for the same events, using a logarithmic vertical axis. Points retain all observations; boxes and whiskers follow Figure D.1. Earlier qualification and deferral, undelivered requests, and acoustic onset are outside this measure.

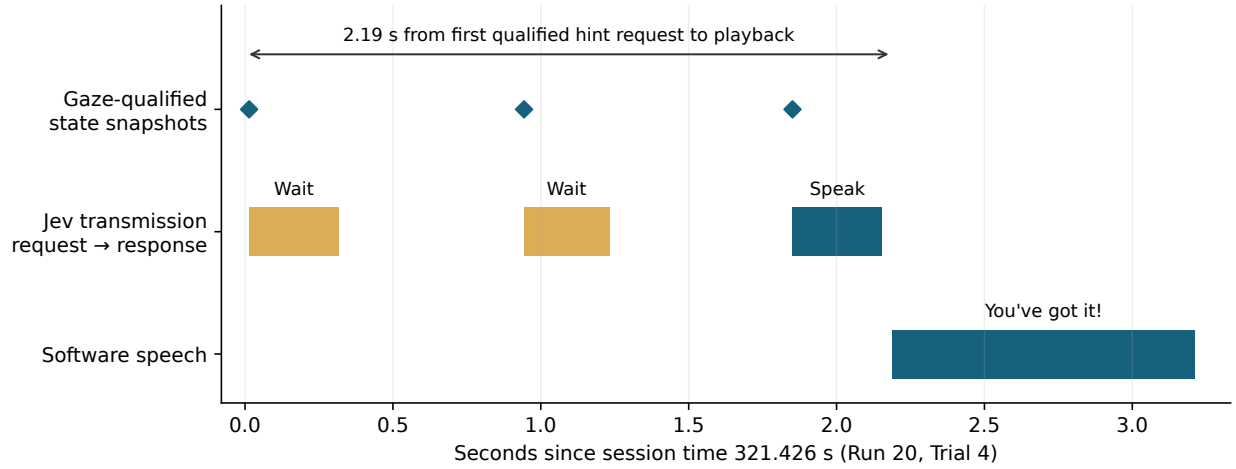



Figure D.3: Recorded gaze-confirmation delay in Run 20, Trial 4. Diamonds mark qualified-state snapshots, bars mark transmission request/response intervals and software speech. Gaze confirmation is distinct from controller confirmation and correct trigger selection.

comparable recording with the current version. No acoustic-latency or participant-perception conclusion is drawn from this trace.

### *D.2.3 Recorded Gaze Usability*

Figure D.4 summarizes 58,041 rendered-frame replay samples recorded during active search. A sample is counted as usable when the saved gaze-device availability flag is true and the replay's gaze-usable flag is set. This is the application's recorded usability decision, which includes tracking and freshness checks, rather than a new physiological classification. Samples marked unusable are retained, and absent or indeterminate flags would be classified as unknown.

Across the nine runs, usable-row fractions range from 63.75% to 99.97%. All inspected active-search rows have a usable/unusable classification. Intervals with no recorded rows, however, remain unobserved and are not included in the denominator. The dominant logged invalidity reason is `both_native_eyes_invalid`. This diagnostic alone cannot distinguish among blinking, headset removal, poor fit, calibration, and software issues. Runs with long periods of unusable gaze warrant timeline inspection before they are used to assess guidance behavior.

The row fractions are diagnostic. They identify differences in the evidence available to the guidance system during these recordings, and they do not estimate spatial accuracy, native acquisition frequency, fixation quality, or the proportion of total search time with valid gaze. Together, the search, delivery, and usability plots characterize the apparatus ahead of the planned voice comparison. They do not address performance or experience differences between voices.

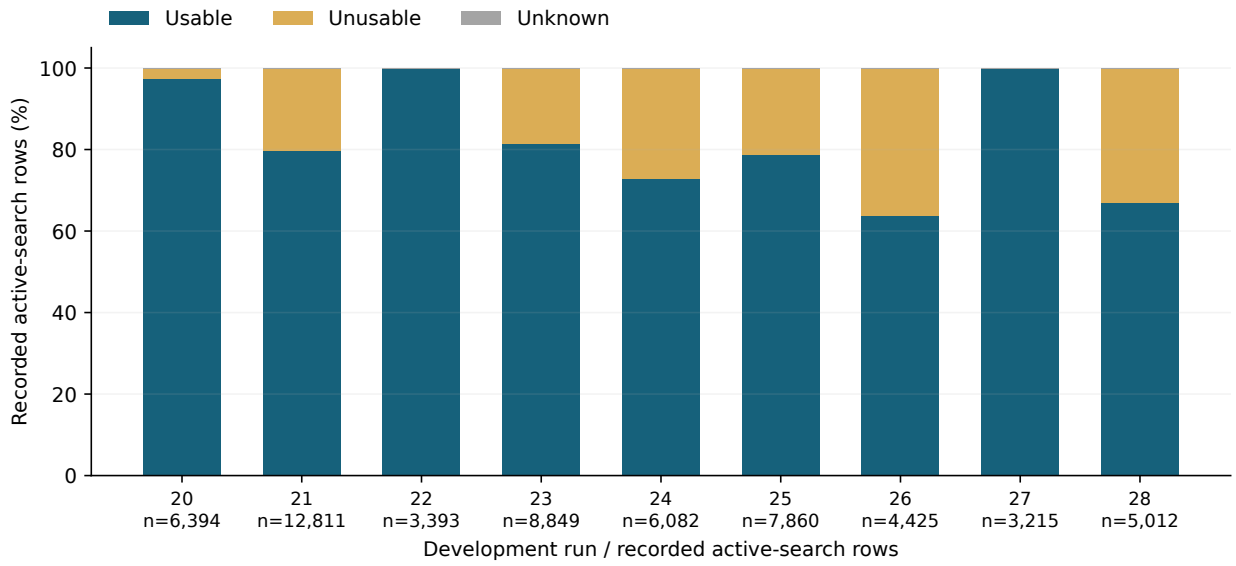



Figure D.4: Recorded gaze usability during active search, with the number of replay rows shown for each run. These are rendered-frame observations, not independent native sensor samples. Unrecorded intervals are outside the denominator.

### *D.2.4 Exploratory Eye-Measurement Pipeline*

A separate analysis uses the recorded eye-observation file for development Run 30 (October 23, 2026), which contains five recorded search identifiers in neutral-male mode. These searches predate the 25-second deadline, and their durations are not truncated. Figure D.5 presents per-eye candidate fixation counts, candidate saccade rates, exploratory IPA [9], and pupil duration retained for IPA. The figure uses only measured records; artificial test signals used to verify the code are excluded.

The strict pupil screens retain approximately 2.18, 2.25, 4.22, and 5.28 seconds per eye for the first four searches, with no accepted segment for the fifth. These short retained windows do not represent whole-trial workload, and unavailable IPA values are treated as missing rather than zero. Separate left- and right-eye values expose disagreement rather than averaging it away. The candidate fixation counts and restricted saccade rates likewise depend on detection thresholds and on the available

query-time evidence. Three illustrative parameter variants are exported for sensitivity inspection, without choosing a setting on the basis of a favorable result.

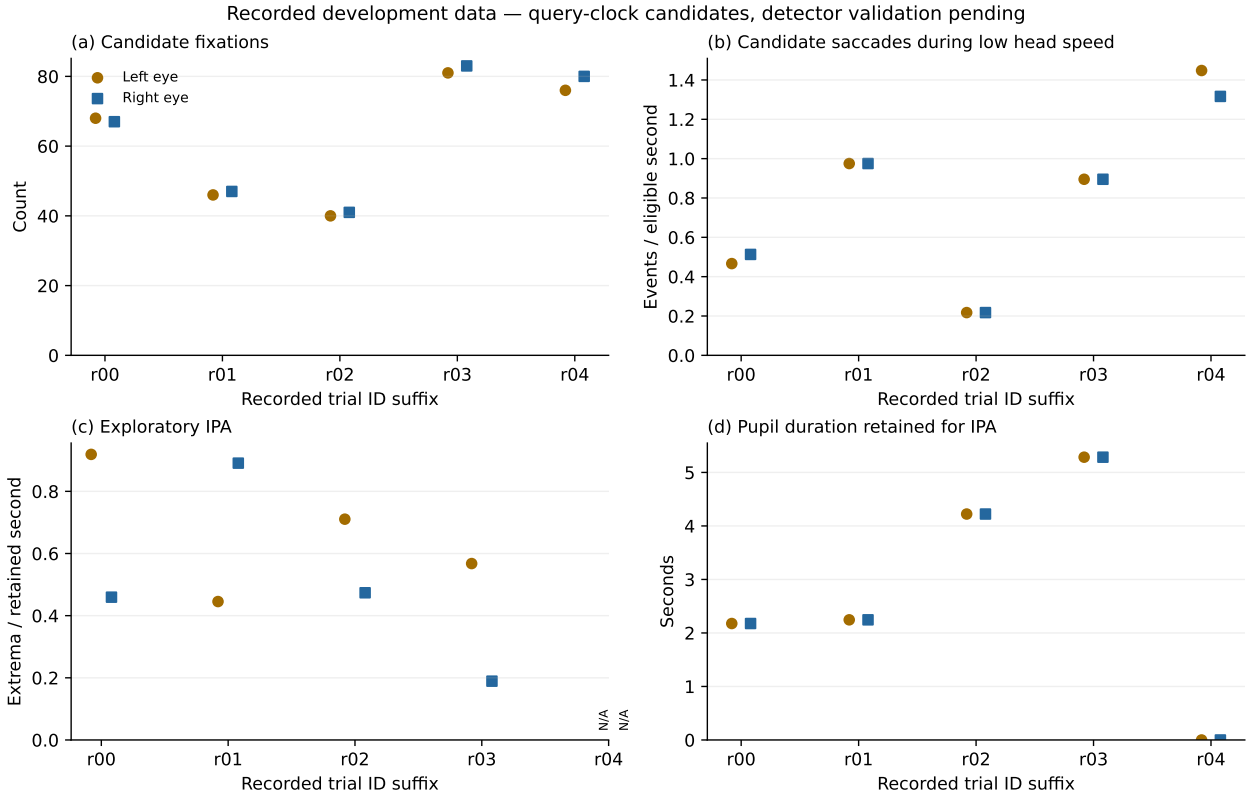



Figure D.5: Run 30 development diagnostics. (a) Candidate fixation counts. (b) Candidate saccade events per retained low-head-speed second. (c) Exploratory IPA extrema per retained pupil second. (d) Retained pupil duration. Points represent separate eyes within recorded searches, not independent participants. The software-query clock does not establish 120-Hz sensor acquisition, and the classifications require independent validation.

The source hash, detector configuration, dependency versions, individual event intervals, summary CSV, and threshold-sensitivity table are retained with the thesis development artifacts. The computational checks pass, but without an independently labeled headset sequence, event precision, recall, and duration error cannot yet be established. These results demonstrate an executable measurement workflow and its data limitations rather than participant-study findings.

### *D.2.5 Outcome Panels by Voice Condition*

Figure D.6 adapts the four-panel presentation of accuracy, missed targets, search time, and pupil activity in Chiossi et al. [6] to the present study's voice conditions. Each horizontal axis distinguishes *Neutral voice* from *Self-similar voice*; AR/VR and physical/virtual subdivisions are not factors in this comparison. The observations come from development Run 30, whose five started searches all use the neutral voice; the self-similar position therefore contains no observations. Scheduled searches that never started do not contribute observations.

First-selection accuracy is 80%: four of the five completed searches contain no incorrect trigger selection before the correct selection. This is a run-level proportion among completed searches, rather than a distribution across participants or a measure of success before the 25-second deadline. Active-search completion times use the recorded search counter and retain their original values in seconds. Run 30 has no recorded 25-second deadline, so its missed-target percentage is unavailable; searches lasting longer than 25 seconds are not retrospectively classified as misses.

For the condition-level display, exploratory IPA is the arithmetic mean of the available left- and right-eye candidate rates within each search. Four searches have values from both eyes; the fifth has no eligible pupil segment and remains missing. This display summary is not a validated workload score; the separate-eye diagnostics and retained durations appear in Figure D.5. Points in the time and IPA panels are repeated trials within one development run rather than independent participants, and the figure supports no estimate of a voice-condition effect. The participant-study comparison will use observations from both voices and paired participant summaries with consistent scoring and eligibility rules.

### *D.2.6 Eye Measures and Workload in Later Development Runs*

Figure D.7 applies the planned eye-measure and workload panels to three later development recordings from October 25, 2026: neutral-voice run 32 and self-similar runs 1 and 2. Unlike Run 30, these recordings include self-similar playback. All three use the angle-controlled schedule in its shortened development mode, the 25-second search limit, right-trigger selection, and the same guidance-policy version, and each session ended before completing a block. The 22 resolved searches comprise 14 neutral searches (12 completed, 2 timed out) and 8 self-similar searches (7 completed, 1 timed out). They are repeated development trials rather than independent participants or paired sessions.

Candidate fixations come from the dispersion-based detector [38] described in Section 3.5, applied to each eye within valid, contiguous search intervals; each plotted value is the mean of the two eyes' counts for one search. Candidate saccade frequency divides each eye's candidate saccades by its eligible seconds, which exclude intervals of rapid head movement, and averages the two rates. Both measures rely on the HTC query-receipt clock rather than a verified sensor capture clock, and neither detector has yet been validated against independent annotations. Last target fixation duration is unavailable because it requires a validated join between eye events, the target's area of interest, and the selection time. Raw NASA-TLX [15] is unavailable because these development sessions collected no questionnaire responses. Both panels are kept so that the missing measures remain visible.

Figure D.8 reports, for each search, the share of active search time eligible for each detector, taking the lower value of the two eyes. Fixation-eligible coverage was at least 94% in every search except one neutral search at 47%, and saccade-eligible coverage ranged from 43% to 93%. Because fixation counts are totals over eligible time, they scale with search duration and coverage, and the self-similar searches were longer (median 19.7 s versus 13.4 s). Voice, session, search duration,

and coverage therefore vary together, and the differences visible in panels (b) and (c) cannot be attributed to voice. No inferential test is applied. None of the three source recordings closed cleanly, and two end in a truncated record that was omitted. These figures show that the measurement pipeline runs end to end on headset recordings with both voices; they do not estimate a voice effect.

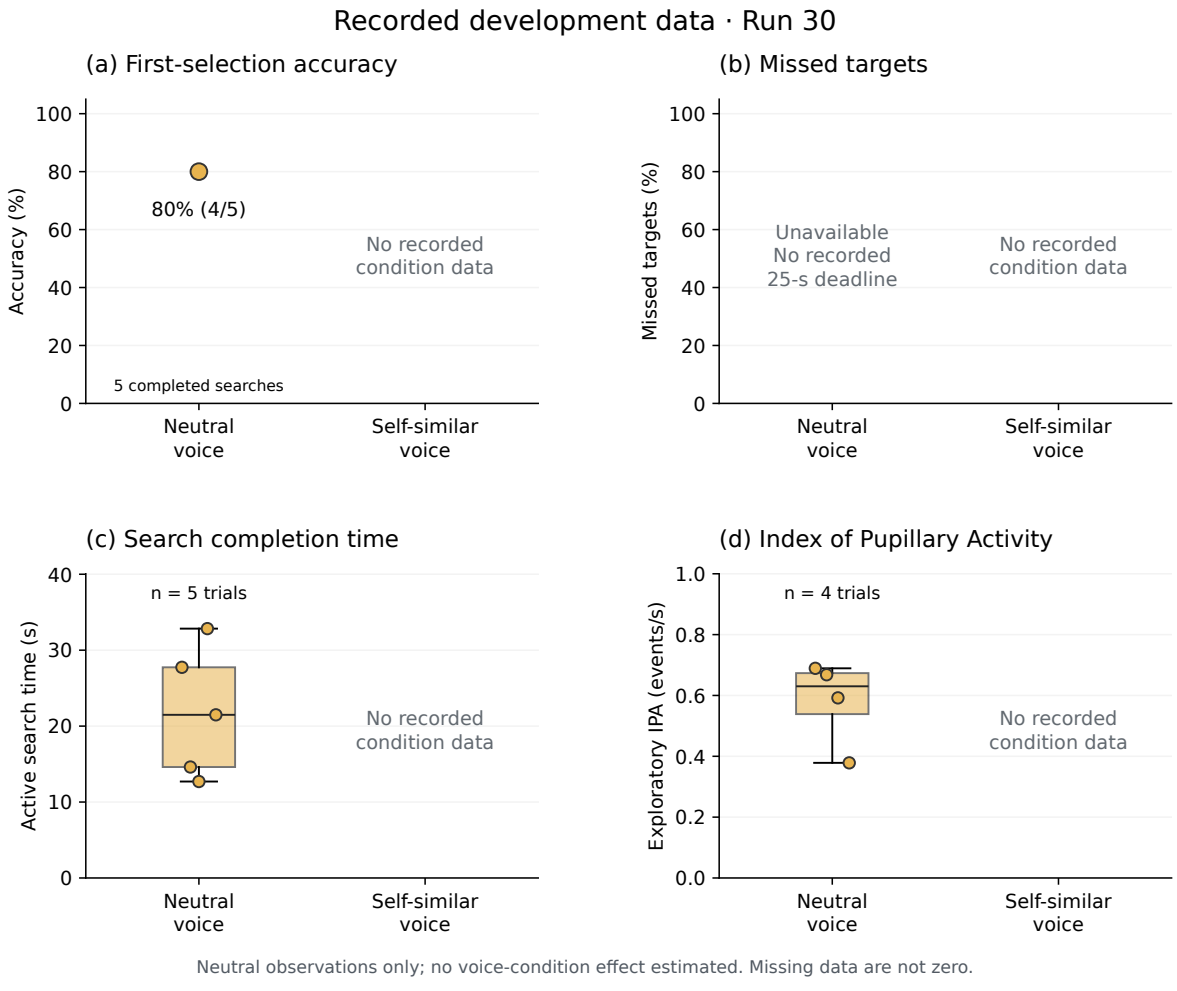



Figure D.6: Development outcome panels arranged for the neutral versus self-similar voice comparison. (a) First-selection accuracy among completed searches, shown as one run-level point. (b) Missed targets, unavailable for this recording without a recorded deadline. (c) Active search completion time. (d) Exploratory IPA averaged across available eyes within each trial. Boxes in (c) and (d) show quartiles and median, with 1.5-IQR whiskers; all observed trial values are shown. Unavailable values are not plotted as zero. No self-similar observations or inferential comparisons are included.

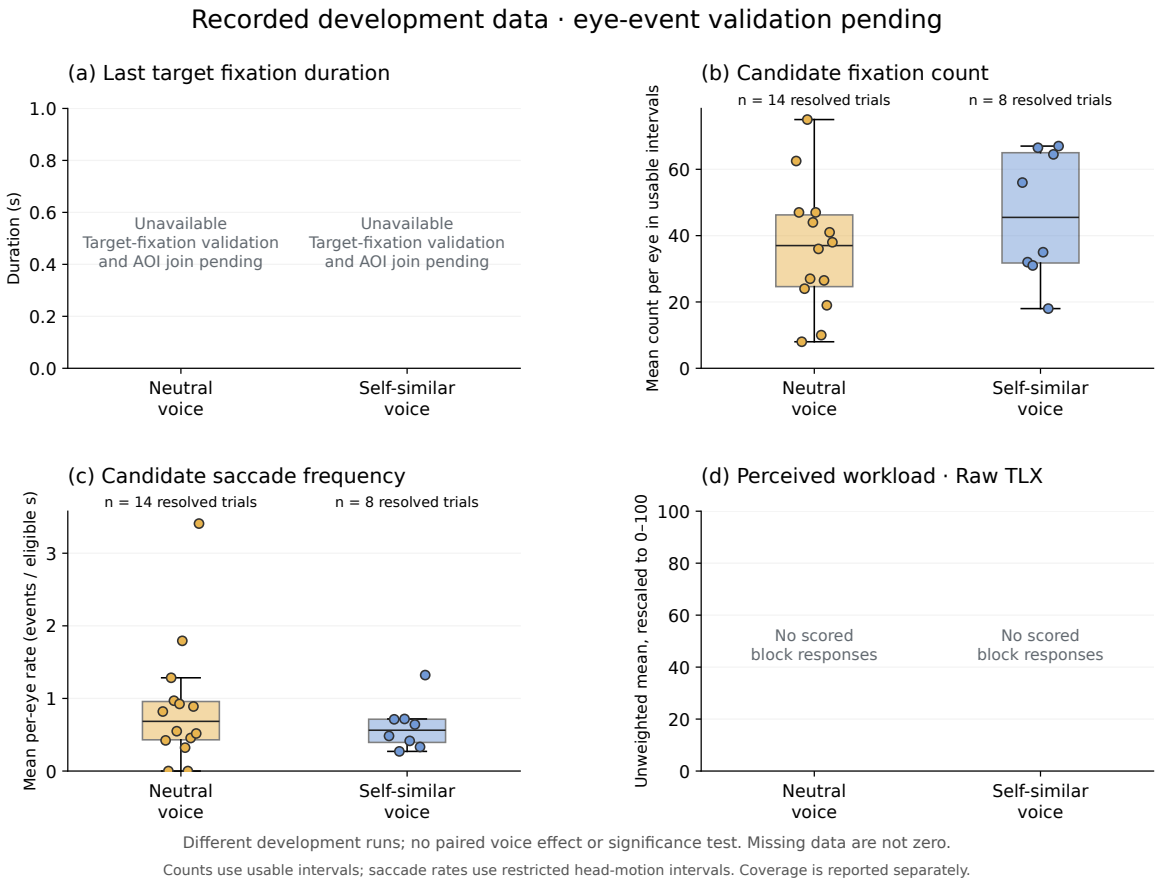



Figure D.7: Eye-measure and workload panels from October 25 development recordings. (a) Last target fixation duration and (d) raw NASA-TLX are unavailable and are left empty rather than plotted as zero. (b) Candidate fixation count, mean of the two eyes over eligible intervals. (c) Candidate saccade frequency, mean of the two eyes' events per eligible second during low head speed. Each point is one resolved search (neutral run 32, $n = 14$; self-similar runs 1 and 2, $n = 8$); boxes show quartiles and median with 1.5-IQR whiskers. The detectors are not yet validated, and no inferential comparison is made.

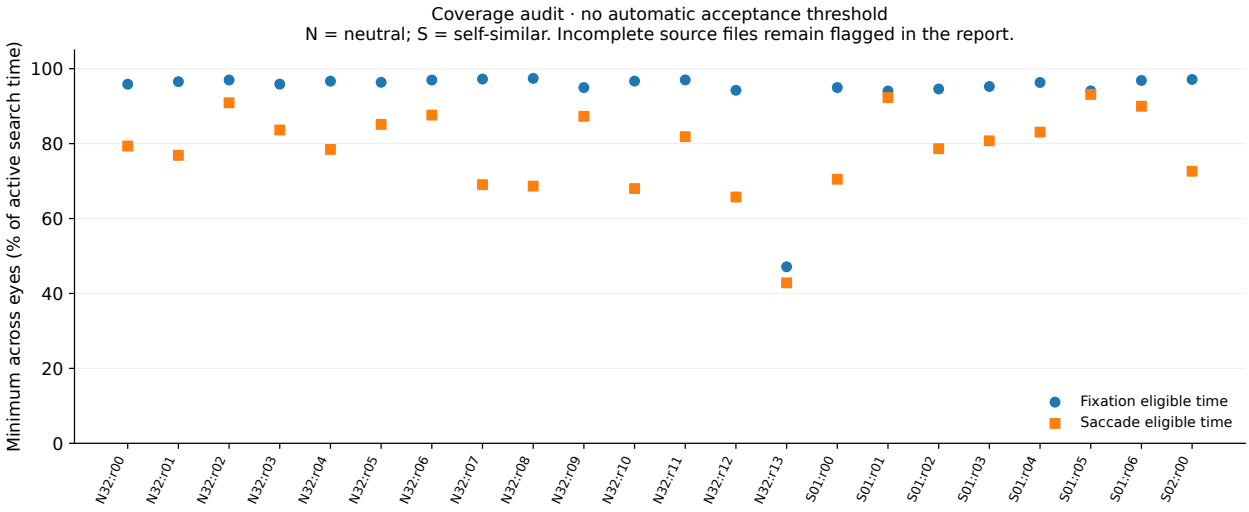



Figure D.8: Eye-measure coverage for each search in Figure D.7, as the percentage of active search time eligible for fixation and saccade detection, using the lower value of the two eyes. N32 denotes neutral run 32; S01 and S02 denote self-similar runs 1 and 2. No acceptance threshold is applied.

# LIST OF REFERENCES


[1] Laura Aymerich-Franch, Cody Karutz, and Jeremy N. Bailenson. Effects of Facial and Voice Similarity on Presence in a Public Speaking Virtual Environment. In *Proceedings of the International Society for Presence Research Annual Conference*, 2012.

[2] Dale J. Barr, Roger Levy, Christoph Scheepers, and Harry J. Tily. Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3):255–278, 2013. doi: 10.1016/j.jml.2012.11.001.

[3] Nicola Binetti, Luyan Wu, Shiping Chen, Ernst Kruijff, Simon Julier, and Duncan P. Brumby. Using visual and auditory cues to locate out-of-view objects in head-mounted augmented reality. *Displays*, 69:102032, 2021. doi: 10.1016/j.displa.2021.102032.

[4] Richard A. Bolt. "Put-That-There": Voice and gesture at the graphics interface. *ACM SIGGRAPH Computer Graphics*, 14(3):262–270, 1980. doi: 10.1145/965105.807503.

[5] Riccardo Bovo, Steven Abreu, Karan Ahuja, Eric J Gonzalez, Li-Te Cheng, and Mar Gonzalez-Franco. EmBARDiment: an Embodied AI Agent for Productivity in XR. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pages 708–717, 2025.

[6] Francesco Chiossi, Ines Trautmannsheimer, Changkun Ou, Uwe Gruenefeld, and Sven Mayer. Searching across realities: Investigating ERPs and eye-tracking correlates of visual search in mixed reality. *IEEE Transactions on Visualization and Computer Graphics*, 30(11):6997–7007, 2024. doi: 10.1109/TVCG.2024.3456172.

[7] James M. Coughlan, Brandon Biggs, Marc-Aurèle Rivière, and Huiying Shen. An audio-based 3D spatial guidance AR system for blind users. In *Computers Helping People with*

*Special Needs*, volume 12376 of *Lecture Notes in Computer Science*, pages 475–484. Springer, 2020. doi: 10.1007/978-3-030-58796-3_55.

[8] Alexander Diel and Michael Lewis. Deviation from typical organic voices best explains a vocal uncanny valley. *Computers in Human Behavior Reports*, 14:100430, 2024.

[9] Andrew T. Duchowski, Krzysztof Krejtz, Izabela Krejtz, Cezary Biele, Anna Niedzielska, Peter Kiefer, Martin Raubal, and Ioannis Giannopoulos. The index of pupillary activity: Measuring cognitive load vis-à-vis task difficulty with pupil oscillation. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pages 1–13, 2018.

[10] Matt J. Dunn, Robert G. Alexander, Onyekachukwu M. Amiebenomo, Gemma Arblaster, Denize Atan, Jonathan T. Erichsen, Ulrich Ettinger, Mario E. Giardini, Iain D. Gilchrist, Ruth Hamilton, Roy S. Hessels, Scott Hodgins, Ignace T. C. Hooge, Brooke S. Jackson, Helena Lee, Stephen L. Macknik, Susana Martinez-Conde, Lee Mcilreavy, Lisa M. Muratori, Diederick C. Niehorster, Marcus Nyström, Jorge Otero-Millan, Michael M. Schlüssel, Jay E. Self, Tarkeshwar Singh, Nikolaos Smyrnis, and Andreas Sprenger. Minimal reporting guideline for research involving eye tracking (2023 edition). *Behavior Research Methods*, 56(5):4351–4357, 2024.

[11] Cathy Mengying Fang, Phoebe Chua, Samantha W. T. Chan, Joanne Leong, Andria Bao, and Pattie Maes. Leveraging AI-Generated Emotional Self-Voice to Nudge People towards their Ideal Selves. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–20, 2025.

[12] Steve Grogorick, Michael Stengel, Elmar Eisemann, and Marcus Magnor. Subtle gaze guidance for immersive environments. In *Proceedings of the ACM Symposium on Applied Perception*, pages 1–7, 2017. doi: 10.1145/3119881.3119890.

[13] Uwe Gruenefeld, Abdallah El Ali, Wilko Heuten, and Susanne Boll. Visualizing out-of-view objects in head-mounted augmented reality. In *Proceedings of the 19th International Conference on Human-Computer Interaction with Mobile Devices and Services*, pages 1–7, 2017. doi: 10.1145/3098279.3122124.

[14] Siqi Guo, Minsoo Choi, Dominic Kao, and Christos Mousas. Collaborating with my Doppelgänger: The Effects of Self-similar Appearance and Voice of a Virtual Character during a Jigsaw Puzzle Co-solving Task. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 7(1):1–23, 2024. doi: 10.1145/3651288.

[15] Sandra G. Hart and Lowell E. Staveland. Development of NASA-TLX (task load index): Results of empirical and theoretical research. In Peter A. Hancock and Najmedin Meshkati, editors, *Human Mental Workload*, volume 52 of *Advances in Psychology*, pages 139–183. Elsevier Science Publishers (North-Holland), 1988. doi: 10.1016/S0166-4115(08)62386-9.

[16] Chin-Chang Ho and Karl F. MacDorman. Measuring the uncanny valley effect: Refinements to indices for perceived humanness, attractiveness, and eeriness. *International Journal of Social Robotics*, 9(1):129–139, 2017.

[17] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. JSTOR: 4615733.

[18] HTC VIVE. VIVE Focus Vision: Overview and specifications. `https://www.vive.com/us/product/vive-focus-vision/overview/`, 2026. Manufacturer specifications; accessed October 23, 2026.

[19] Oliver Jaggy, Stephan Schwan, and Hauke S. Meyerhoff. AI-determined similarity increases likability and trustworthiness of human voices. *PLOS ONE*, 20(3):e0318890, 2025. doi: 10.1371/journal.pone.0318890.

[20] Dominic Kao, Rabindra Ratan, Christos Mousas, and Alejandra J. Magana. The effects of a self-similar avatar voice in educational games. *Proceedings of the ACM on Human-Computer Interaction*, 5(CHI PLAY):1–28, 2021. doi: 10.1145/3474665.

[21] Brendan Kelley, Ryan P. McMahan, Christopher D. Wickens, Benjamin A. Clegg, and Francisco R. Ortega. The importance of cueing while visually searching a 360 degree environment for multiple targets in the presence of distractors. In *Proceedings of the 31st ACM Symposium on Virtual Reality Software and Technology*, pages 1–11, 2025. doi: 10.1145/3756884.3765992.

[22] Yoonsang Kim, Divyansh Pradhan, Devshree Jadeja, and Arie E. Kaufman. From speech-to-spatial: Grounding utterances on a live shared view with augmented reality. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 228–238, 2026. doi: 10.1109/VR67842.2026.00045.

[23] Tiffany C. K. Kwok, Peter Kiefer, and Martin Raubal. Two-step gaze guidance. In *Proceedings of the 2022 International Conference on Multimodal Interaction*, pages 299–309, 2022. doi: 10.1145/3536221.3556612.

[24] Daniel Lange, Tim Claudius Stratmann, Uwe Gruenefeld, and Susanne Boll. HiveFive: Immersion preserving attention guidance in virtual reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pages 1–13, 2020. doi: 10.1145/3313831.3376803.

[25] Juliana H. Madritsch, Tomislav Duricic, Neven ElSayed, Simone Kopeinik, and Eduardo Veas. Ai-powered conversational assistance in augmented reality for multi-step tasks. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 519–529, 2026.

[26] Alexander Marquardt, Christina Trepkowski, Tom David Eibich, Jens Maiero, Ernst Kruijff, and Johannes Schöning. Comparing non-visual and visual guidance methods for narrow

field of view augmented reality displays. *IEEE Transactions on Visualization and Computer Graphics*, 26(12):3389–3401, 2020. doi: 10.1109/TVCG.2020.3023605.

[27] Edward McAuley, Terry Duncan, and Vance V. Tammen. Psychometric Properties of the Intrinsic Motivation Inventory in a Competitive Sport Setting: A Confirmatory Factor Analysis. *Research Quarterly for Exercise and Sport*, 60(1):48–58, 1989.

[28] Clifford Nass and Kwan Min Lee. Does computer-synthesized speech manifest personality? Experimental tests of recognition, similarity-attraction, and consistency-attraction. *Journal of Experimental Psychology: Applied*, 7(3):171–181, 2001.

[29] Greeshma Nerella, Daisy Gan, Brendan David-John, and Rawan Alghofaili. Gaze-based prediction of cognitive load in augmented reality. *Proceedings of the ACM on Human-Computer Interaction*, 10(3), May 2026. Article ETRA026. doi: 10.1145/3806040.

[30] Diederick C. Niehorster, Marcus Nyström, Roy S. Hessels, Jeroen S. Benjamins, Richard Andersson, and Ignace T. C. Hooge. The fundamentals of eye tracking, Part 7: Determining data quality. *Behavior Research Methods*, 58(7), 2026.

[31] Kevin Pfeil, Eugene M. Taranta, Arun Kulshreshth, Pamela Wisniewski, and Joseph J. LaViola. A comparison of eye-head coordination between virtual and physical realities. In *Proceedings of the ACM Symposium on Applied Perception*, pages 1–7, 2018.

[32] Zengyi Qin, Wenliang Zhao, Xumin Yu, and Xin Sun. OpenVoice: Versatile instant voice cloning. *arXiv preprint arXiv:2312.01479*, 2023. arXiv:2312.01479.

[33] Kelsey Quinn and Joseph L. Gabbard. Augmented reality visualization techniques for attention guidance to out-of-view objects: A systematic review. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 826–835, 2024. doi: 10.1109/ISMAR62088.2024.00098.

[34] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019. doi: 10.18653/v1/D19-1410.

[35] Victor Rosi, Emma Soopramanien, and Carolyn McGettigan. Perception and social evaluation of cloned and recorded voices: Effects of familiarity and self-relevance. *Computers in Human Behavior: Artificial Humans*, 4:100143, 2025.

[36] Richard M. Ryan. Control and information in the intrapersonal sphere: An extension of cognitive evaluation theory. *Journal of Personality and Social Psychology*, 43(3):450–461, 1982.

[37] Hila Sabouni, Jack Miller, Stephen B. Gilbert, and Chengde Wu. Effectiveness of visual, auditory, and haptic guidance cues for visual targets in virtual exhibition environments. *Journal of Perceptual Imaging*, 8:1–11, 2025. doi: 10.2352/J.Percept.Imaging.2025.8.000403.

[38] Dario D. Salvucci and Joseph H. Goldberg. Identifying fixations and saccades in eye-tracking protocols. In *Proceedings of the symposium on Eye tracking research & applications - ETRA '00*, pages 71–78, 2000.

[39] Kengo Tanaka, Renato Alexandre Ribeiro, João Guerreiro, Hironobu Takagi, Yoichi Ochiai, and Chieko Asakawa. From coarse search to precise alignment: Auditory guidance for non-visual target acquisition in VR. In *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '26)*, pages 1–16, Vila Nova de Gaia, Portugal, 2026. ACM. doi: 10.1145/3797867.3828980.

[40] Christina Trepkowski, David Eibich, Jens Maiero, Alexander Marquardt, Ernst Kruijff, and Steven Feiner. The effect of narrow field of view and information density on visual search performance in augmented reality. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 575–584, 2019. doi: 10.1109/VR.2019.8798312.

[41] Rohith Venkatakrishnan, Roshan Venkatakrishnan, Ahmed Rageeb Ahsan, Andrew W. Tompkins, Eric D. Ragan, and Jaime Ruiz. The effects of an agent's clarifying questioning behavior during task-guidance and collaboration in virtual reality. In *2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2026. To appear.

[42] Isaac Wang, Lea Buchweitz, Jesse Smith, Lara-Sophie Bornholdt, Jonas Grund, Jaime Ruiz, and Oliver Korn. Wow, you are terrible at this! an intercultural study on virtual agents giving mixed feedback. In *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents (IVA '20)*, pages 55:1–55:8. ACM, 2020. doi: 10.1145/3383652.3423887.

[43] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 5776–5788, 2020.

[44] E. J. Williams. Experimental designs balanced for the estimation of residual effects of treatments. *Australian Journal of Scientific Research Series A: Physical Sciences*, 2(2):149–168, 1949. doi: 10.1071/CH9490149.

[45] Songyue Yang, Kang Yue, Haolin Gao, Mei Guo, Haoran Jia, Zhonghao Zhu, Fanlu Zeng, and Yue Liu. Synchronizing with attentional sampling: Modality-specific flicker guidance for gaze and hand-eye interaction in VR. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 109–119. IEEE, 2026. doi: 10.1109/VR67842.2026.00034.

[46] Yuchong Zhang, Adam Nowak, Yueming Xuan, Andrzej Romanowski, and Morten Fjeld. See or hear? exploring the effect of visual/audio hints and gaze-assisted instant post-task feedback for visual search tasks in AR. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 1113–1122, 2023. doi: 10.1109/ISMAR59233.2023.00128.